

2_EDA

2025-06-11

Data Cleaning and Exploratory Data Analysis

This R markdown document will be used to perform Exploratory Data Analysis (EDA) on the merged_df_raw that was developed in the previous workbook 1_data_extraction.Rmd.

Data Importing

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.4
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(readr)
library(ggplot2)
library(grid)
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(here)
```

```
## here() starts at /home/rstudio/Documents/ads503_project_g1
```

```
library(naniar)
library(VIM)
```

```
## Loading required package: colorspace
## VIM is ready to use.
##
```

```

## Suggestions and bug-reports can be submitted at: https://github.com/statistikat/VIM/issues
##
## Attaching package: 'VIM'
##
## The following object is masked from 'package:datasets':
##
##     sleep
library(skimr)

##
## Attaching package: 'skimr'
##
## The following object is masked from 'package:naniar':
##
##     n_complete
library(stringr)
library(rcompanion)

merged_df <- read_csv(here('Data/Raw', 'merged_df_raw.csv'))

## Rows: 899 Columns: 77
## -- Column specification -----
## Delimiter: ","
## chr (2): name, location
## dbl (75): id, ccf, age, sex, painloc, painexer, relrest, pncaden, cp, trestb...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
head(merged_df)

## # A tibble: 6 x 77
##       id    ccf    age    sex painloc painexer relrest pncaden    cp trestbps    htn
##   <dbl> <dbl> <dbl> <dbl>   <dbl>   <dbl>   <dbl>   <dbl> <dbl>   <dbl> <dbl>
## 1     1     0    63     1     -9     -9     -9     -9     1    145     1
## 2     2     0    67     1     -9     -9     -9     -9     4    160     1
## 3     3     0    67     1     -9     -9     -9     -9     4    120     1
## 4     4     0    37     1     -9     -9     -9     -9     3    130     0
## 5     6     0    41     0     -9     -9     -9     -9     2    130     1
## 6     7     0    56     1     -9     -9     -9     -9     2    120     1
## # i 66 more variables: chol <dbl>, smoke <dbl>, cigs <dbl>, years <dbl>,
## #   fbs <dbl>, dm <dbl>, famhist <dbl>, restecg <dbl>, ekgmo <dbl>,
## #   ekgday <dbl>, ekgyr <dbl>, dig <dbl>, prop <dbl>, nitr <dbl>, pro <dbl>,
## #   diuretic <dbl>, proto <dbl>, thaldur <dbl>, thaltime <dbl>, met <dbl>,
## #   thalach <dbl>, thalrest <dbl>, tpeakbps <dbl>, tpeakbpd <dbl>, dummy <dbl>,
## #   trestbpd <dbl>, exang <dbl>, xhypo <dbl>, oldpeak <dbl>, slope <dbl>,
## #   rldv5 <dbl>, rldv5e <dbl>, ca <dbl>, restckm <dbl>, exerckm <dbl>, ...

```

Categorical Conversions

```

## categorical features and their levels and labels
#sex
merged_df$sex <- factor(merged_df$sex,
                        levels = c(0, 1),

```

```

        labels = c("Female", "Male")
    )

#painloc
#unique(merged_df$painloc)
merged_df$painloc <- factor(merged_df$painloc,
    levels = c(0, 1),
    labels = c("Otherwise", "Substernal")
)

#painexer
#unique(merged_df$painexer)
merged_df$painexer <- factor(merged_df$painexer,
    levels = c(0, 1),
    labels = c("Otherwise", "Provoked by Exertion")
)

#relrest
#unique(merged_df$relrest)
merged_df$relrest <- factor(merged_df$relrest,
    levels = c(0, 1),
    labels = c("Otherwise", "Relieved after rest")
)

#cp
#unique(merged_df$cp)
merged_df$cp <- factor(merged_df$cp,
    levels = c(1, 2, 3, 4),
    labels = c("Typical Angina", "Atypical Angina",
        "Non-anginal Pain", "Asymptomatic"),
    ordered = TRUE)

#smoke
#unique(merged_df$smoke)
merged_df$smoke <- factor(merged_df$smoke,
    levels = c(0, 1),
    labels = c("No", "Yes")
)

#fbs
#unique(merged_df$fbs)
merged_df$fbs <- factor(merged_df$fbs,
    levels = c(0, 1),
    labels = c("No", "Yes")
)

#dm
#unique(merged_df$dm)
merged_df$dm <- factor(merged_df$dm,
    levels = c(0, 1),
    labels = c("No", "Yes")
)

#famhist
#unique(merged_df$famhist)
merged_df$famhist <- factor(merged_df$famhist,
    levels = c(0, 1),
    labels = c("No", "Yes")
)

#restecg
#unique(merged_df$restecg)

```

```

merged_df$restecg <- factor(merged_df$restecg,
  levels = c(0, 1, 2),
  labels = c("Normal", "Abnormal", "Hypertrophy"),
  ordered = TRUE)

#dig
#unique(merged_df$dig)
merged_df$dig <- factor(merged_df$dig,
  levels = c(0, 1),
  labels = c("No", "Yes")
)

#prop
#unique(merged_df$prop)
# find the number of 22's a feature outside the range/
#sum(merged_df$prop == 22, na.rm = TRUE)
# convert the 22 to NA
merged_df <- merged_df %>%
  mutate(prop = ifelse(prop == 22, NA, prop))
#create factor
merged_df$prop <- factor(merged_df$prop,
  levels = c(0, 1),
  labels = c("No", "Yes")
)

#nitr
#unique(merged_df$nitr)
merged_df$nitr <- factor(merged_df$nitr,
  levels = c(0, 1),
  labels = c("No", "Yes")
)

#pro
#unique(merged_df$pro)
merged_df$pro <- factor(merged_df$pro,
  levels = c(0, 1),
  labels = c("No", "Yes")
)

#diuretic
#unique(merged_df$diuretic)
merged_df$diuretic <- factor(merged_df$diuretic,
  levels = c(0, 1),
  labels = c("No", "Yes")
)

#proto
#unique(merged_df$proto)
merged_df$proto <- factor(merged_df$proto,
  levels = c(0, 1),
  labels = c("No", "Yes")
)

#exang
#unique(merged_df$exang)
merged_df$exang <- factor(merged_df$exang,
  levels = c(0, 1),
  labels = c("No", "Yes")
)

#xhypo

```

```

#unique(merged_df$xhypo)
merged_df$xhypo <- factor(merged_df$xhypo,
  levels = c(0, 1),
  labels = c("No", "Yes")
)

#slope
#unique(merged_df$slope)
# check count of 0 valuyes
#sum(merged_df$slope == 0, na.rm = TRUE)
# replace 0 with NA
merged_df <- merged_df %>%
  mutate(slope = ifelse(slope == 0, NA, slope))
# factor
merged_df$slope <- factor(merged_df$slope,
  levels = c(1, 2, 3),
  labels = c("Upsloping", "Flat", "Downsloping"),
  ordered = TRUE
)

#restwm
#unique(merged_df$restwm)
# check count of 0 valuyes
# factor
merged_df$restwm <- factor(merged_df$restwm,
  levels = c(0, 1, 2, 3),
  labels = c("None", "Mild/Moderate", "Moderate/Severe",
    "Akinesis"),
  ordered = TRUE
)

#thal
#unique(merged_df$thal)
# examine 1, 2, 5 values which are not in the documentation
# check count of 1,2,5 values
#sum(merged_df$thal == 1, na.rm = TRUE)
#sum(merged_df$thal == 2, na.rm = TRUE)
#sum(merged_df$thal == 5, na.rm = TRUE)
# replace 1, 2, 5 with na
merged_df <- merged_df %>%
  mutate(thal = ifelse(thal == 1, NA, thal)) %>%
  mutate(thal = ifelse(thal == 2, NA, thal)) %>%
  mutate(thal = ifelse(thal == 4, NA, thal)) %>%
  mutate(thal = ifelse(thal == 5, NA, thal))

```

Data Cleaning

This below section of code starts by removing columns that are unused as indicated by the `heart-disease.names` text file which serves as a data dictionary and source for information on all the data sets. Once the explicitly unnecessary columns are removed, we convert the -9's and -9.0's into NA values and set a threshold for feature removal of if the feature is missing more than 60% of the observations it is immediately dropped. Once these features have been removed we focus on removing near zero variance features.

```

# remove unnecessary columns as indicated in heart-disease.names
# cols marked as not used or dummy variables
not_used_dummy_var <- c(
  "id", "ccf", "dummy", "restckm", "exerckm", "thalsev", "thalpul",

```

```

"earlobe", "lvx1", "lvx2", "lvx3", "lvx4", "lvf", "cathef", "junk", "name"
)

df_cleaning <- merged_df %>%
  select(-all_of(not_used_dummy_var))

# convert -9 to NA
df_cleaning <- df_cleaning %>%
  mutate(across(where(is.character), ~ na_if(.x, "-9"))) %>% # character columns
  mutate(across(where(is.numeric), ~ na_if(.x, -9))) # numeric columns
#Get NA count
sum(is.na(df_cleaning))

## [1] 15881

#get dimensions of the data
df_cleaning %>% dim

## [1] 899 61

#reexamine the head
df_cleaning %>% head()

## # A tibble: 6 x 61
##   age sex   painloc painexer relrest pncaden cp   trestbps   htn   chol smoke
##   <dbl> <fct> <fct> <fct> <fct> <dbl> <ord> <dbl> <dbl> <dbl> <fct>
## 1 63 Male <NA> <NA> <NA> NA Typi~ 145 1 233 <NA>
## 2 67 Male <NA> <NA> <NA> NA Asym~ 160 1 286 <NA>
## 3 67 Male <NA> <NA> <NA> NA Asym~ 120 1 229 <NA>
## 4 37 Male <NA> <NA> <NA> NA Non~ 130 0 250 <NA>
## 5 41 Female <NA> <NA> <NA> NA Atyp~ 130 1 204 <NA>
## 6 56 Male <NA> <NA> <NA> NA Atyp~ 120 1 236 <NA>
## # i 50 more variables: cigs <dbl>, years <dbl>, fbs <fct>, dm <fct>,
## # famhist <fct>, restecg <ord>, ekgmo <dbl>, ekgday <dbl>, ekgyr <dbl>,
## # dig <fct>, prop <fct>, nitr <fct>, pro <fct>, diuretic <fct>, proto <fct>,
## # thaldur <dbl>, thaltime <dbl>, met <dbl>, thalach <dbl>, thalrest <dbl>,
## # tpeakbps <dbl>, tpeakbpd <dbl>, trestbpd <dbl>, exang <fct>, xhypo <fct>,
## # oldpeak <dbl>, slope <ord>, rldv5 <dbl>, rldv5e <dbl>, ca <dbl>,
## # restef <dbl>, restwm <ord>, exeref <dbl>, exerwm <dbl>, thal <dbl>, ...

# calculate percentage missing
missing_threshold <- 60
missing_pct <- df_cleaning %>%
  summarise(across(everything(), ~ mean(is.na(.)) * 100)) %>%
  pivot_longer(everything(), names_to = "column", values_to = "missing_percent") %>%
  arrange(desc(missing_percent))

# filter by col missing
missing_cols <- missing_pct %>%
  filter(missing_percent > missing_threshold)
var_missing_gt_60 <- missing_cols$column

missing_cols

## # A tibble: 12 x 2
##   column missing_percent

```

```
##      <chr>          <dbl>
## 1 pncaden          100
## 2 exeref           99.8
## 3 exerwm           99.4
## 4 restef           96.9
## 5 restwm           96.7
## 6 dm               89.4
## 7 smoke            74.4
## 8 ca                67.6
## 9 proto            64.0
## 10 om2              63.6
## 11 ramus            63.1
## 12 diag             62.1

# drop cols missing greater than threshold
df_cleaning <- df_cleaning %>%
  select(-all_of(var_missing_gt_60))
dim(df_cleaning)

## [1] 899 49

# check near zero variance vars
nzv_cols <- nearZeroVar(df_cleaning %>%
  select(-num), saveMetrics = TRUE)
nzv_cols %>% filter(nzv == TRUE)

##      freqRatio percentUnique zeroVar  nzv
## dig      27.65517      0.2224694  FALSE TRUE
## xhypo     37.22727      0.2224694  FALSE TRUE

# get column names of nzv
nzv_names <- nzv_cols %>%
  filter(nzv == TRUE) %>%
  rownames()
nzv_names

## [1] "dig"  "xhypo"

# remove nzv variables
df_cleaning <- df_cleaning %>%
  select(-all_of(nzv_names))

# add back ca column for EDA
df_cleaning$ca <- merged_df$ca
df_cleaning$ca <- na_if(df_cleaning$ca, -9)

dim(df_cleaning)

## [1] 899 48
```

In the process of removing variables with >60% missing data, the `ca` variable, which represents *Major vessels via fluoroscopy* was removed due to having 67.6% missing data. This variable was part of the *cleveland-14* set. It has re-added to the final data set after *nzv* variables were removed.

Export cleaned dataset

```
# export to file
write.csv(df_cleaning, here("Data/Clean", "heart_disease_cleaned.csv"), row.names = FALSE)
saveRDS(df_cleaning, here("Data/Clean", "heart_disease_cleaned.rds"))
```

Exploratory Data Analysis

```
df <- readRDS(here("Data/Clean", "heart_disease_cleaned.rds"))
# head(df)

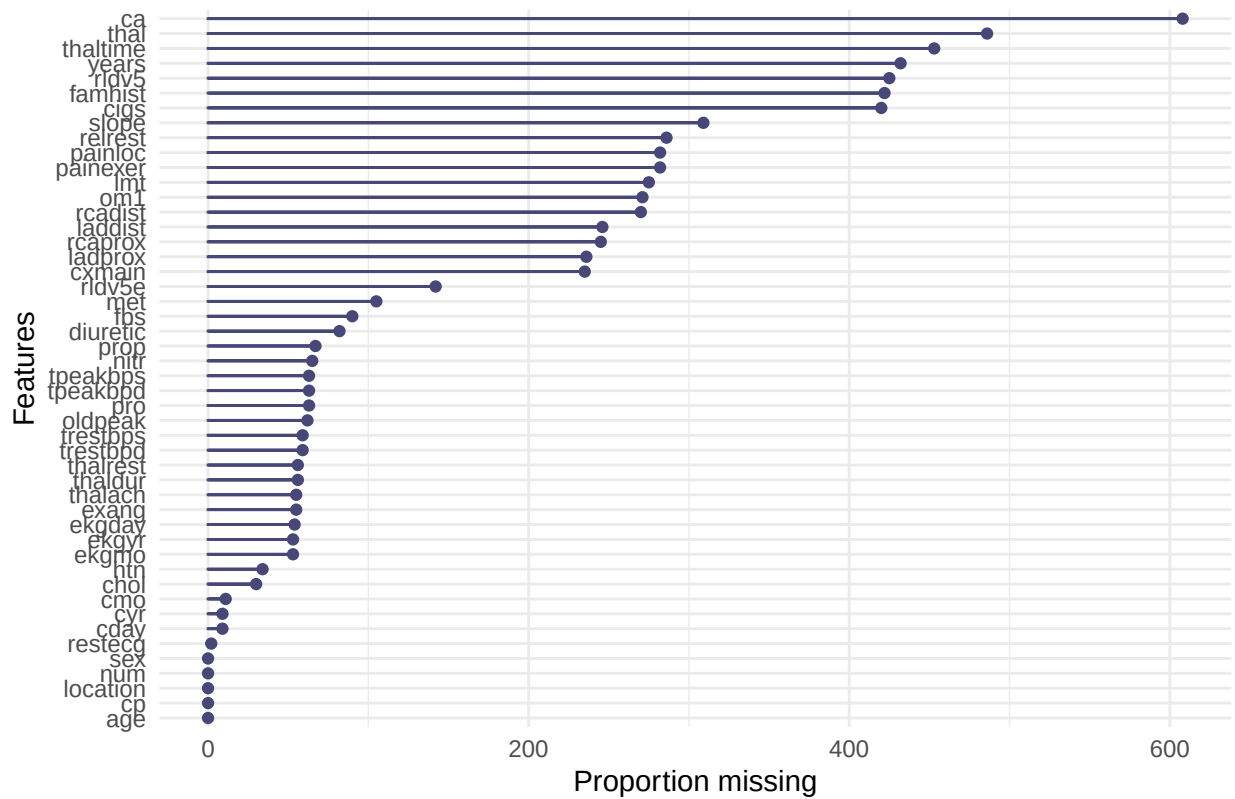
missing_var <- df %>%
  summarise(across(everything(), ~ mean(is.na(.)))) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "missing_pct")

missing_var %>%
  filter(missing_pct > 0) %>%
  arrange(desc(missing_pct))
```

```
## # A tibble: 43 x 2
##   variable missing_pct
##   <chr>          <dbl>
## 1 ca              0.676
## 2 thal            0.541
## 3 thaltime        0.504
## 4 years           0.481
## 5 rldv5           0.473
## 6 famhist         0.469
## 7 cigs            0.467
## 8 slope           0.344
## 9 relrest         0.318
## 10 painloc        0.314
## # i 33 more rows
```

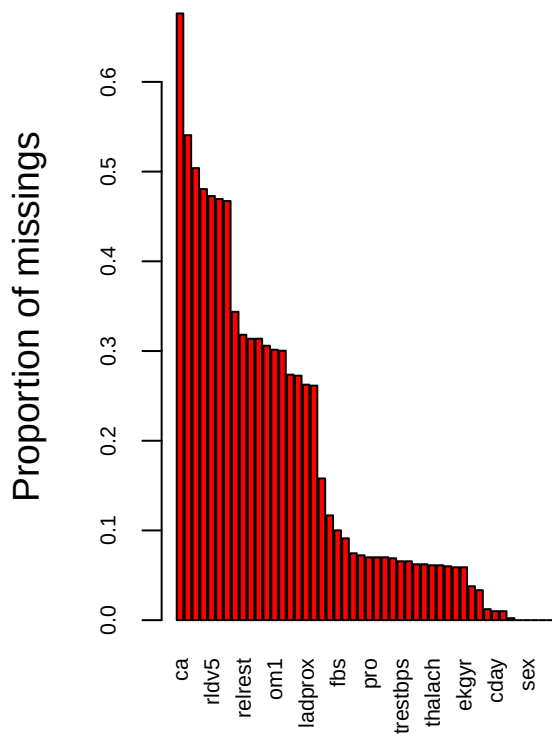
```
# bar chart of missingness per feature
gg_miss_var(df) +
  labs(
    title = "Missing Data per Feature",
    x = "Features",
    y = "Proportion missing"
  )
```


Missing Data per Feature

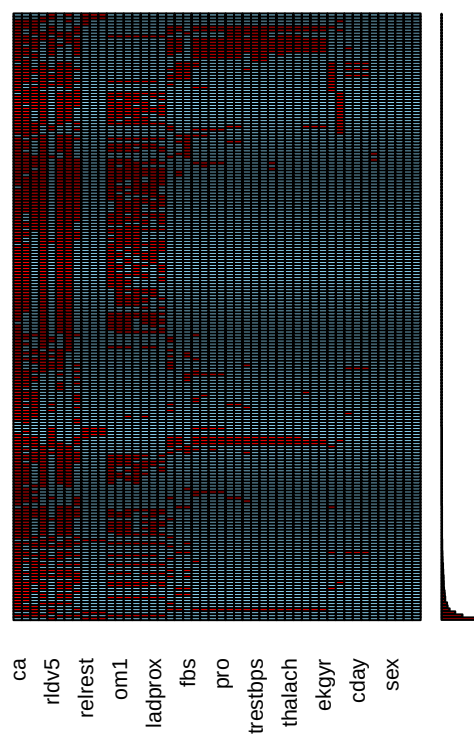


```
# missingness matrix
aggr(df, numbers = TRUE, sortVars = TRUE, cex.axis = 0.7,
     main = "Missingness Matrix",
     sub = "Red = missing, Blue = observed")
```

```
## Warning in plot.aggr(res, ...): not enough vertical space to display
## frequencies (too many combinations)
```



Combinations



```
##
## Variables sorted by number of missings:
## Variable      Count
##      ca 0.676307008
##      thal 0.540600667
## thaltime 0.503893215
##      years 0.480533927
##      rldv5 0.472747497
## famhist 0.469410456
##      cigs 0.467185762
##      slope 0.343715239
## relrest 0.318131257
## painloc 0.313681869
## painexer 0.313681869
##      lmt 0.305895439
##      om1 0.301446051
## rcadist 0.300333704
## laddist 0.273637375
## rcaprox 0.272525028
## ladprox 0.262513904
## cxmain 0.261401557
## rldv5e 0.157953281
##      met 0.116796440
##      fbs 0.100111235
## diuretic 0.091212458
##      prop 0.074527253
```

```
##      nitr 0.072302558
##      pro 0.070077864
## tpeakbps 0.070077864
## tpeakbpd 0.070077864
##   oldpeak 0.068965517
## trestbps 0.065628476
## trestbpd 0.065628476
##   thaldur 0.062291435
## thalrest 0.062291435
##   thalach 0.061179088
##     exang 0.061179088
##   ekgday 0.060066741
##   ekgmo 0.058954394
##   ekgyr 0.058954394
##     htn 0.037819800
##   chol 0.033370412
##     cmo 0.012235818
##   cday 0.010011123
##   cyr 0.010011123
## restecg 0.002224694
##   age 0.000000000
##   sex 0.000000000
##   cp 0.000000000
##   num 0.000000000
## location 0.000000000
```

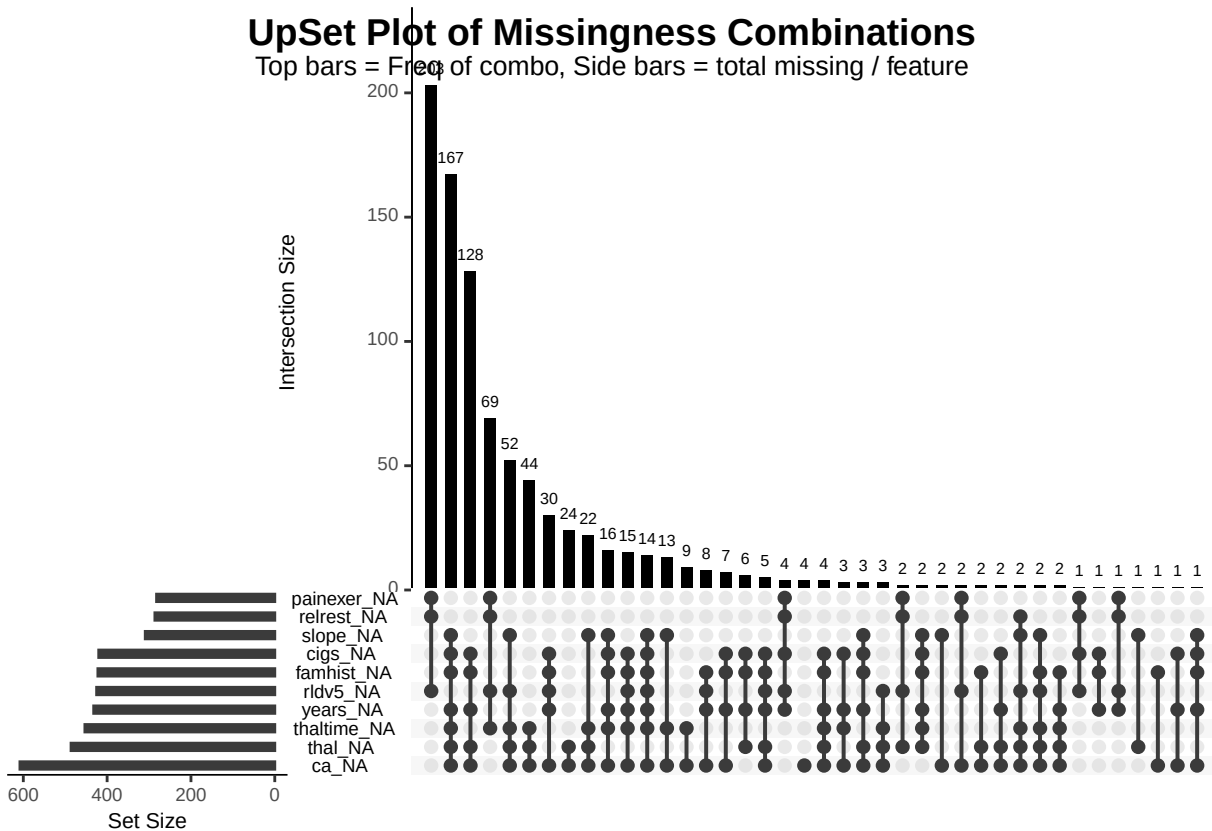
```
# missingness correlation / upset plot
```

```
gg_miss_upset(df, nsets = 10, main.bar.color = "black")
```

```
# Add a title manually
```

```
grid.text("UpSet Plot of Missingness Combinations", y = 0.95, gp = gpar(fontsize = 14, fontface = "bold"
```

```
grid.text("Top bars = Freq of combo, Side bars = total missing / feature", y = 0.91, gp = gpar(fontsize
```



Variables Missing > 40%

- ca: cleveland-14 variable - represents # of major vessels (0 - 3) colored by fluoroscopy.
- thal: cleveland-14 variable - results from thallium stress test
- thaltime: time duration for thallium stress test
- years: number of years as a smoker
- rldv5: height at rest
- famhist: family history of coronary artery disease
- cigs: cigarettes smoked per day

Upset

The UpSet plot shows the intersections of variable sets, similar to a Venn Diagram. This plot shows the combinations of variables that have missing values and the total of those combinations.

- based on upset table, ca, thal, and thaltime are missing together
- missingness might not be random

Custom Functions

```
# function for showing histogram
show_histogram <- function(dataframe, feature) {
  ggplot(dataframe, aes(x = {{feature}})) +
    geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
    labs(
      title = paste("Histogram of", rlang::as_label(enquo(feature))),

```

```

    x = rlang::as_label(enquo(feature)),
    y = "Count"
  ) +
  theme_minimal()
}
# function for showing density plot
show_densityplot <- function(dataframe, feature) {
  ggplot(df, aes(x = {{feature}})) +
    geom_density(fill = "darkorange", alpha = 0.5) +
    labs(
      title = paste("Density Plot of", rlang::as_label(enquo(feature))),
      x = rlang::as_label(enquo(feature)),
      y = "Count"
    ) +
    theme_minimal()
}

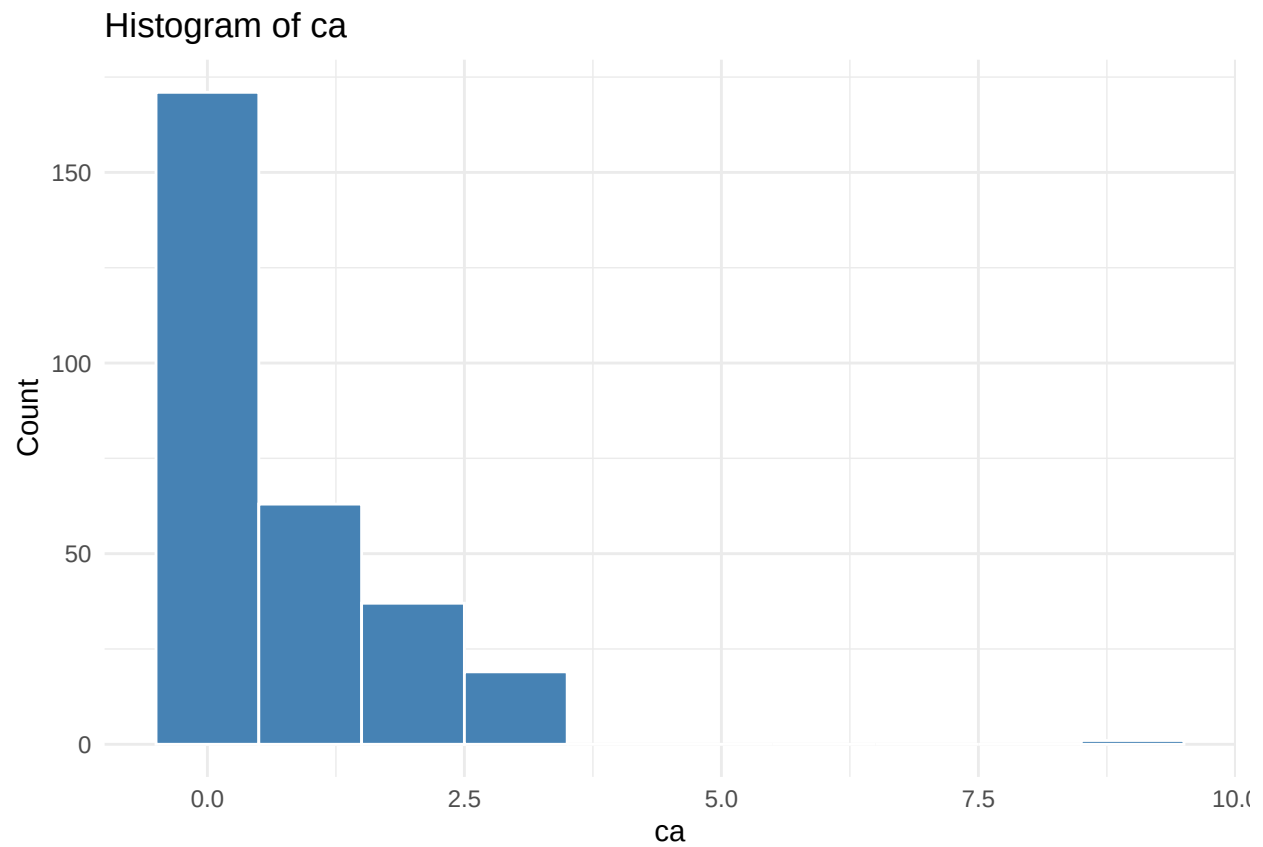
show_barplot <- function(dataframe, feature) {
  ggplot(dataframe, aes(x = {{feature}})) +
    geom_bar(fill = "steelblue") +
    labs(
      title = paste("Bar Plot of", rlang::as_label(enquo(feature))),
      x = rlang::as_label(enquo(feature)),
      y = "Count"
    ) +
    theme_minimal()
}

```

Feature: ca

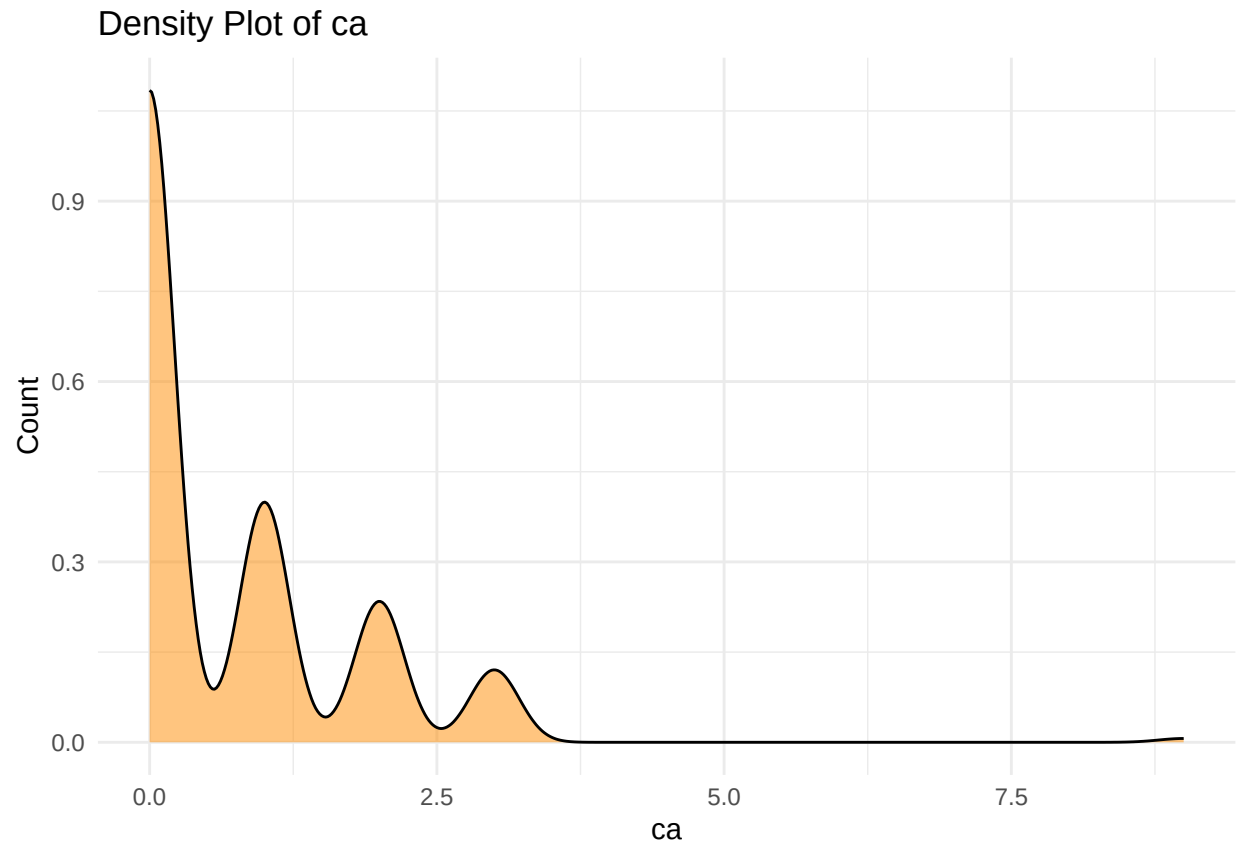
```
show_histogram(df, ca)
```

```
## Warning: Removed 608 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



```
show_densityplot(df, ca)
```

```
## Warning: Removed 608 rows containing non-finite outside the scale range  
## (`stat_density()`).
```



```
# impute with median and missingness flag
df <- df %>%
  mutate(
    ca_missing = ifelse(is.na(ca), 1, 0),
    ca_imputed = ifelse(is.na(ca), median(ca, na.rm = TRUE), ca)
  )

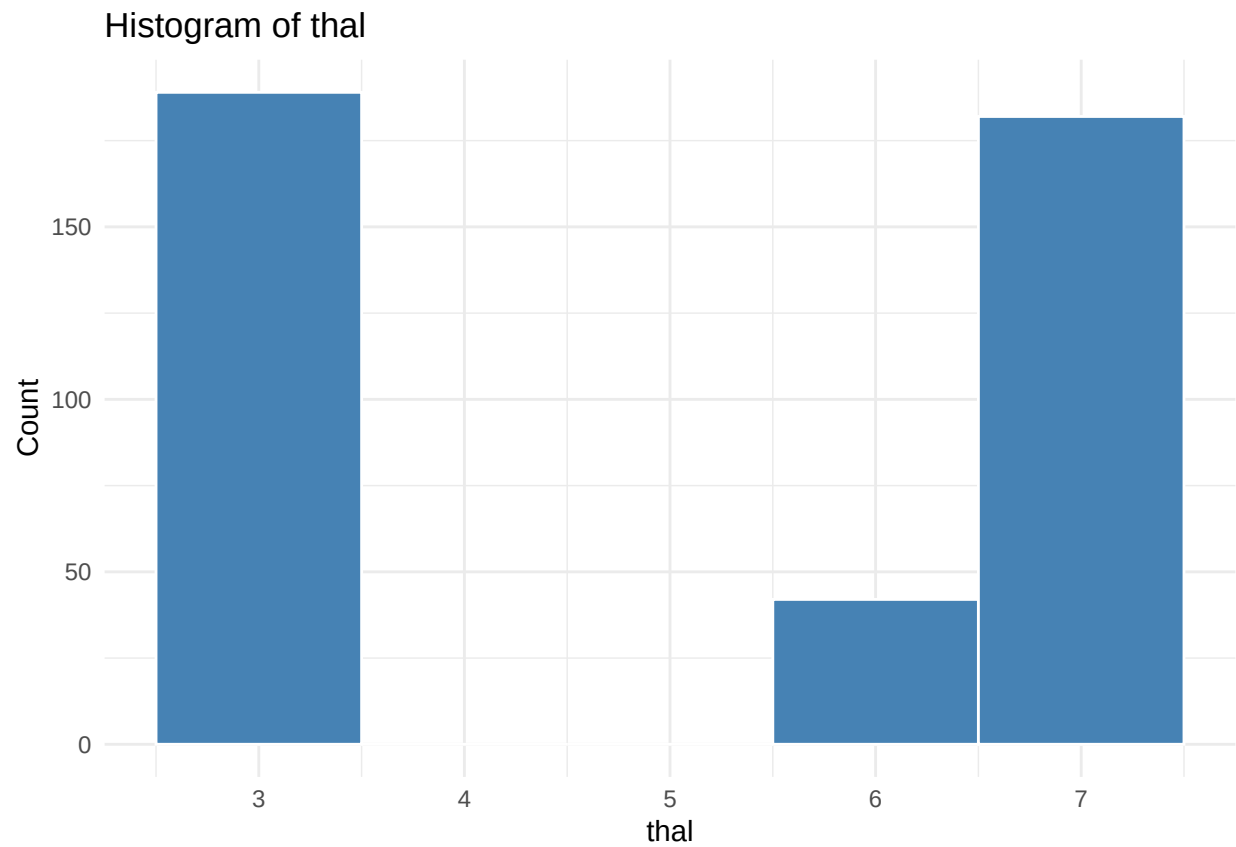
# drop imputed column
df <- df %>%
  select(-ca)
```

Feature `ca` has a right skew therefore median was used for imputation.

Feature: `thal`

```
show_histogram(df, thal)
```

```
## Warning: Removed 486 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



```
show_densityplot(df, thal)
```

```
## Warning: Removed 486 rows containing non-finite outside the scale range  
## (`stat_density()`).
```


Density Plot of thal



```
table(df$thal)
```

```
##
##    3    6    7
## 189   42  182
```

```
# impute with mode and missingness flag
```

```
df <- df %>%
  mutate(
    thal_missing = ifelse(is.na(df$thal), 1, 0),
    thal_imputed = ifelse(is.na(df$thal), 3, df$thal) # mode imputation
  )
```

```
# drop imputed column
```

```
df <- df %>%
  select(-thal)
```

Feature: thaltime

```
summary(df$thaltime)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##      0.00   3.00   6.00   5.69   8.00   20.00    453
```

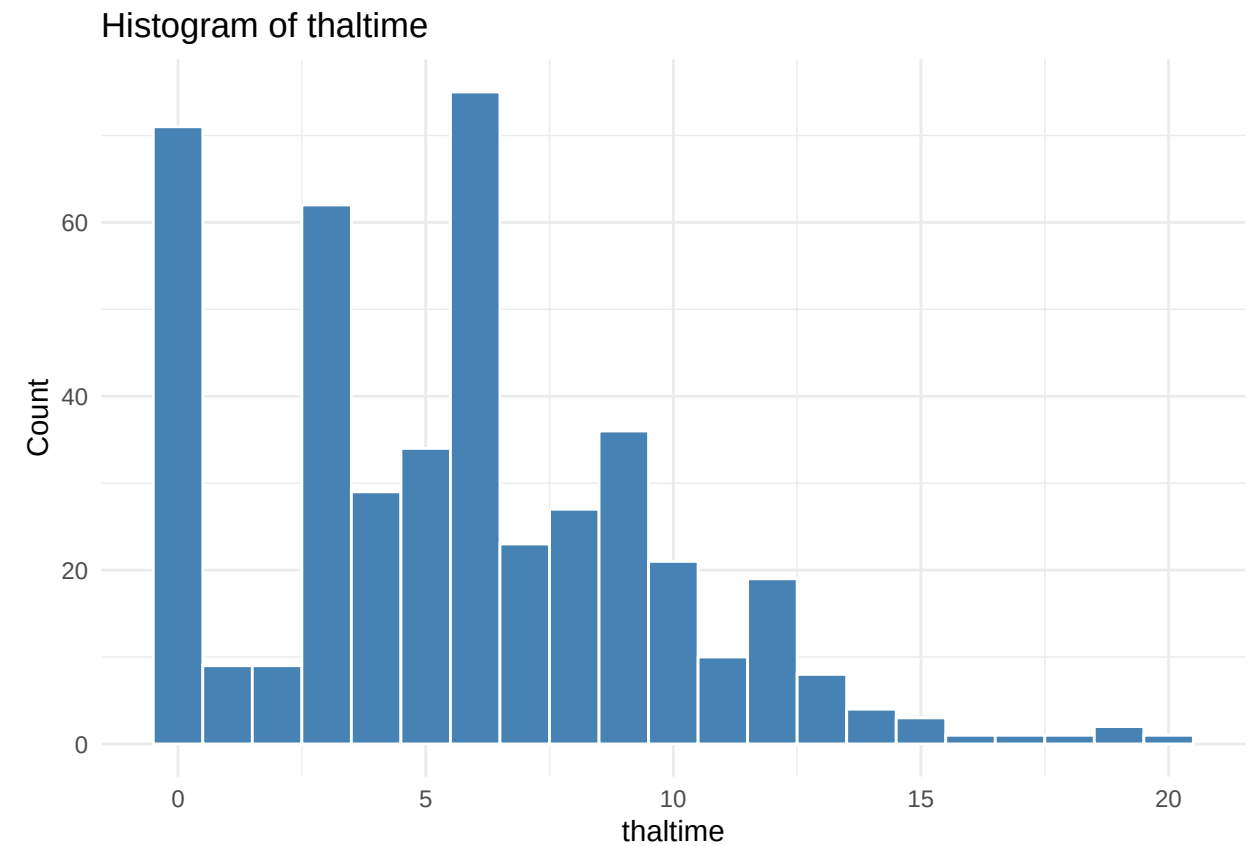
```
sort(table(df$thaltime), decreasing = TRUE)
```

```
##
##      0      6      3      9      8     10      4     12      5     7.5     5.5      7     4.5      2     11      1
```

```
## 70 65 57 28 20 19 17 17 16 11 10 9 8 7 6 5
## 3.5 6.5 8.5 13 1.5 8.6 15 2.5 3.6 4.8 5.1 5.8 6.3 7.4 7.8 9.3
## 5 5 4 4 3 3 3 2 2 2 2 2 2 2 2 2
## 9.5 13.2 14 19 0.5 0.7 3.8 4.2 4.6 4.7 5.2 5.4 6.4 6.8 7.9 9.4
## 2 2 2 2 1 1 1 1 1 1 1 1 1 1 1 1
## 10.2 10.5 10.6 10.7 11.1 11.3 11.7 11.8 12.6 13.5 14.3 14.5 15.8 17.5 17.8 20
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

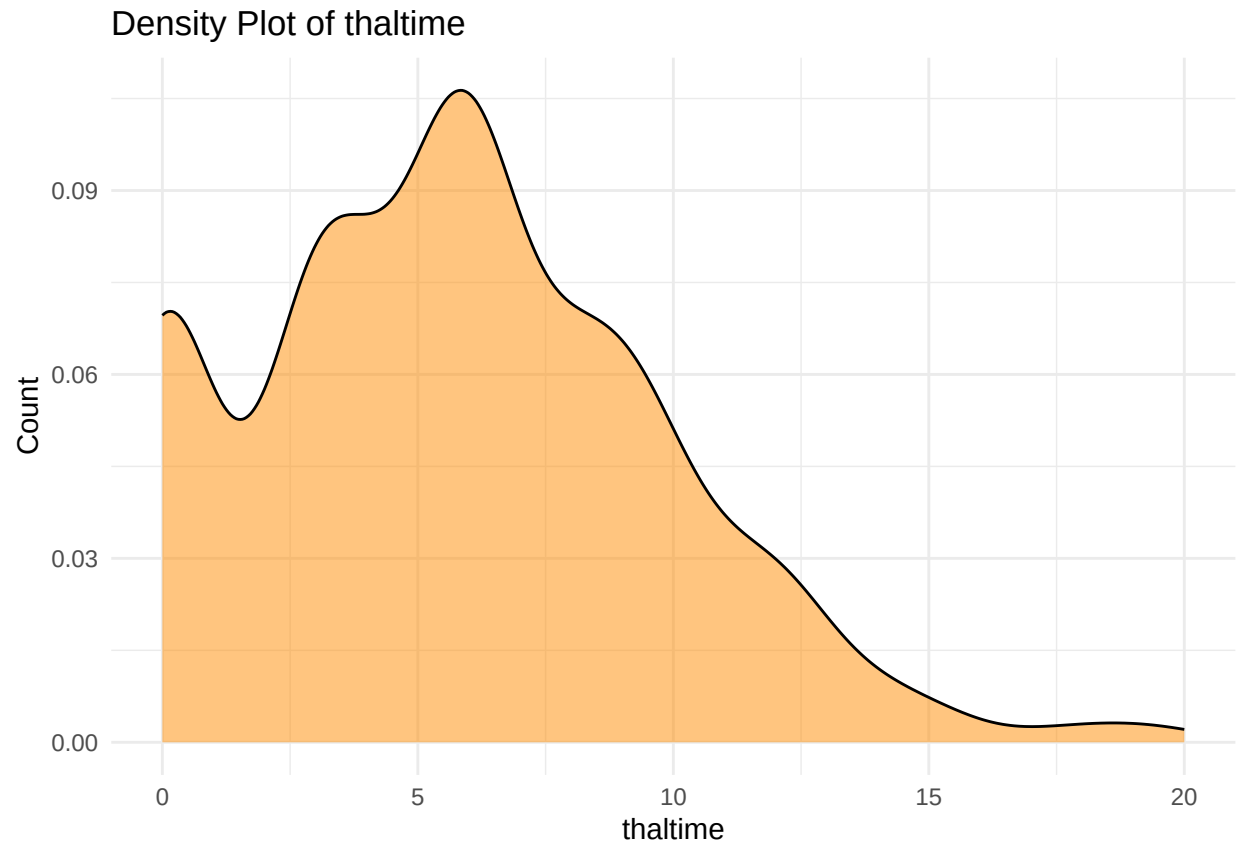
```
# histogram
show_histogram(df, thaltime)
```

```
## Warning: Removed 453 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



```
show_densityplot(df, thaltime)
```

```
## Warning: Removed 453 rows containing non-finite outside the scale range
## (`stat_density()`).
```



```
# impute with median + missingness flag
df <- df %>%
  mutate(
    thaltime_missing = ifelse(is.na(thaltime), 1, 0),
    thaltime_imputed = ifelse(is.na(thaltime), median(thaltime, na.rm = TRUE), thaltime)
  )

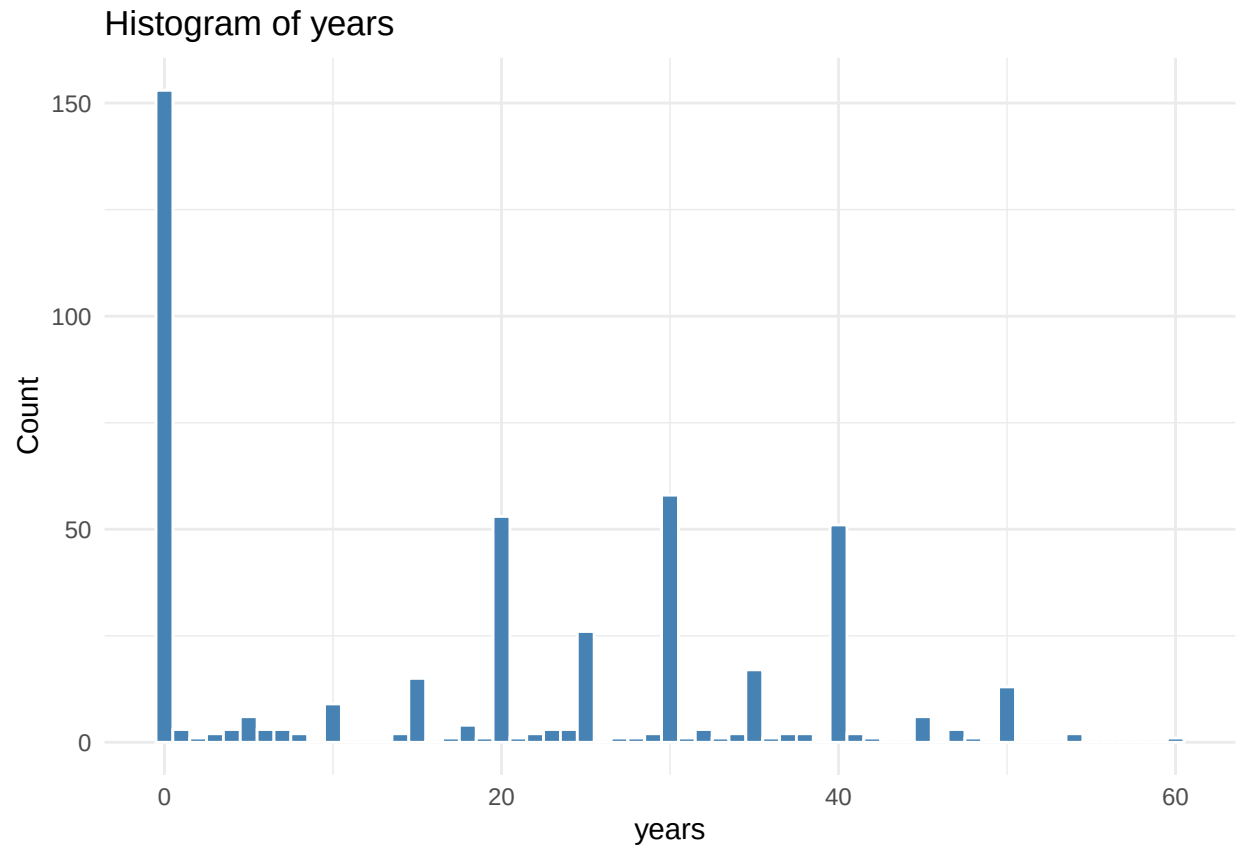
# drop imputed column
df <- df %>%
  select(-thaltime)
```

Feature thaltime showed a right skew. Imputed with median.

Features: years and cigs

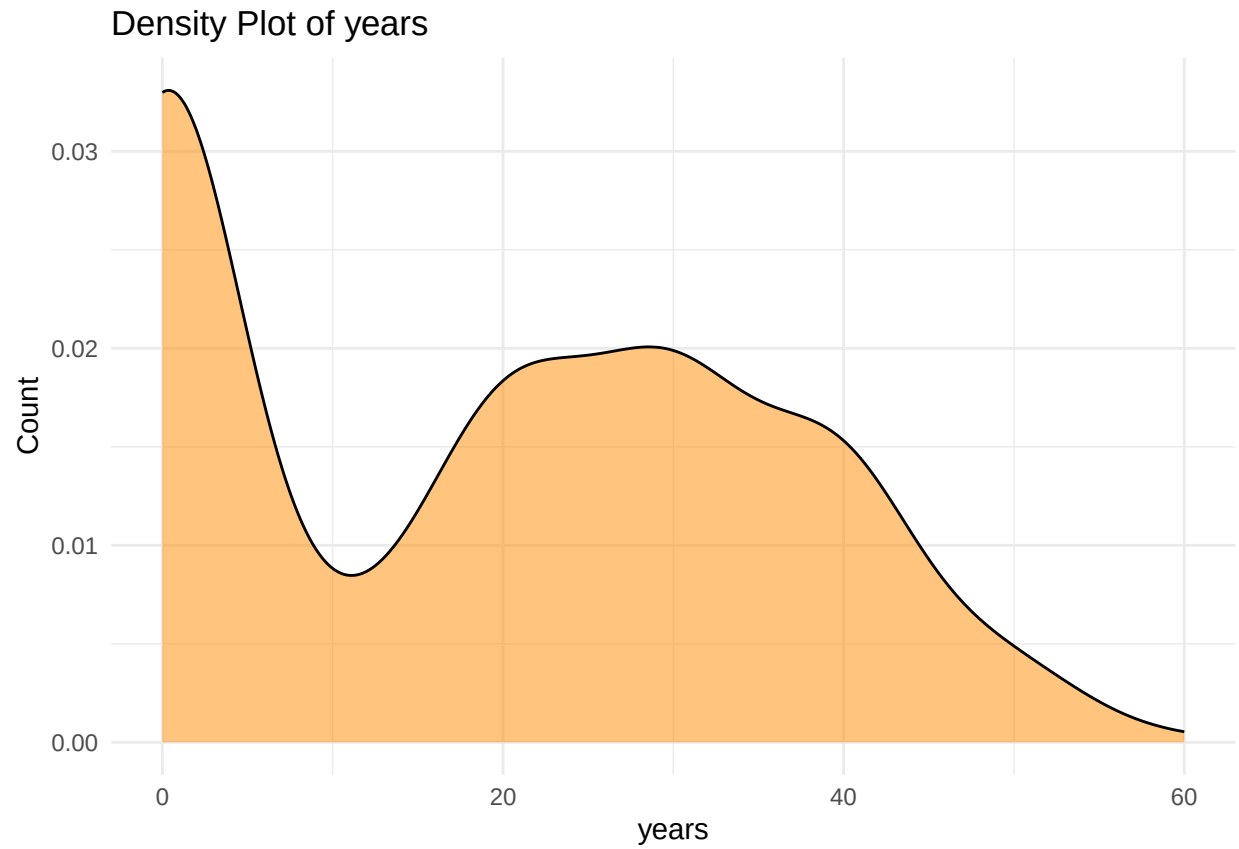
```
show_histogram(df, years)
```

```
## Warning: Removed 432 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



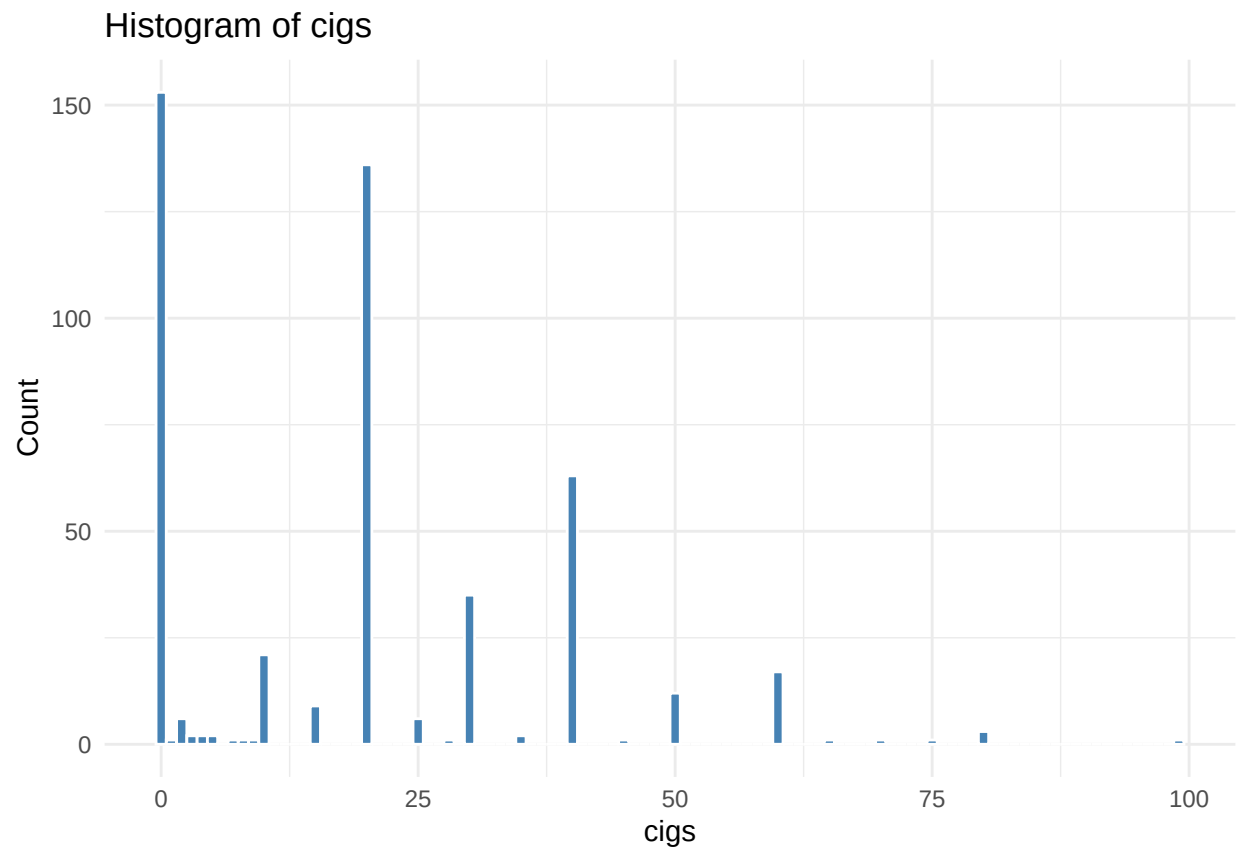
```
show_densityplot(df, years)
```

```
## Warning: Removed 432 rows containing non-finite outside the scale range  
## (`stat_density()`).
```



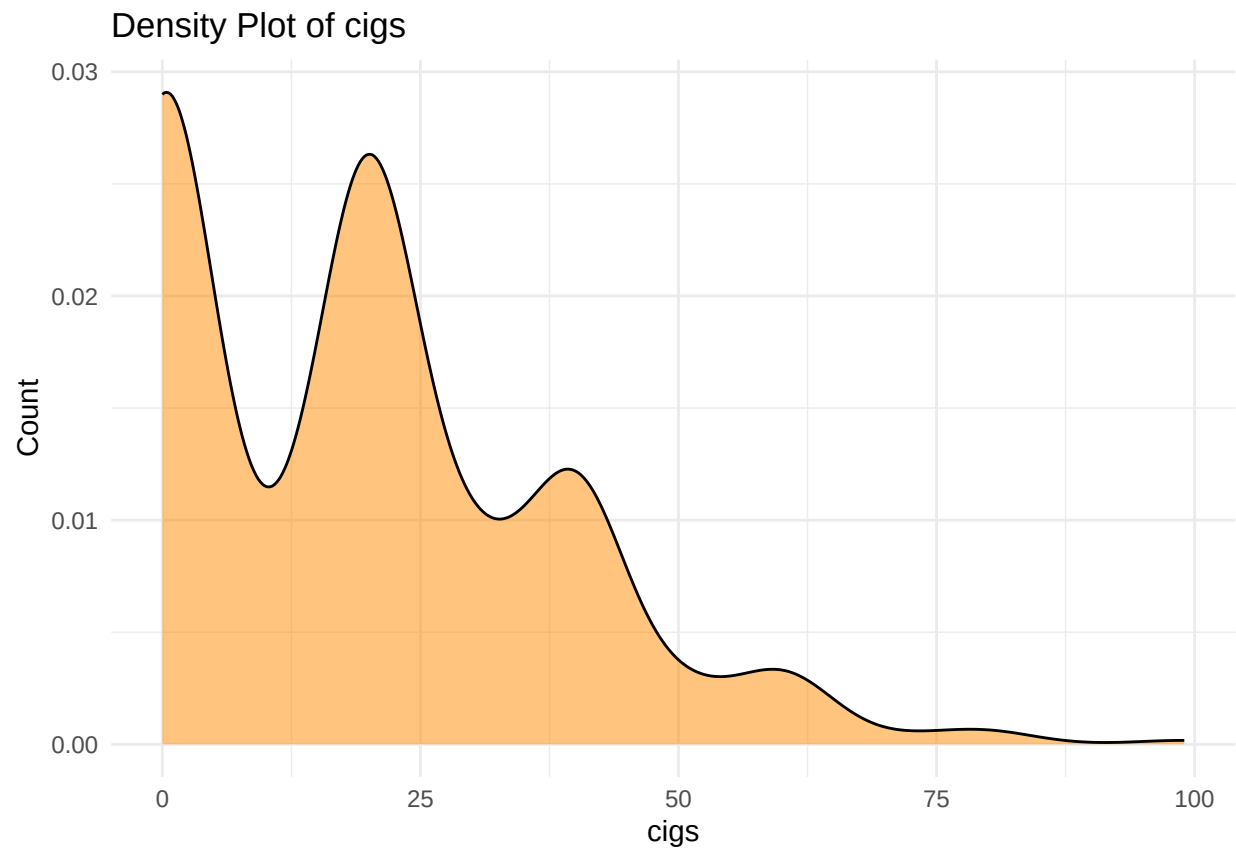
```
show_histogram(df, cigs)
```

```
## Warning: Removed 420 rows containing non-finite outside the scale range  
## (`stat_bin()`).
```



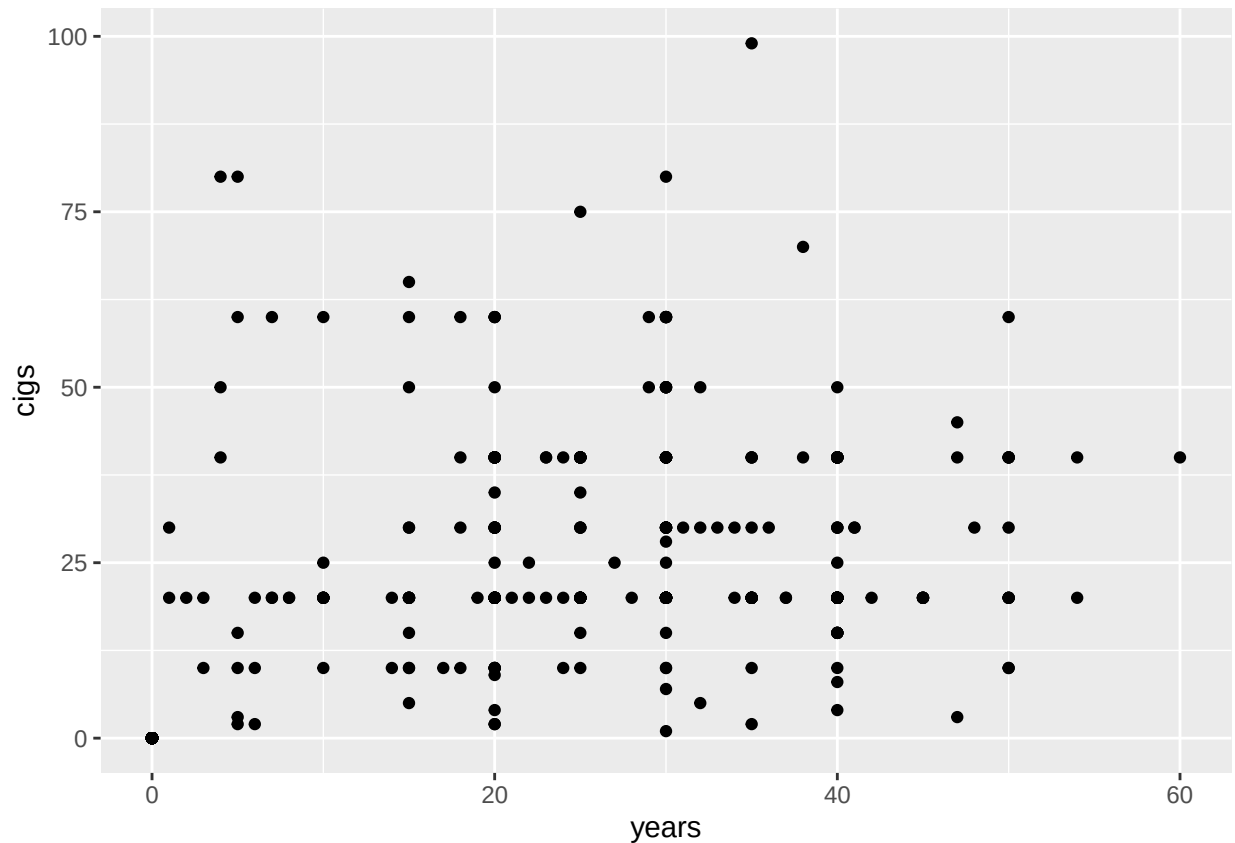
```
show_densityplot(df, cigs)
```

```
## Warning: Removed 420 rows containing non-finite outside the scale range
## (`stat_density()`).
```



```
ggplot(df, aes(x = years, y = cigs)) + geom_point()
```

```
## Warning: Removed 435 rows containing missing values or values outside the scale range  
## (`geom_point()`).
```



```
corr_val <- cor(df$years, df$cigs, use = "complete.obs", method = "pearson")
cat("Correlation value:", corr_val)
```

```
## Correlation value: 0.5890923
```

```
# create smoke_profile feature
# generative AI was used to create this code to create the smoke_profile feature
df <- df %>%
```

```
  mutate(
    # Categorize smoking duration
    years_cat = case_when(
      is.na(years) ~ "missing",
      years == 0   ~ "non",
      years <= 10  ~ "short",
      years <= 20  ~ "medium",
      years <= 40  ~ "long",
      TRUE        ~ "very_long"
    ),
```

```
    # Categorize cigarette intensity
    cigs_cat = case_when(
      is.na(cigs) ~ "missing",
      cigs == 0   ~ "non",
      cigs <= 10  ~ "light",
      cigs <= 20  ~ "moderate",
      cigs <= 40  ~ "heavy",
      TRUE        ~ "very_heavy"
    )
  )
```



```

),

# Combine into single profile
smoke_profile = case_when(
  years_cat == "non" & cigs_cat == "non" ~ "non",
  years_cat == "missing" | cigs_cat == "missing" ~ "missing",
  TRUE ~ paste0(cigs_cat, "_", years_cat)
),

# Optional: convert to factor
smoke_profile = factor(smoke_profile)
)

# drop helper columns
df <- df %>%
  select(-c(cigs_cat, years_cat))

# drop imputed column
df <- df %>%
  select(-c(years, cigs))

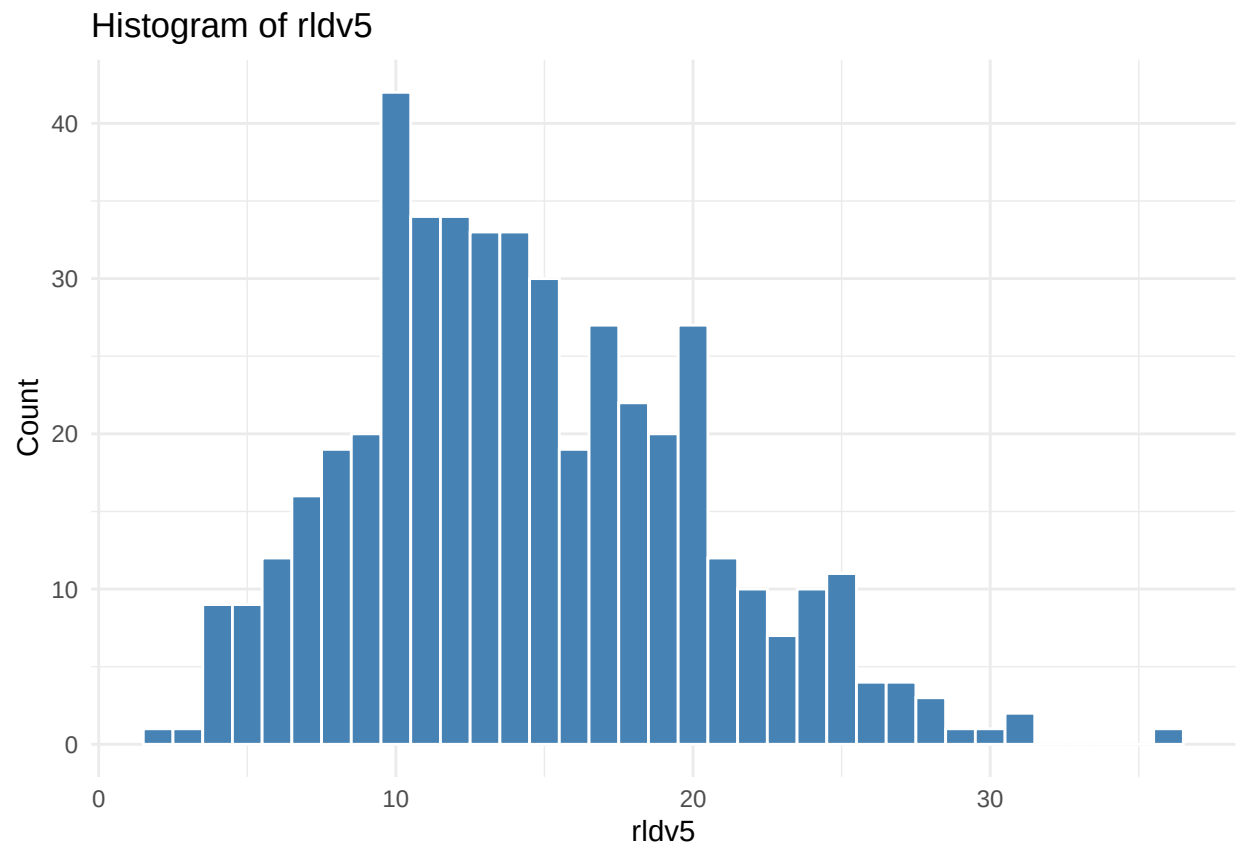
```

Features `years` and `cigs` show moderate collinearity (Pearson correlation = 0.59). Due to missing data, a `smoke_profile` was engineered to combine the two columns.

Feature: `rldv5`

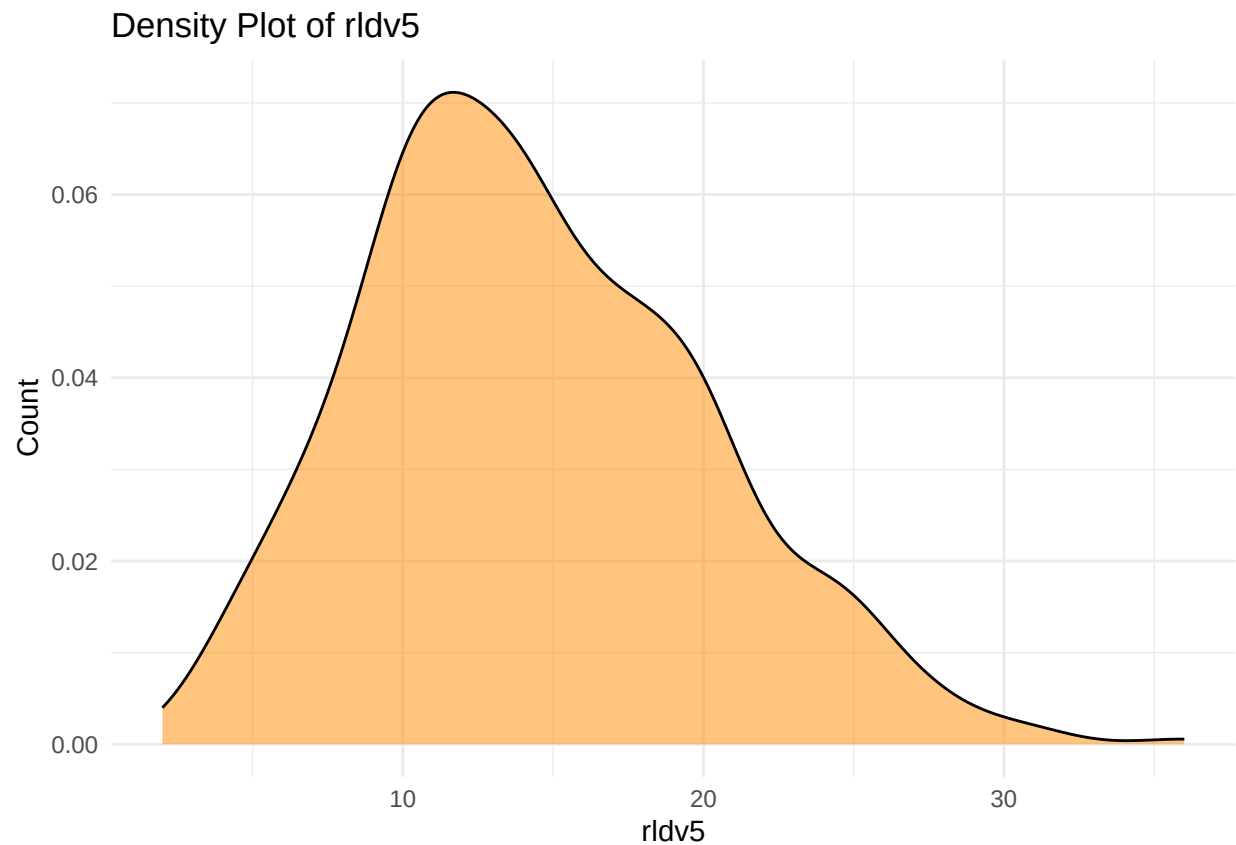
```
show_histogram(df, rldv5)
```

```
## Warning: Removed 425 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



```
show_densityplot(df, rldv5)
```

```
## Warning: Removed 425 rows containing non-finite outside the scale range  
## (`stat_density()`).
```

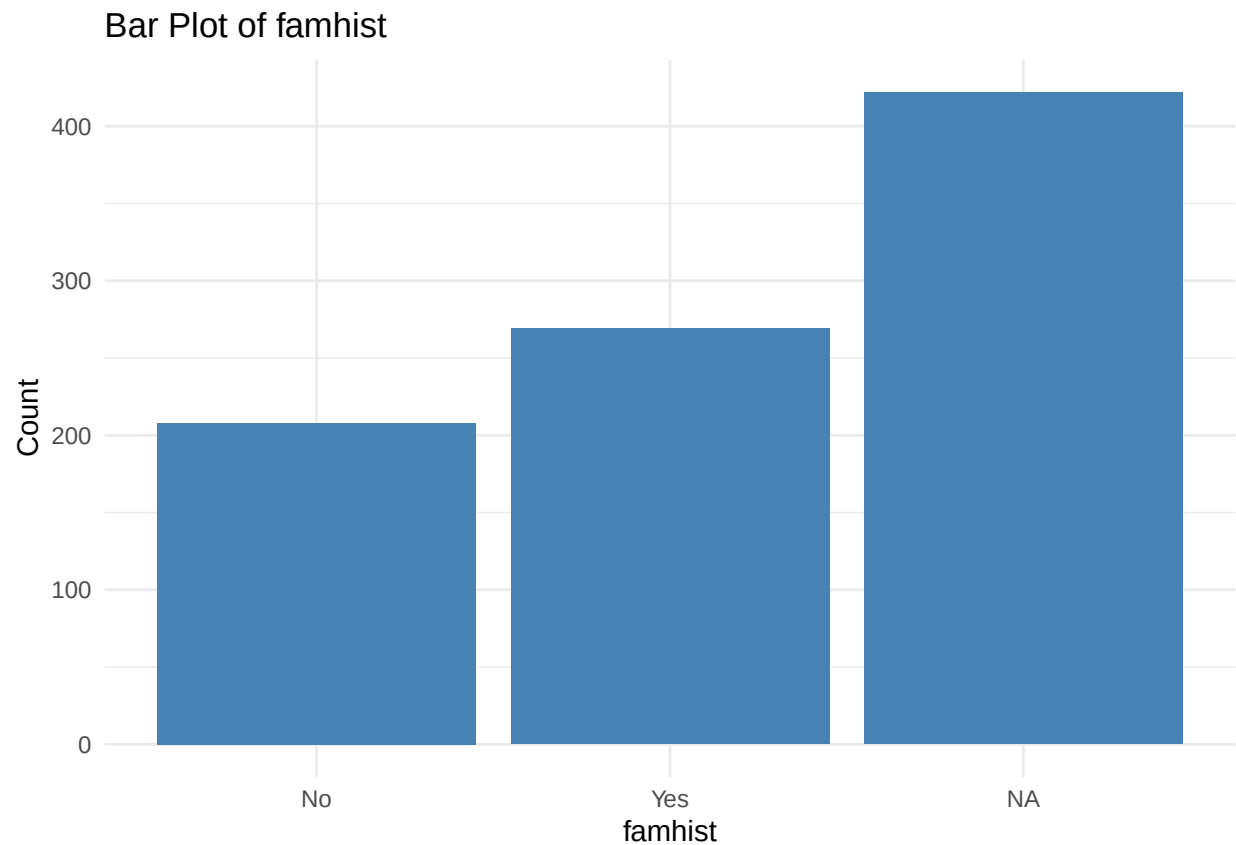


```
df <- df %>%  
  mutate(  
    rldv5_missing = ifelse(is.na(rldv5), 1, 0),  
    rldv5_imputed = ifelse(is.na(rldv5), median(rldv5, na.rm = TRUE), rldv5)  
  )  
  
# drop imputed column  
df <- df %>%  
  select(-rldv5)
```

Feature `rldv5` has a right skew therefore imputed with median.

Feature: famhist

```
show_barplot(df, famhist)
```



```
unique(df$famhist)
```

```
## [1] Yes No <NA>
```

```
## Levels: No Yes
```

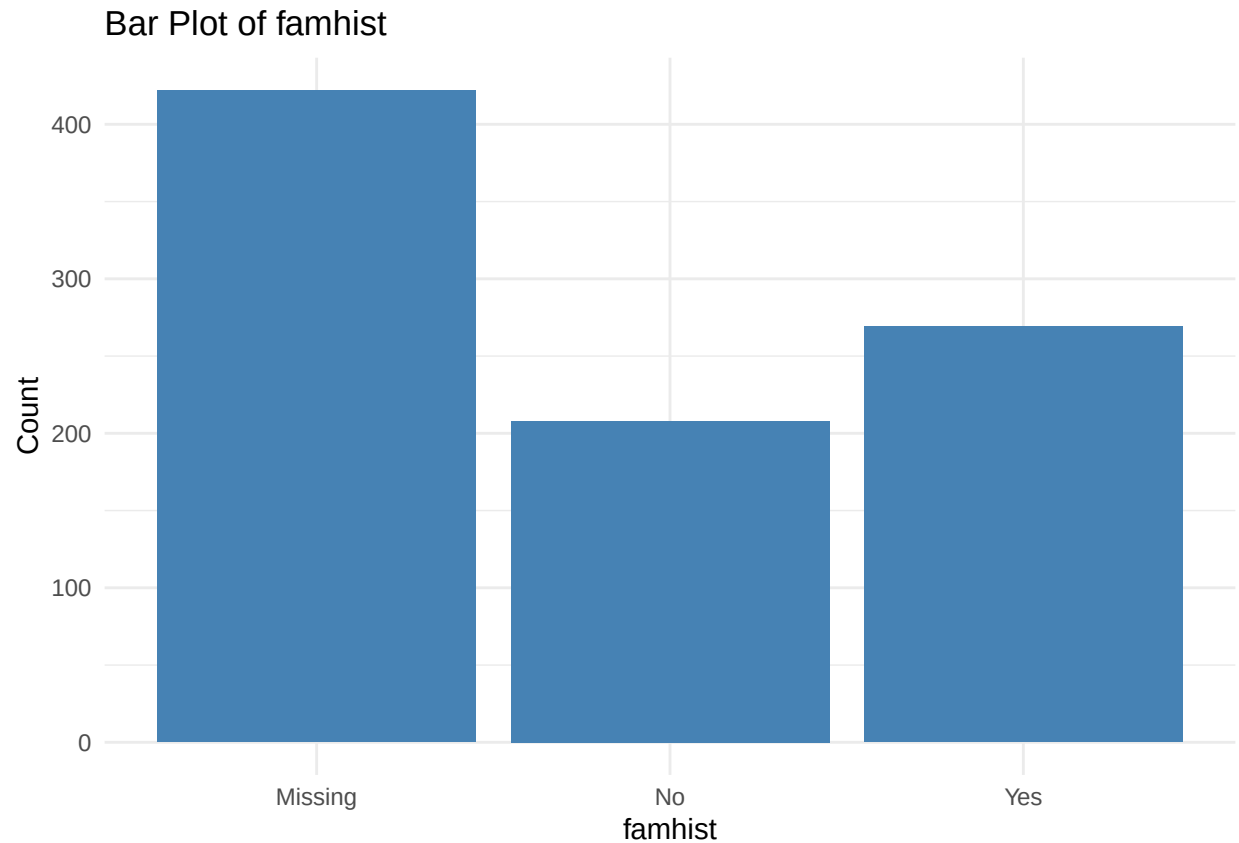
```
# replace NA with "Missing"
```

```
df$famhist <- as.character(df$famhist)
```

```
df$famhist[is.na(df$famhist)] <- "Missing"
```

```
df$famhist <- factor(df$famhist)
```

```
show_barplot(df, famhist)
```



Missing values in `famhist` were imputed with 'missing'.

```
# list of cols
# missing_gt40 <- c('ca', 'thal', 'thalttime', 'years', 'rldv5', 'famhist', 'cigs')
imputed_cols <-c('ca_imputed', 'ca_missing', 'thal_imputed', 'thal_missing',
                'thalttime_missing', 'thalttime_imputed', 'smoke_profile',
                'rldv5_missing', 'rldv5_imputed', 'famhist')
# drop date / time columns
drop_cols <- c('cmo', 'cday', 'cyr', 'ekgmo', 'ekgday', 'ekgyr')
df1 <- df %>%
  select(-all_of(drop_cols))

# compare dimensions
dim(df_cleaning)
```

Dropping original columns after imputation

```
## [1] 899  48
```

```
dim(df)
```

```
## [1] 899  51
```

```
dim(df1)
```

```
## [1] 899  45
```

```
colnames(df1)
```

```
## [1] "age"          "sex"          "painloc"      "painexer"
## [5] "relrest"      "cp"           "trestbps"     "htn"
## [9] "chol"         "fbs"          "famhist"      "restecg"
## [13] "prop"         "nitr"         "pro"          "diuretic"
## [17] "thaldur"      "met"          "thalach"      "thalrest"
## [21] "tpeakbps"     "tpeakbpd"     "trestbpd"     "exang"
## [25] "oldpeak"      "slope"        "rldv5e"       "num"
## [29] "lmt"          "ladprox"      "laddist"      "cxmain"
## [33] "om1"          "rcaprox"      "rcadist"      "location"
## [37] "ca_missing"   "ca_imputed"   "thal_missing" "thal_imputed"
## [41] "thalttime_missing" "thalttime_imputed" "smoke_profile" "rldv5_missing"
## [45] "rldv5_imputed"
```

```
head(df1)
```

```
## # A tibble: 6 x 45
##   age sex   painloc painexer relrest cp   trestbps htn chol fbs   famhist
##   <dbl> <fct> <fct> <fct> <fct> <ord> <dbl> <dbl> <dbl> <fct> <fct>
## 1  63 Male <NA> <NA> <NA> Typi~ 145 1 233 Yes Yes
## 2  67 Male <NA> <NA> <NA> Asym~ 160 1 286 No Yes
## 3  67 Male <NA> <NA> <NA> Asym~ 120 1 229 No Yes
## 4  37 Male <NA> <NA> <NA> Non~ 130 0 250 No Yes
## 5  41 Female <NA> <NA> <NA> Atyp~ 130 1 204 No Yes
## 6  56 Male <NA> <NA> <NA> Atyp~ 120 1 236 No Yes
## # i 34 more variables: restecg <ord>, prop <fct>, nitr <fct>, pro <fct>,
## # diuretic <fct>, thaldur <dbl>, met <dbl>, thalach <dbl>, thalrest <dbl>,
## # tpeakbps <dbl>, tpeakbpd <dbl>, trestbpd <dbl>, exang <fct>, oldpeak <dbl>,
## # slope <ord>, rldv5e <dbl>, num <dbl>, lmt <dbl>, ladvprox <dbl>,
## # laddist <dbl>, cxmain <dbl>, om1 <dbl>, rcaprox <dbl>, rcadist <dbl>,
## # location <chr>, ca_missing <dbl>, ca_imputed <dbl>, thal_missing <dbl>,
## # thal_imputed <dbl>, thalttime_missing <dbl>, thalttime_imputed <dbl>, ...
```

Missing < 40%

Binary Variables

```
# list binary var
df1 %>%
  summarise(across(everything(), ~ n_distinct(na.omit(.)))) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "n_unique") %>%
  filter(n_unique == 2)
```

```
## # A tibble: 21 x 2
##   variable n_unique
##   <chr>      <int>
## 1 sex        2
## 2 painloc    2
## 3 painexer   2
## 4 relrest    2
## 5 htn        2
## 6 fbs        2
## 7 prop       2
## 8 nitr       2
```

```
## 9 pro 2
## 10 diuretic 2
## # i 11 more rows

# get binary var names
binary_vars <- df1 %>%
  select(where(~ n_distinct(na.omit(.)) == 2)) %>%
  names()

# convert binary vars to factors
df1[binary_vars] <- lapply(df1[binary_vars], as.factor)
# convert sex to male/female
#df$sex <- factor(df$sex, levels = c(0, 1), labels = c("female", "male"))

head(df1)

## # A tibble: 6 x 45
##   age sex painloc painexer relrest cp trestbps htn chol fbs famhist
##   <dbl> <fct> <fct> <fct> <fct> <ord> <dbl> <fct> <dbl> <fct> <fct>
## 1 63 Male <NA> <NA> <NA> Typi~ 145 1 233 Yes Yes
## 2 67 Male <NA> <NA> <NA> Asym~ 160 1 286 No Yes
## 3 67 Male <NA> <NA> <NA> Asym~ 120 1 229 No Yes
## 4 37 Male <NA> <NA> <NA> Non~ 130 0 250 No Yes
## 5 41 Female <NA> <NA> <NA> Atyp~ 130 1 204 No Yes
## 6 56 Male <NA> <NA> <NA> Atyp~ 120 1 236 No Yes
## # i 34 more variables: restecg <ord>, prop <fct>, nitr <fct>, pro <fct>,
## # diuretic <fct>, thaldur <dbl>, met <dbl>, thalach <dbl>, thalrest <dbl>,
## # tpeakbps <dbl>, tpeakbpd <dbl>, trestbpd <dbl>, exang <fct>, oldpeak <dbl>,
## # slope <ord>, rldv5e <dbl>, num <dbl>, lmt <dbl>, ladprox <fct>,
## # laddist <fct>, cxmain <fct>, om1 <fct>, rcaprox <fct>, readist <fct>,
## # location <chr>, ca_missing <fct>, ca_imputed <dbl>, thal_missing <fct>,
## # thal_imputed <dbl>, thaltime_missing <fct>, thaltime_imputed <dbl>, ...
```

Numerical Variables

```
numeric_vars <- df1 %>%
  select(where(is.numeric)) %>%
  names()

numeric_na <- df1 %>%
  select(where(is.numeric)) %>%
  select(where(~ any(is.na(.)))) %>%
  names()

numeric_vars

## [1] "age" "trestbps" "chol" "thaldur"
## [5] "met" "thalach" "thalrest" "tpeakbps"
## [9] "tpeakbpd" "trestbpd" "oldpeak" "rldv5e"
## [13] "num" "lmt" "ca_imputed" "thal_imputed"
## [17] "thaltime_imputed" "rldv5_imputed"

df1[numeric_na]
```

```
## # A tibble: 899 x 12
##   trestbps chol thaldur met thalach thalrest tpeakbps tpeakbpd trestbpd
```

```
##      <dbl> <dbl>    <dbl> <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1      145  233     10.5   13      150      60      190      90      85
## 2      160  286      9.5   13      108      64      160      90      90
## 3      120  229      8.5   10      129      78      140      80      80
## 4      130  250      13     17      187      84      195      68      78
## 5      130  204       7     9       172      71      160      74      86
## 6      120  236     11.3   16      178      73      165      70      75
## 7      140  268       6     7       160      83      180      84     100
## 8      120  354       9     10      163      84      165      80      80
## 9      130  254       8     9       147      75      120      70     105
## 10     140  203      5.5    7       155      86      185     120     100
## # i 889 more rows
## # i 3 more variables: oldpeak <dbl>, rldv5e <dbl>, lmt <dbl>

# impute w median and create missingness columns
for (var in numeric_na) {
  median_val <- median(df1[[var]], na.rm = TRUE)
  df1[[paste0(var, "_missing")]] <- ifelse(is.na(df1[[var]]), 1, 0)
  df1[[paste0(var, "_imputed")]] <- ifelse(is.na(df1[[var]]), median_val, df1[[var]])
}

# drop imputed columns
df1 <- df1 %>%
  select(-all_of(numeric_na))
dim(df1)

## [1] 899 57
```

Numerical variables were imputed with the median value. Missing flags were also created for these variables.

Categorical Variables

```
categorical_na <- df1 %>%
  select(where(~ is.factor(.) || is.character(.))) %>%
  select(where(~ any(is.na(.)))) %>%
  names()

# impute with Missing
for (var in categorical_na) {
  df1[[var]] <- as.character(df1[[var]])
  df1[[var]][is.na(df1[[var]])] <- "Missing"
  df1[[var]] <- factor(df1[[var]])
}
dim(df1)
```

```
## [1] 899 57
```

```
anyNA(df1)
```

```
## [1] FALSE
```

For NAs in categorical variables, NAs were replaced with 'Missing'.

Create binary for target (num)

```
# create binary target 0 = no disease, 1 - 3 = varying levels
df1 <- df1 %>%
```



```
mutate(
  has_disease = factor(ifelse(num == 0, "No", "Yes"), levels = c("No", "Yes"))
)

head(df1)

## # A tibble: 6 x 58
##   age sex   painloc painexer relrest cp   htn   fbs   famhist restecg prop
##   <dbl> <fct> <fct> <fct> <fct> <ord> <fct> <fct> <fct> <fct> <fct>
## 1    63 Male   Missing Missing Missing Typic~ 1    Yes   Yes   Hypert~ No
## 2    67 Male   Missing Missing Missing Asymp~ 1    No    Yes   Hypert~ Yes
## 3    67 Male   Missing Missing Missing Asymp~ 1    No    Yes   Hypert~ Yes
## 4    37 Male   Missing Missing Missing Non-a~ 0    No    Yes   Normal  Yes
## 5    41 Female Missing Missing Missing Atypi~ 1    No    Yes   Hypert~ No
## 6    56 Male   Missing Missing Missing Atypi~ 1    No    Yes   Normal  No
## # i 47 more variables: nitr <fct>, pro <fct>, diuretic <fct>, exang <fct>,
## #   slope <fct>, num <dbl>, ladprox <fct>, laddist <fct>, cxmain <fct>,
## #   om1 <fct>, rcaprox <fct>, rcadist <fct>, location <chr>, ca_missing <fct>,
## #   ca_imputed <dbl>, thal_missing <fct>, thal_imputed <dbl>,
## #   thaltime_missing <fct>, thaltime_imputed <dbl>, smoke_profile <fct>,
## #   rldv5_missing <fct>, rldv5_imputed <dbl>, trestbps_missing <dbl>,
## #   trestbps_imputed <dbl>, chol_missing <dbl>, chol_imputed <dbl>, ...
```

Binary target column for the target variable num was created. The column num was not removed from the dataset.

Export finalized dataset

```
# export final dataset
saveRDS(df1, here("Data/Clean", "heart_disease_final.rds"))
```

Univariate Analysis

```
knitr::opts_chunk$set(echo = TRUE)
```

```
df <- readRDS(here("Data/Clean", "heart_disease_final.rds"))
head(df)
```

```
## # A tibble: 6 x 58
##   age sex   painloc painexer relrest cp   htn   fbs   famhist restecg prop
##   <dbl> <fct> <fct> <fct> <fct> <ord> <fct> <fct> <fct> <fct> <fct>
## 1    63 Male   Missing Missing Missing Typic~ 1    Yes   Yes   Hypert~ No
## 2    67 Male   Missing Missing Missing Asymp~ 1    No    Yes   Hypert~ Yes
## 3    67 Male   Missing Missing Missing Asymp~ 1    No    Yes   Hypert~ Yes
## 4    37 Male   Missing Missing Missing Non-a~ 0    No    Yes   Normal  Yes
## 5    41 Female Missing Missing Missing Atypi~ 1    No    Yes   Hypert~ No
## 6    56 Male   Missing Missing Missing Atypi~ 1    No    Yes   Normal  No
## # i 47 more variables: nitr <fct>, pro <fct>, diuretic <fct>, exang <fct>,
## #   slope <fct>, num <dbl>, ladprox <fct>, laddist <fct>, cxmain <fct>,
## #   om1 <fct>, rcaprox <fct>, rcadist <fct>, location <chr>, ca_missing <fct>,
## #   ca_imputed <dbl>, thal_missing <fct>, thal_imputed <dbl>,
## #   thaltime_missing <fct>, thaltime_imputed <dbl>, smoke_profile <fct>,
## #   rldv5_missing <fct>, rldv5_imputed <dbl>, trestbps_missing <dbl>,
## #   trestbps_imputed <dbl>, chol_missing <dbl>, chol_imputed <dbl>, ...
```

```
# separate target from variables
target_vars <- c("num", "has_disease")
y_df <- df %>%
  select(all_of(target_vars))

X_df <- df %>%
  select(-all_of(target_vars))
```

Numerical vars

```
num_vars <- X_df %>%
  select(where(is.numeric)) %>%
  names()

# summary
X_df %>%
  select(all_of(num_vars)) %>%
  summary()
```

```
##      age      ca_imputed    thal_imputed  thaltime_imputed
##  Min.   :28.00   Min.   :0.0000   Min.    :3.00   Min.    : 0.000
##  1st Qu.:47.00   1st Qu.:0.0000   1st Qu.:3.00   1st Qu.: 6.000
##  Median :54.00   Median :0.0000   Median :3.00   Median : 6.000
##  Mean   :53.48   Mean    :0.2258   Mean    :3.95   Mean    : 5.846
##  3rd Qu.:60.00   3rd Qu.:0.0000   3rd Qu.:3.00   3rd Qu.: 6.000
##  Max.   :77.00   Max.    :9.0000   Max.    :7.00   Max.    :20.000
##  rldv5_imputed  trestbps_missing  trestbps_imputed  chol_missing
##  Min.    : 2.00   Min.    :0.00000   Min.    : 0      Min.    :0.00000
##  1st Qu.:13.00   1st Qu.:0.00000   1st Qu.:120      1st Qu.:0.00000
##  Median :14.00   Median :0.00000   Median :130      Median :0.00000
##  Mean    :14.21   Mean    :0.06563   Mean    :132      Mean    :0.03337
##  3rd Qu.:14.00   3rd Qu.:0.00000   3rd Qu.:140      3rd Qu.:0.00000
##  Max.    :36.00   Max.    :1.00000   Max.    :200      Max.    :1.00000
##  chol_imputed  thaldur_missing  thaldur_imputed  met_missing
##  Min.    : 0.0   Min.    :0.00000   Min.    : 1.000   Min.    :0.0000
##  1st Qu.:177.5   1st Qu.:0.00000   1st Qu.: 6.000   1st Qu.:0.0000
##  Median :224.0   Median :0.00000   Median : 8.100   Median :0.0000
##  Mean    :199.6   Mean    :0.06229   Mean    : 8.621   Mean    :0.1168
##  3rd Qu.:267.0   3rd Qu.:0.00000   3rd Qu.:10.300   3rd Qu.:0.0000
##  Max.    :603.0   Max.    :1.00000   Max.    :24.000   Max.    :1.0000
##  met_imputed  thalach_missing  thalach_imputed  thalrest_missing
##  Min.    : 2.00   Min.    :0.00000   Min.    : 60.0   Min.    :0.00000
##  1st Qu.: 6.00   1st Qu.:0.00000   1st Qu.:120.0   1st Qu.:0.00000
##  Median : 7.00   Median :0.00000   Median :140.0   Median :0.00000
##  Mean    :15.38   Mean    :0.06118   Mean    :137.5   Mean    :0.06229
##  3rd Qu.:10.00   3rd Qu.:0.00000   3rd Qu.:155.0   3rd Qu.:0.00000
##  Max.    :200.00  Max.    :1.00000   Max.    :202.0   Max.    :1.00000
##  thalrest_imputed  tpeakbps_missing  tpeakbps_imputed  tpeakbpd_missing
##  Min.    : 37.00   Min.    :0.00000   Min.    : 84.0   Min.    :0.00000
##  1st Qu.: 65.00   1st Qu.:0.00000   1st Qu.:158.0   1st Qu.:0.00000
##  Median : 74.00   Median :0.00000   Median :170.0   Median :0.00000
##  Mean    : 75.39   Mean    :0.07008   Mean    :171.5   Mean    :0.07008
##  3rd Qu.: 84.00   3rd Qu.:0.00000   3rd Qu.:190.0   3rd Qu.:0.00000
```

```
## Max. :139.00 Max. :1.00000 Max. :240.0 Max. :1.00000
## tpeakbpd_imputed trestbpd_missing trestbpd_imputed oldpeak_missing
## Min. : 11.00 Min. :0.00000 Min. : 0.00 Min. :0.00000
## 1st Qu.: 80.00 1st Qu.:0.00000 1st Qu.: 80.00 1st Qu.:0.00000
## Median : 88.00 Median :0.00000 Median : 80.00 Median :0.00000
## Mean : 87.34 Mean :0.06563 Mean : 83.29 Mean :0.06897
## 3rd Qu.: 98.00 3rd Qu.:0.00000 3rd Qu.: 90.00 3rd Qu.:0.00000
## Max. :134.00 Max. :1.00000 Max. :120.00 Max. :1.00000
## oldpeak_imputed rldv5e_missing rldv5e_imputed lmt_missing
## Min. : -2.6000 Min. :0.000 Min. : 2.00 Min. :0.0000
## 1st Qu.: 0.0000 1st Qu.:0.000 1st Qu.: 13.00 1st Qu.:0.0000
## Median : 0.5000 Median :0.000 Median : 19.00 Median :0.0000
## Mean : 0.8449 Mean :0.158 Mean : 49.24 Mean :0.3059
## 3rd Qu.: 1.5000 3rd Qu.:0.000 3rd Qu.: 88.00 3rd Qu.:1.0000
## Max. : 6.2000 Max. :1.000 Max. :270.00 Max. :1.0000
## lmt_imputed
## Min. : 0.000
## 1st Qu.: 1.000
## Median : 1.000
## Mean : 1.225
## 3rd Qu.: 1.000
## Max. :162.000
```

*# Example: just the *_imputed columns (exclude *_missing)*

```
imputed_vars <- names(X_df)[str_detect(names(X_df), "_imputed") & !str_detect(names(X_df), "_missing")]
```

```
X_df %>%
```

```
  select(all_of(imputed_vars)) %>%
```

```
  pivot_longer(cols = everything(), names_to = "variable", values_to = "value") %>%
```

```
  ggplot(aes(x = variable, y = value)) +
```

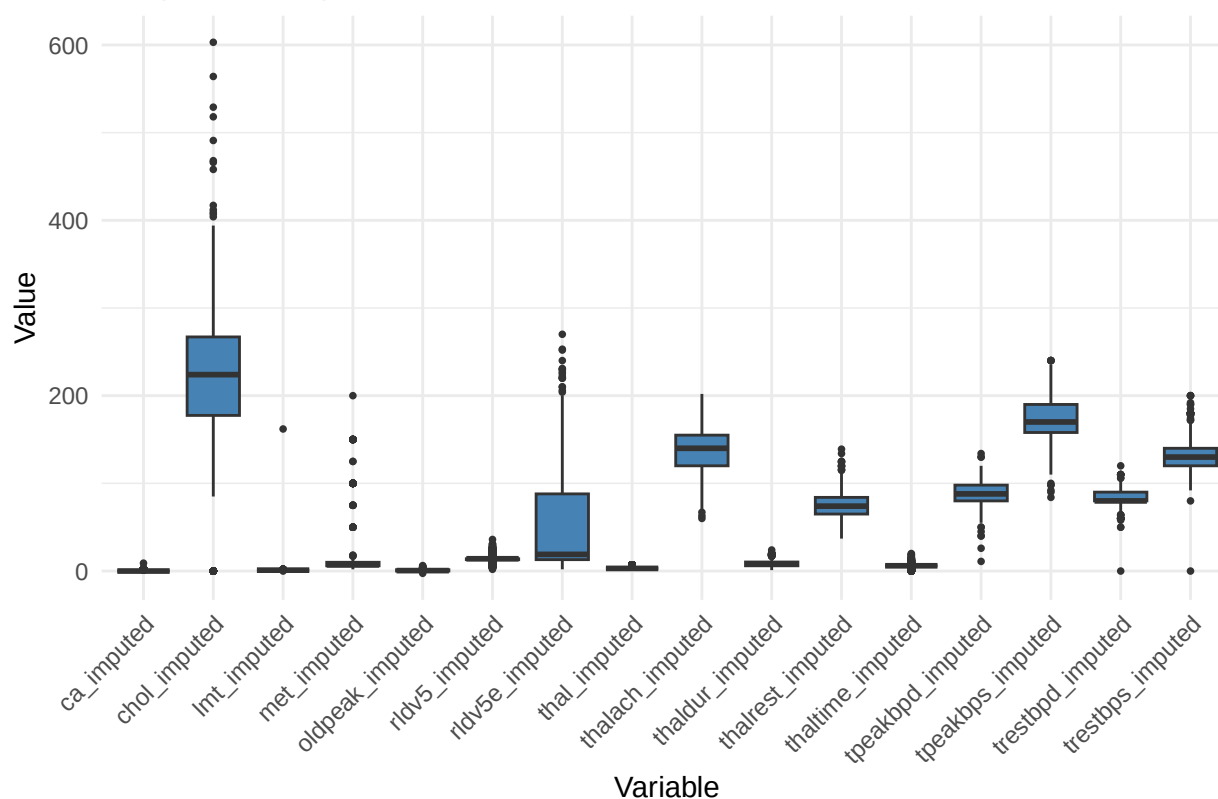
```
  geom_boxplot(fill = "steelblue", outlier.shape = 16, outlier.size = 1) +
```

```
  theme_minimal() +
```

```
  labs(title = "Boxplots of Imputed Numeric Variables", x = "Variable", y = "Value") +
```

```
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Boxplots of Imputed Numeric Variables



```
# summary statistics
X_df %>%
  select(where(is.numeric)) %>%
  skimr::skim()
```

Table 1: Data summary

Name	Piped data
Number of rows	899
Number of columns	29
Column type frequency:	
numeric	29
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
age	0	1	53.48	9.44	28.0	47.0	54.0	60.0	77.0	
ca_imputed	0	1	0.23	0.68	0.0	0.0	0.0	0.0	9.0	
thal_imputed	0	1	3.95	1.66	3.0	3.0	3.0	3.0	7.0	
thaltime_imputed	0	1	5.85	2.82	0.0	6.0	6.0	6.0	20.0	
rldv5_imputed	0	1	14.21	4.14	2.0	13.0	14.0	14.0	36.0	
trestbpd_missing	0	1	0.07	0.25	0.0	0.0	0.0	0.0	1.0	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
trestbps_imputed	0	1	131.96	18.52	0.0	120.0	130.0	140.0	200.0	
chol_missing	0	1	0.03	0.18	0.0	0.0	0.0	0.0	1.0	
chol_imputed	0	1	199.60	110.04	0.0	177.5	224.0	267.0	603.0	
thaldur_missing	0	1	0.06	0.24	0.0	0.0	0.0	0.0	1.0	
thaldur_imputed	0	1	8.62	3.63	1.0	6.0	8.1	10.3	24.0	
met_missing	0	1	0.12	0.32	0.0	0.0	0.0	0.0	1.0	
met_imputed	0	1	15.38	29.08	2.0	6.0	7.0	10.0	200.0	
thalach_missing	0	1	0.06	0.24	0.0	0.0	0.0	0.0	1.0	
thalach_imputed	0	1	137.46	25.17	60.0	120.0	140.0	155.0	202.0	
thalrest_missing	0	1	0.06	0.24	0.0	0.0	0.0	0.0	1.0	
thalrest_imputed	0	1	75.39	14.27	37.0	65.0	74.0	84.0	139.0	
tpeakbps_missing	0	1	0.07	0.26	0.0	0.0	0.0	0.0	1.0	
tpeakbps_imputed	0	1	171.53	24.82	84.0	158.0	170.0	190.0	240.0	
tpeakbpd_missing	0	1	0.07	0.26	0.0	0.0	0.0	0.0	1.0	
tpeakbpd_imputed	0	1	87.34	14.21	11.0	80.0	88.0	98.0	134.0	
trestbpd_missing	0	1	0.07	0.25	0.0	0.0	0.0	0.0	1.0	
trestbpd_imputed	0	1	83.29	9.95	0.0	80.0	80.0	90.0	120.0	
oldpeak_missing	0	1	0.07	0.25	0.0	0.0	0.0	0.0	1.0	
oldpeak_imputed	0	1	0.84	1.05	-2.6	0.0	0.5	1.5	6.2	
rldv5e_missing	0	1	0.16	0.36	0.0	0.0	0.0	0.0	1.0	
rldv5e_imputed	0	1	49.24	56.87	2.0	13.0	19.0	88.0	270.0	
lmt_missing	0	1	0.31	0.46	0.0	0.0	0.0	1.0	1.0	
lmt_imputed	0	1	1.22	5.37	0.0	1.0	1.0	1.0	162.0	

Categorical vars

```
cat_vars <- X_df %>%
  select(where(~ is.factor(.) || is.character(.))) %>%
  names()

# relative frequency tables
for (var in cat_vars) {
  cat("----", var, "----\n")
  print(prop.table(table(X_df[[var]])))
  cat("\n")
}
```

```
## ---- sex ----
##
##      Female      Male
## 0.2091212 0.7908788
##
## ---- painloc ----
##
##      Missing  Otherwise  Substernal
## 0.31368187 0.05450501 0.63181313
##
## ---- painexer ----
##
##              Missing      Otherwise Provoked by Exertion
##              0.3136819      0.2791991      0.4071190
##
```

```

## ---- relrest ----
##
##           Missing           Otherwise Relieved after rest
##           0.3181313         0.2235818         0.4582870
##
## ---- cp ----
##
##   Typical Angina  Atypical Angina Non-anginal Pain  Asymptomatic
##           0.05005562         0.18576196         0.22469410         0.53948832
##
## ---- htn ----
##
##           0           1   Missing
## 0.5038932 0.4582870 0.0378198
##
## ---- fbs ----
##
##   Missing           No           Yes
## 0.1001112 0.7497219 0.1501669
##
## ---- famhist ----
##
##   Missing           No           Yes
## 0.4694105 0.2313682 0.2992214
##
## ---- restecg ----
##
##   Abnormal Hypertrophy   Missing   Normal
## 0.196885428 0.202447164 0.002224694 0.598442714
##
## ---- prop ----
##
##   Missing           No           Yes
## 0.07452725 0.68743048 0.23804227
##
## ---- nitr ----
##
##   Missing           No           Yes
## 0.07230256 0.68075640 0.24694105
##
## ---- pro ----
##
##   Missing           No           Yes
## 0.07007786 0.76974416 0.16017798
##
## ---- diuretic ----
##
##   Missing           No           Yes
## 0.09121246 0.80645161 0.10233593
##
## ---- exang ----
##
##   Missing           No           Yes
## 0.06117909 0.57174638 0.36707453

```

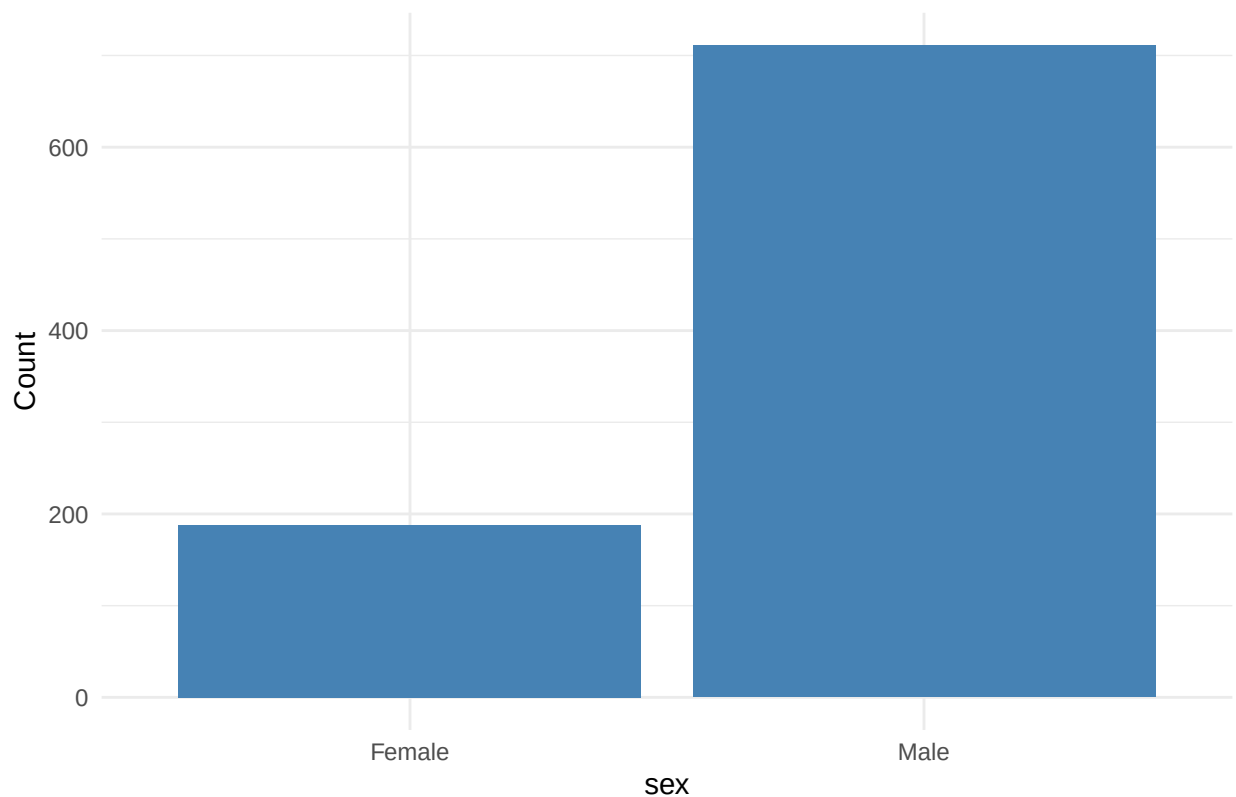
```

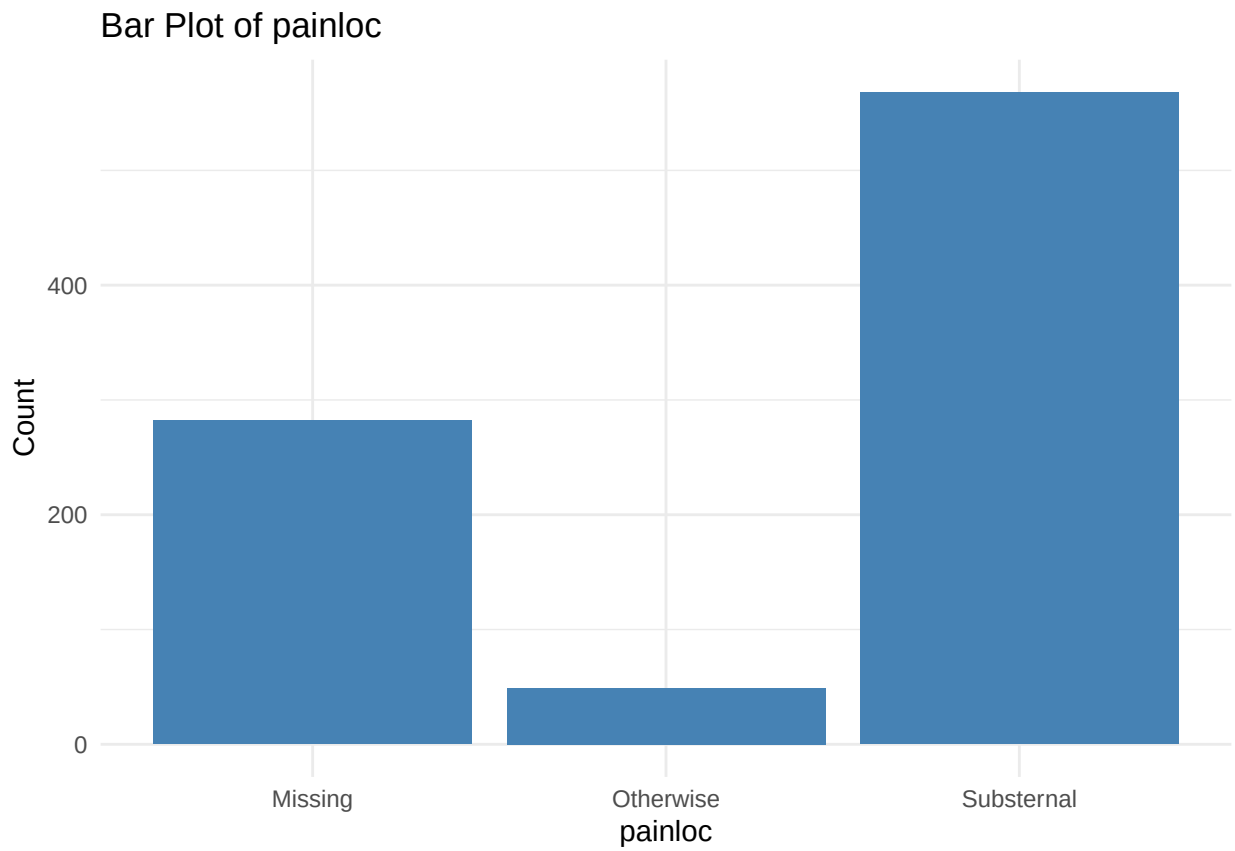
##
## ---- slope ----
##
## Downsloping      Flat      Missing  Upsloping
## 0.06674082 0.37152392 0.34371524 0.21802002
##
## ---- ladprox ----
##
##      1      2  Missing
## 0.4961068 0.2413793 0.2625139
##
## ---- laddist ----
##
##      1      2  Missing
## 0.5461624 0.1802002 0.2736374
##
## ---- cxmain ----
##
##      1      2  Missing
## 0.5194661 0.2191324 0.2614016
##
## ---- om1 ----
##
##      1      2  Missing
## 0.5750834 0.1234705 0.3014461
##
## ---- rcaprox ----
##
##      1      2  Missing
## 0.4783092 0.2491657 0.2725250
##
## ---- rcadist ----
##
##      1      2  Missing
## 0.5795328 0.1201335 0.3003337
##
## ---- location ----
##
##      Cleveland      Hungary  switzerland VA Long Beach
##      0.3136819      0.3270300      0.1368187      0.2224694
##
## ---- ca_missing ----
##
##      0      1
## 0.323693 0.676307
##
## ---- thal_missing ----
##
##      0      1
## 0.4593993 0.5406007
##
## ---- thaltime_missing ----
##
##      0      1

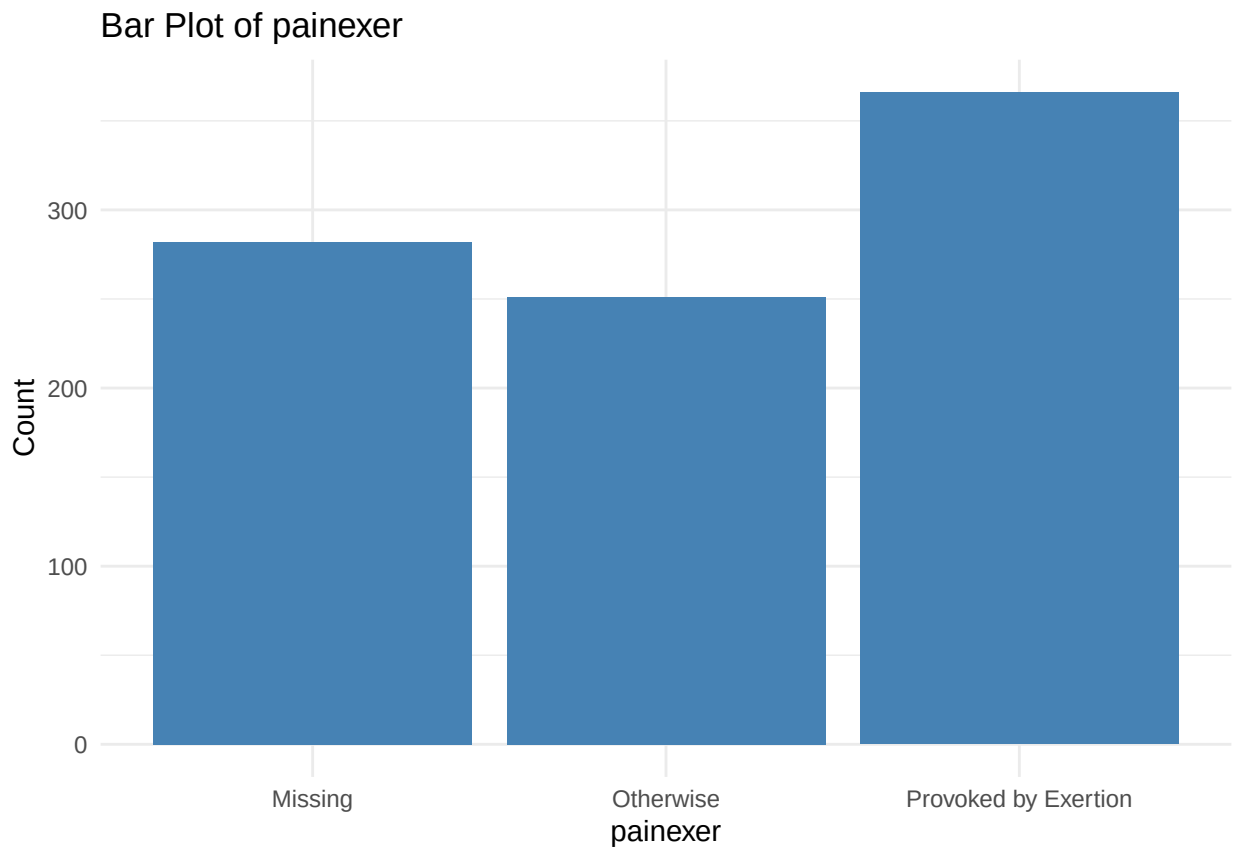
```

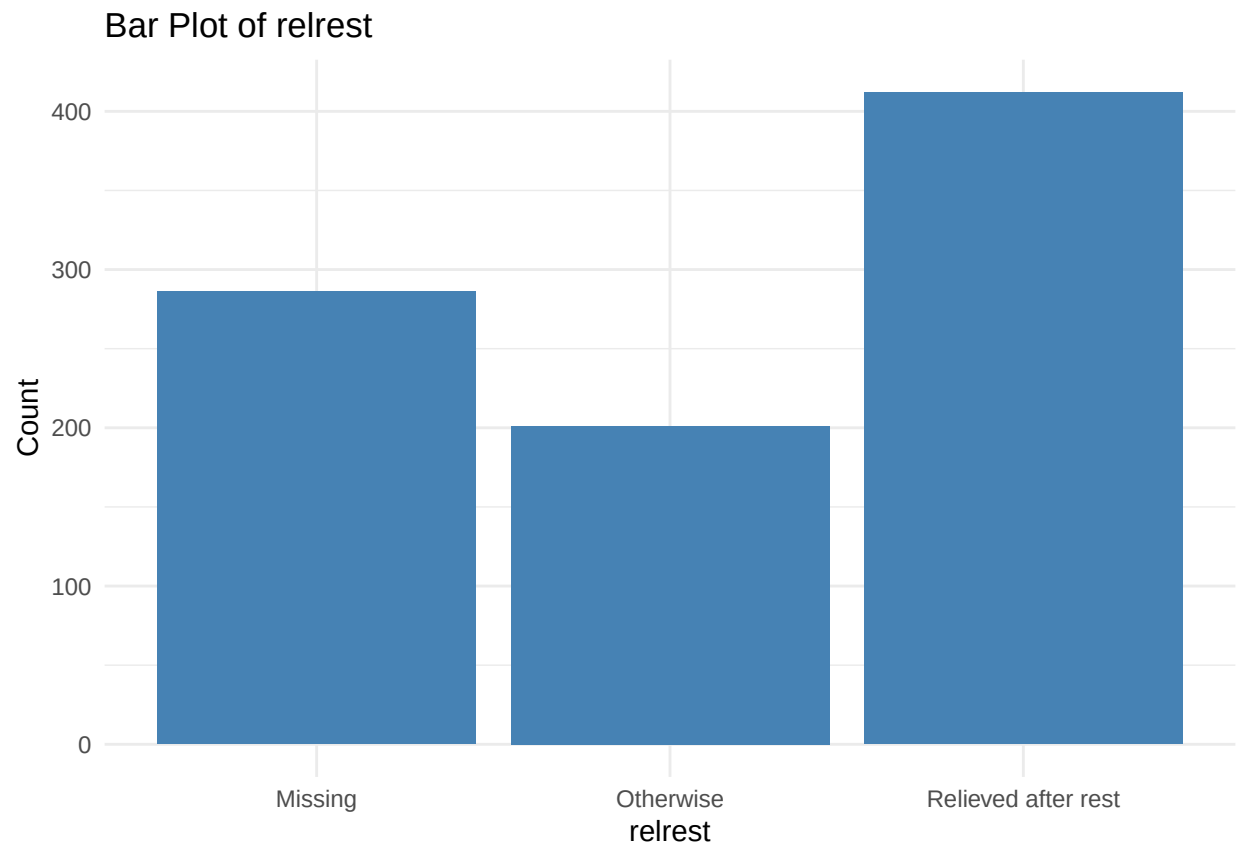
```
## 0.4961068 0.5038932
##
## ---- smoke_profile ----
##
##      heavy_long      heavy_medium      heavy_short
##      0.073414905      0.027808676      0.003337041
##      heavy_very_long      light_long      light_medium
##      0.012235818      0.013348165      0.015572859
##      light_short      light_very_long      missing
##      0.007786429      0.003337041      0.483870968
##      moderate_long      moderate_medium      moderate_short
##      0.087875417      0.032258065      0.016685206
##      moderate_very_long      non      very_heavy_long
##      0.013348165      0.170189099      0.021134594
##      very_heavy_medium      very_heavy_short      very_heavy_very_long
##      0.008898776      0.006674082      0.002224694
##
## ---- rldv5_missing ----
##
##      0      1
## 0.5272525 0.4727475
```

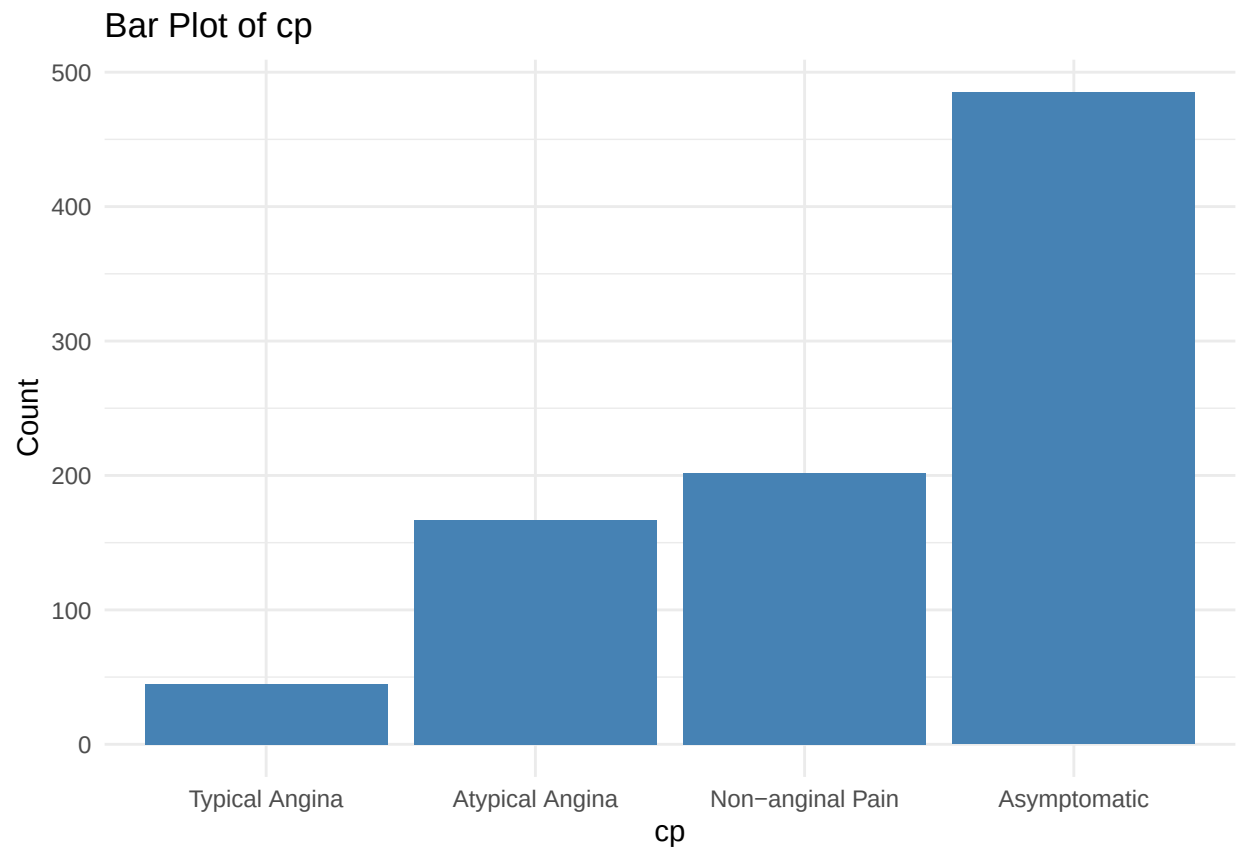
Bar Plot of sex

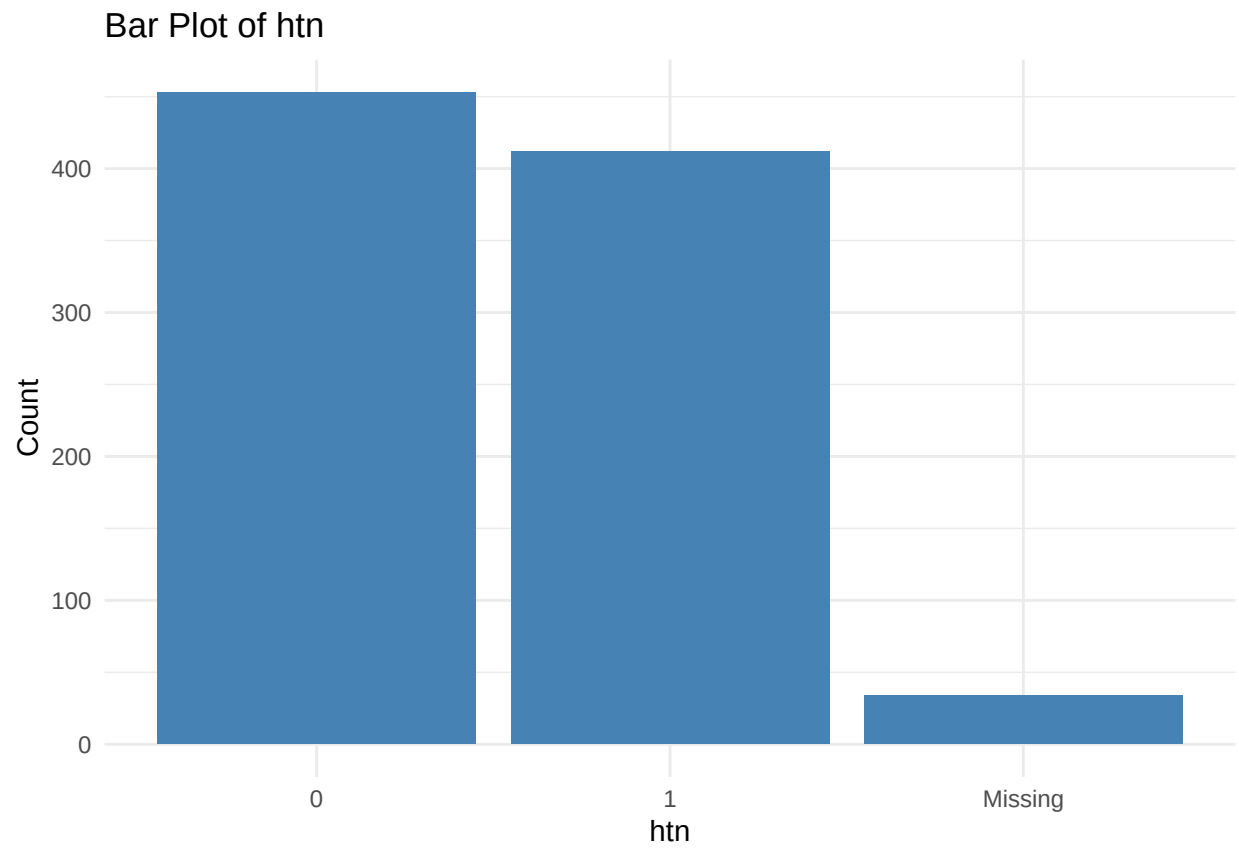


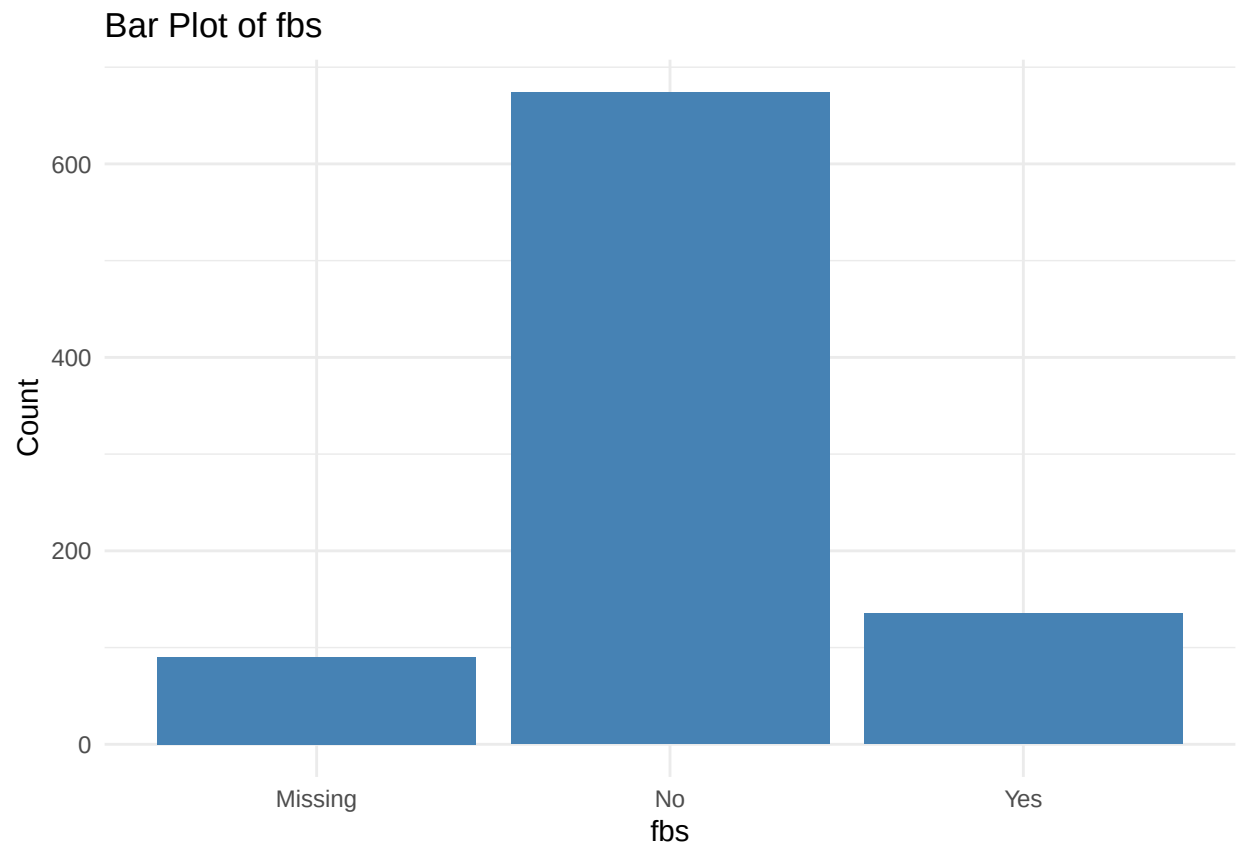


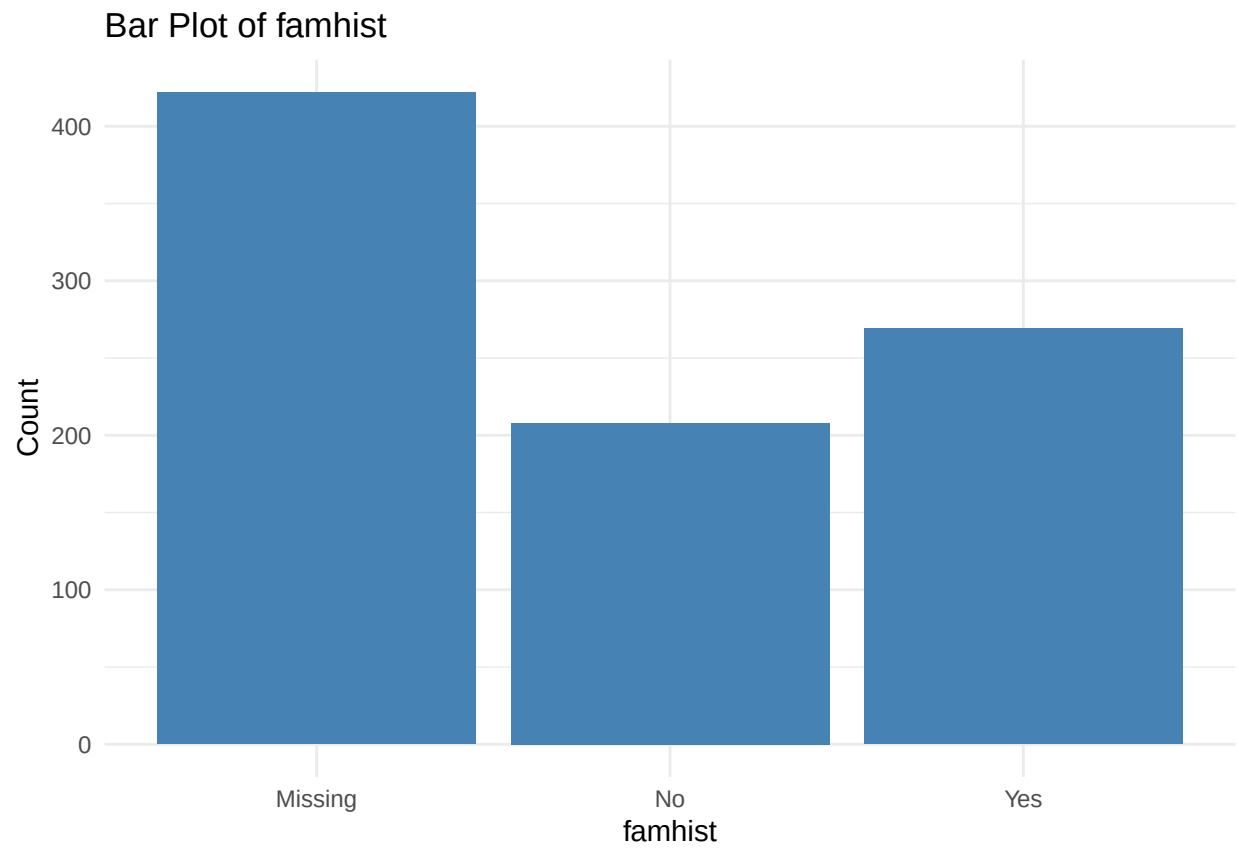


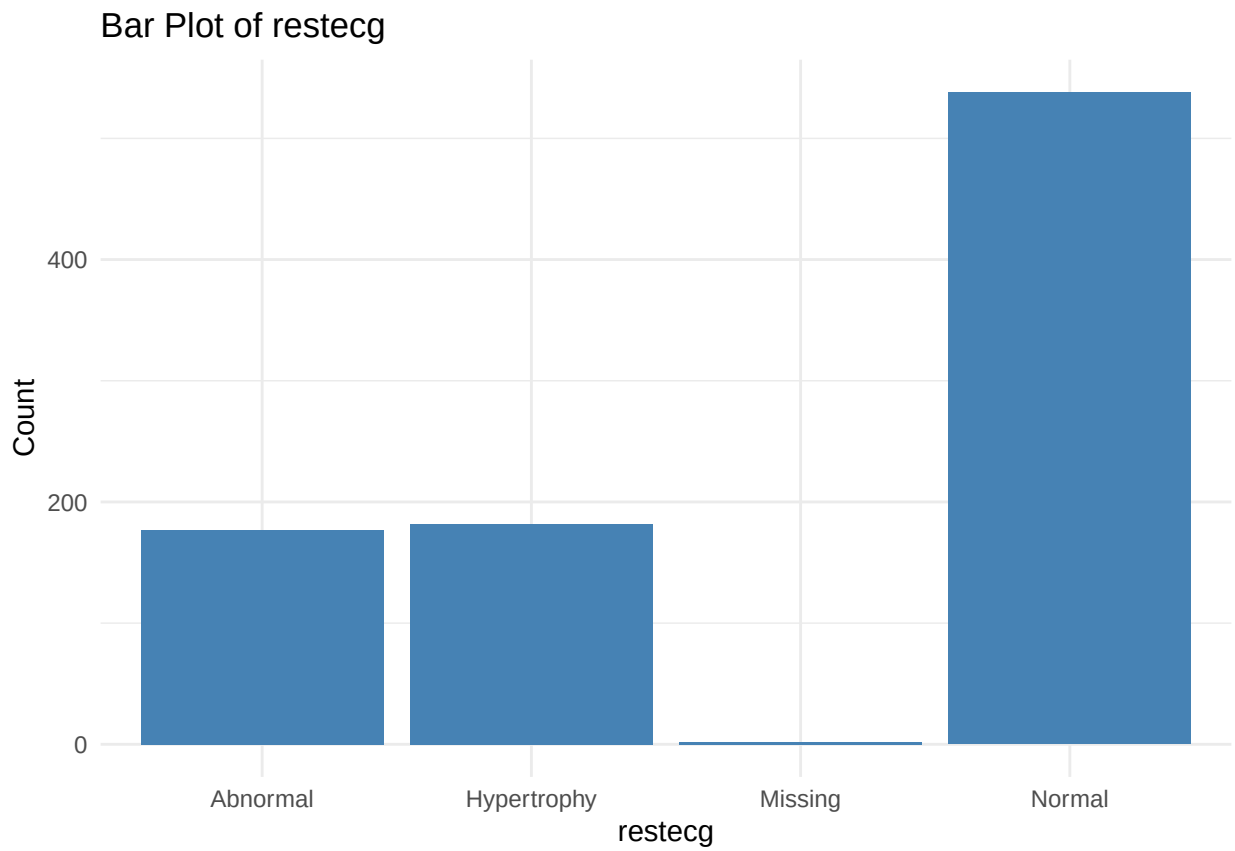


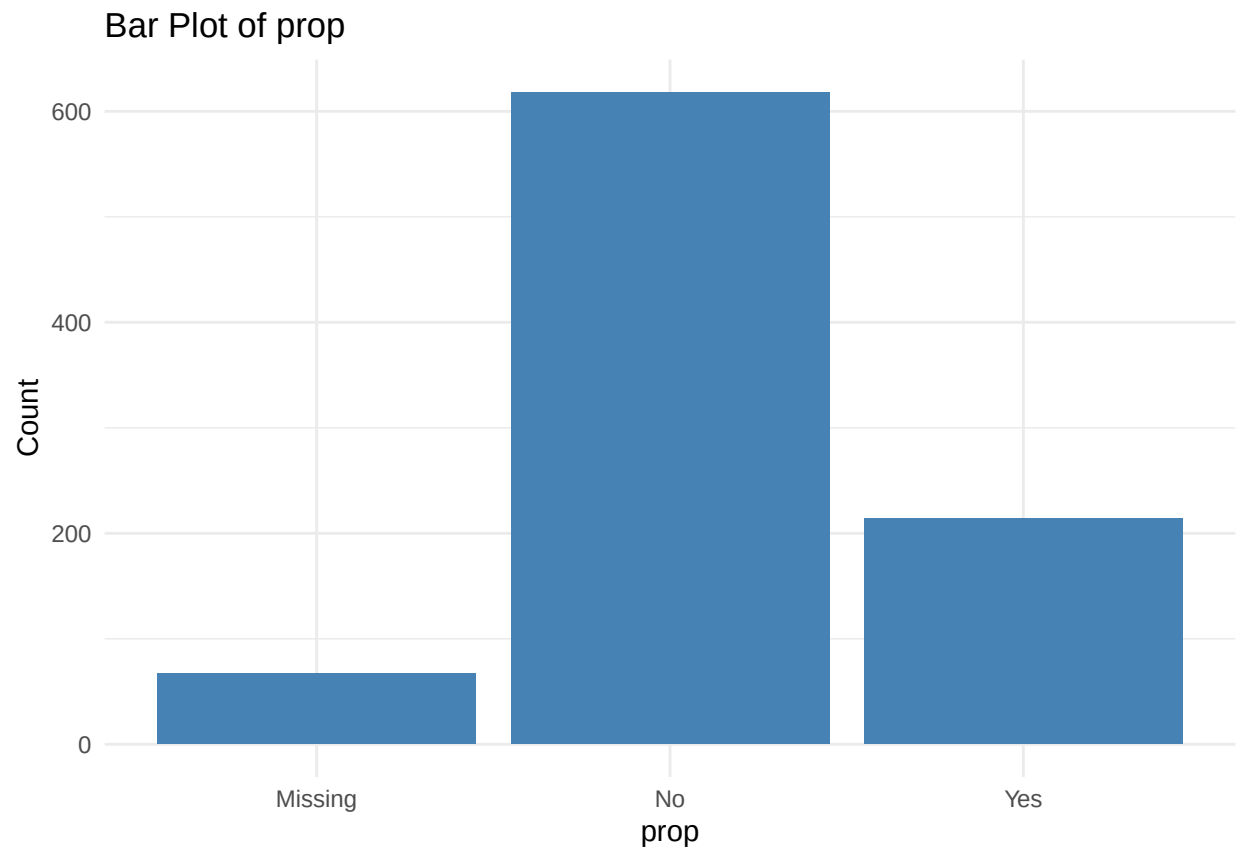


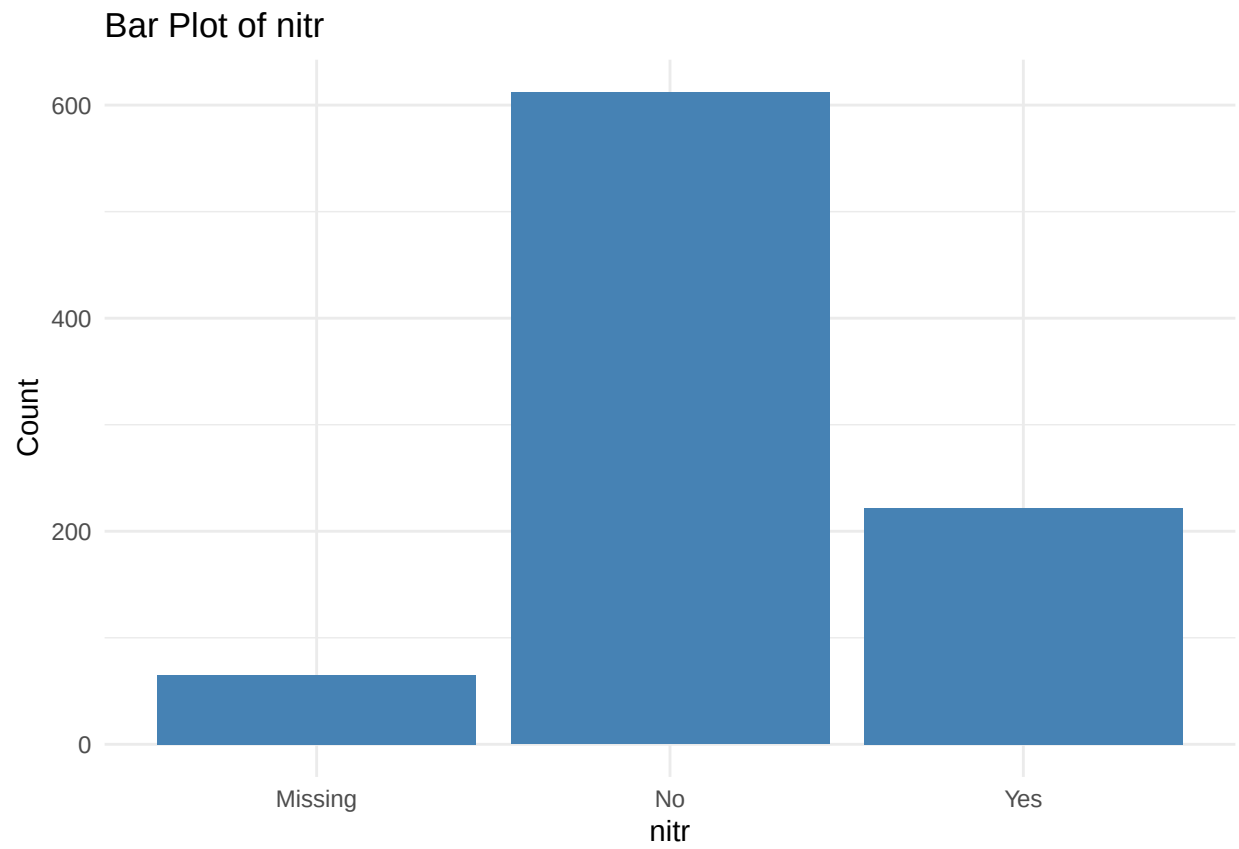


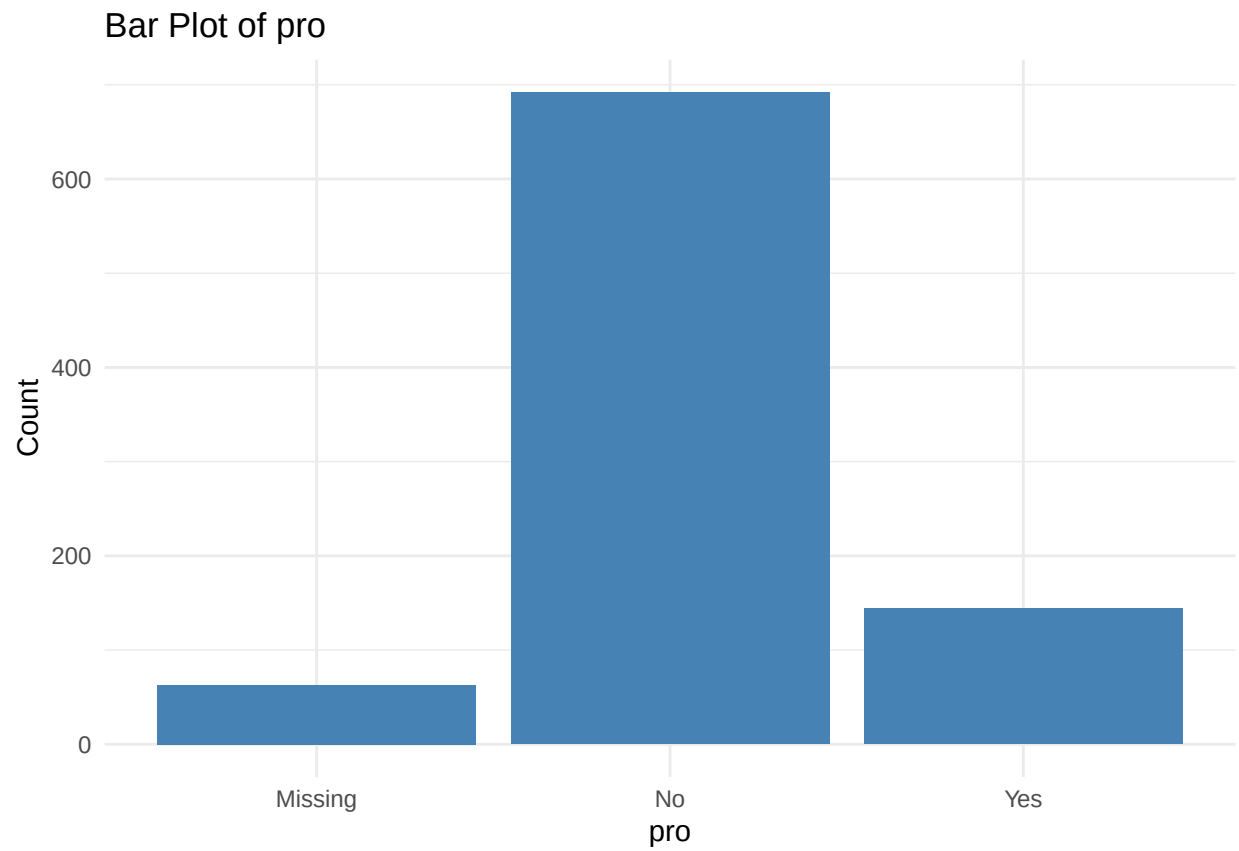


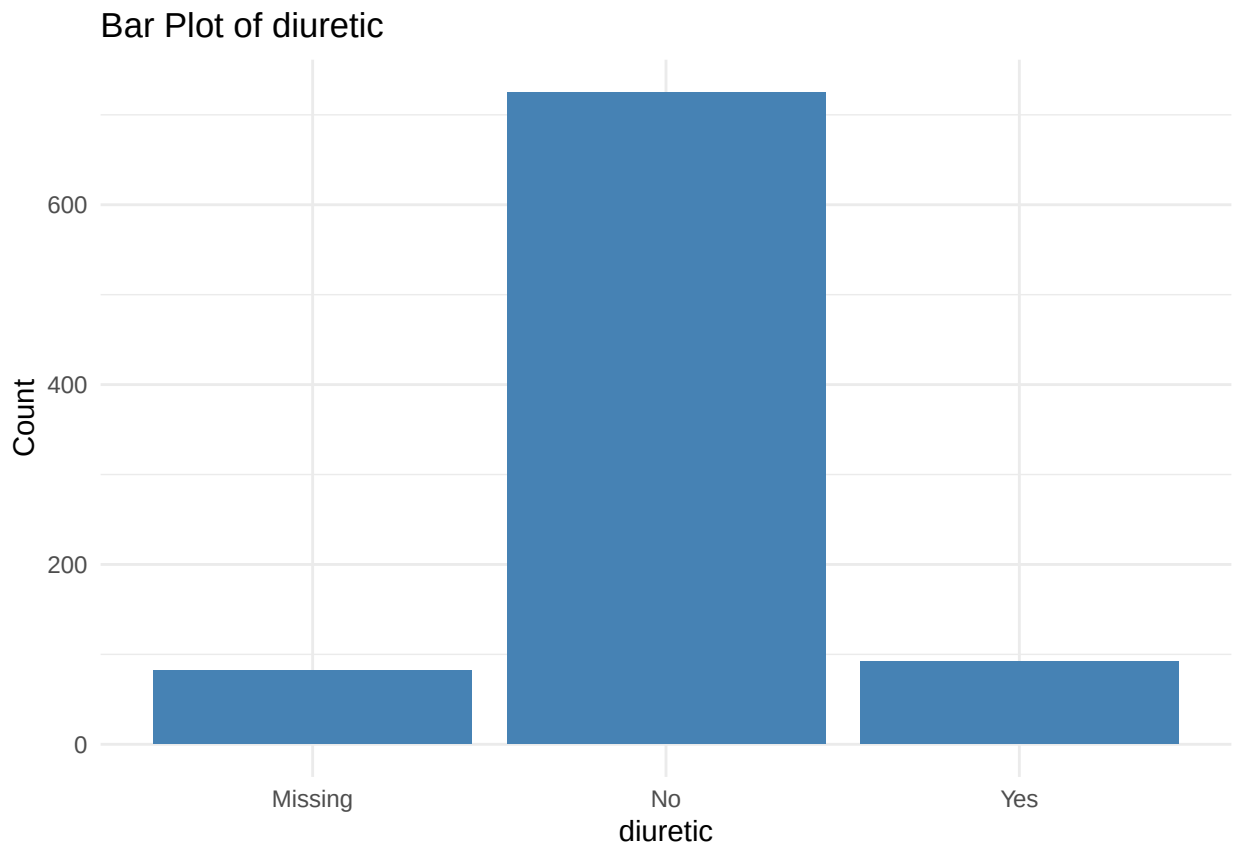


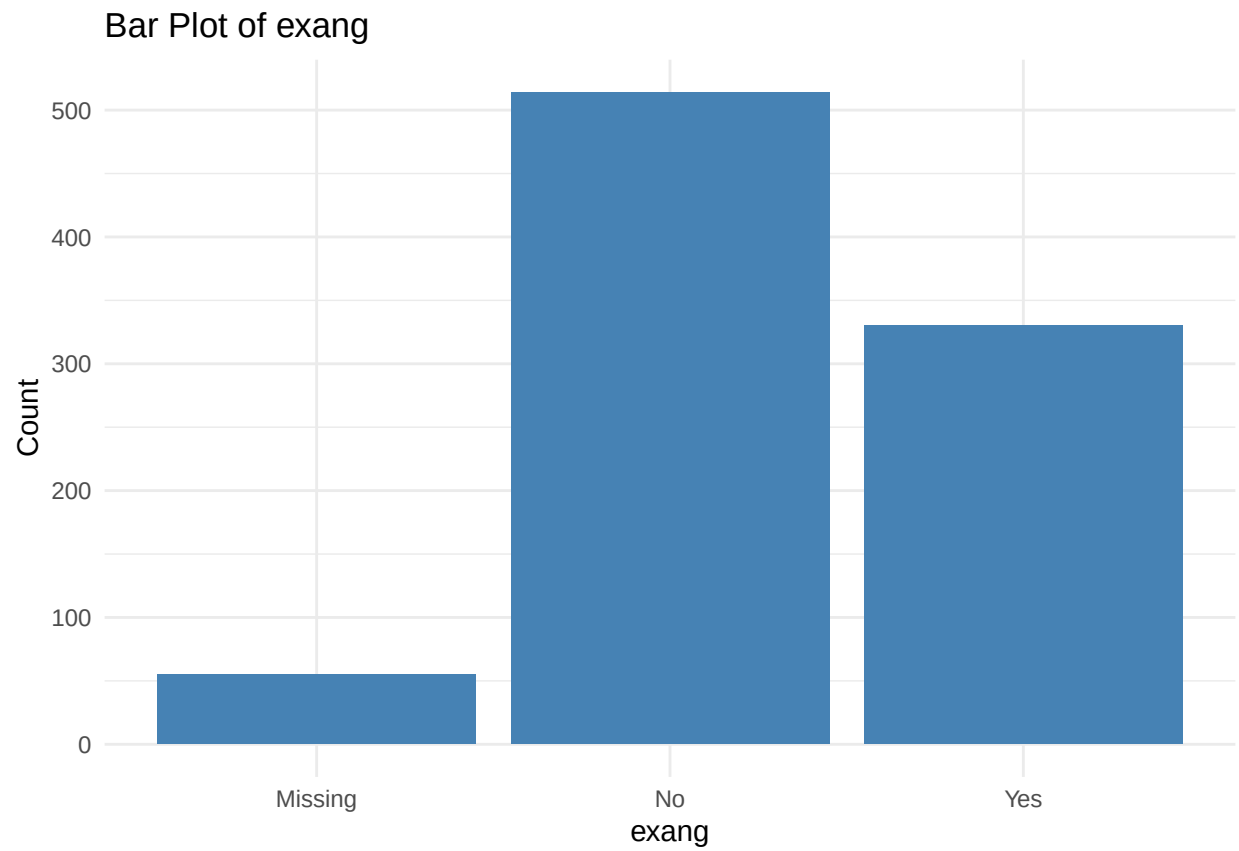


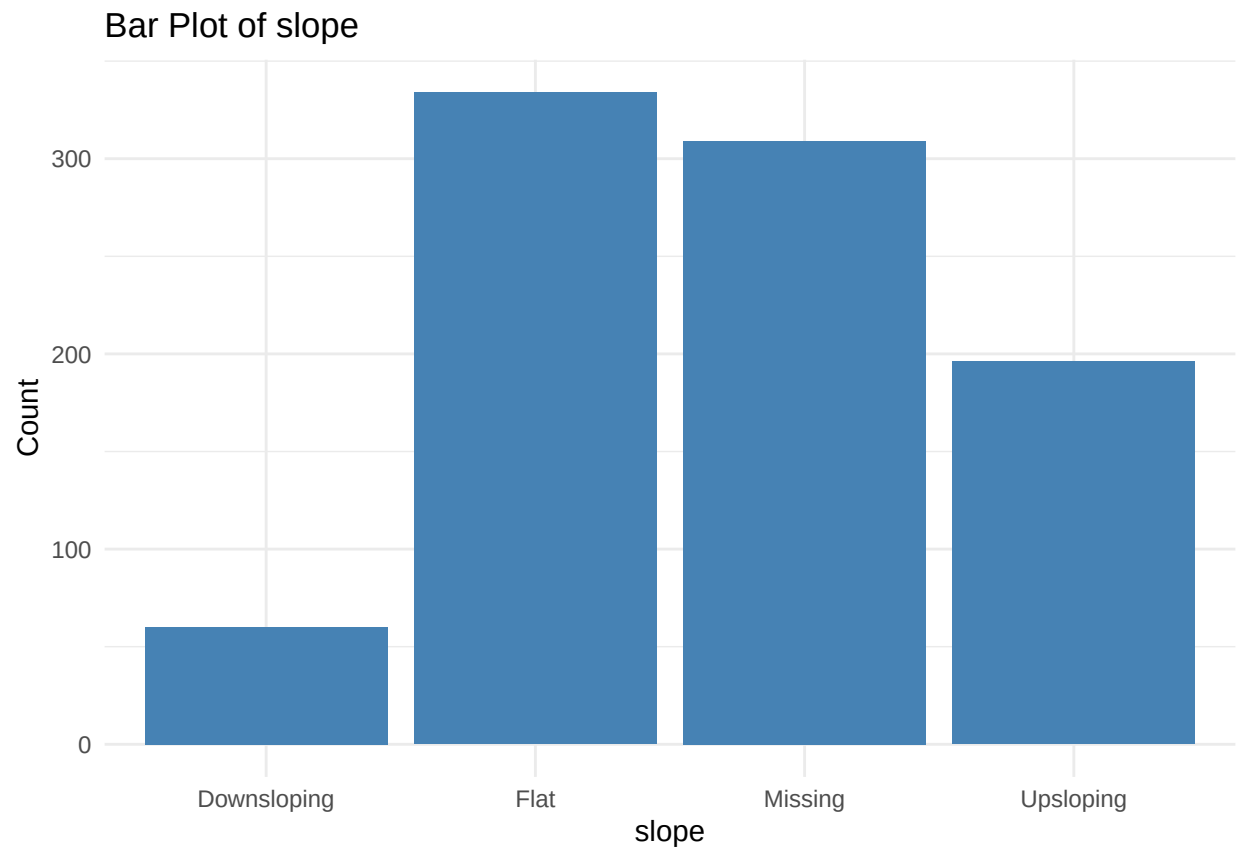


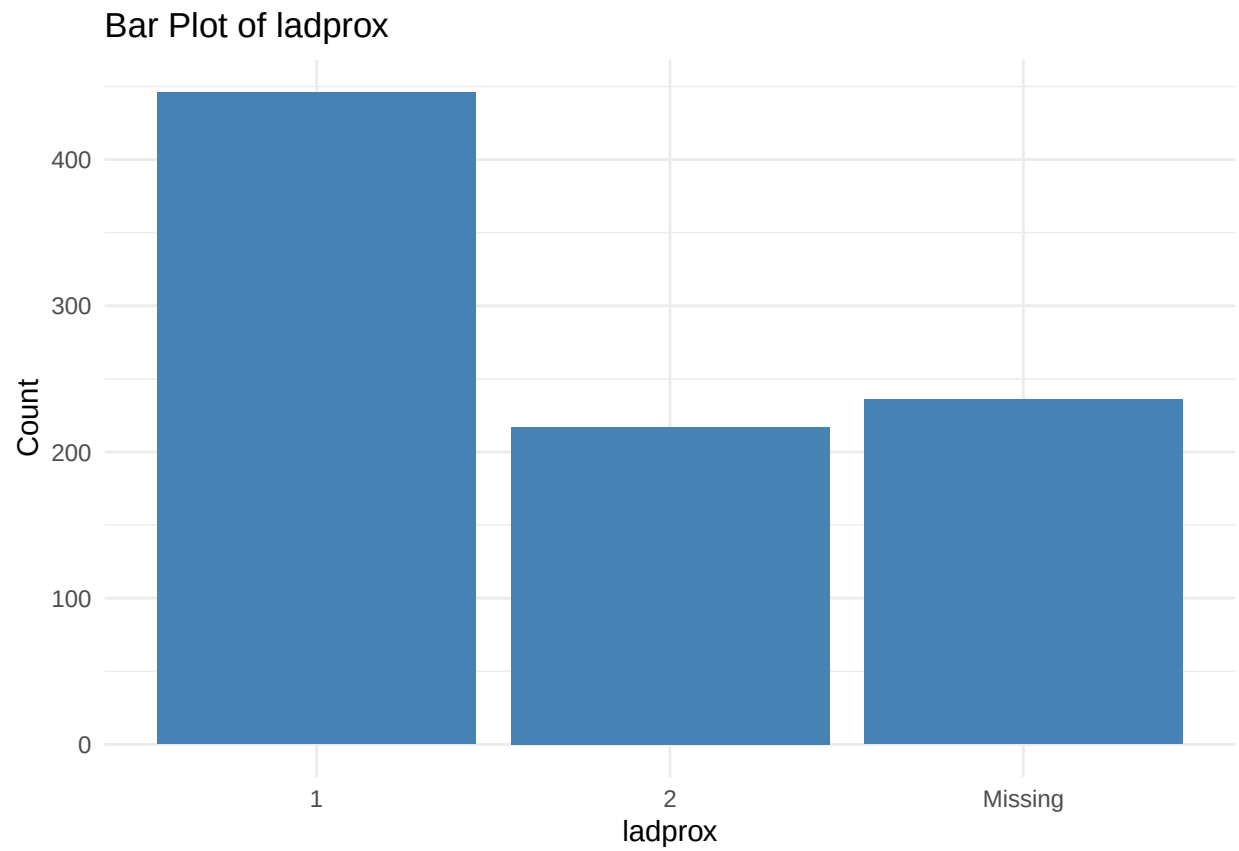


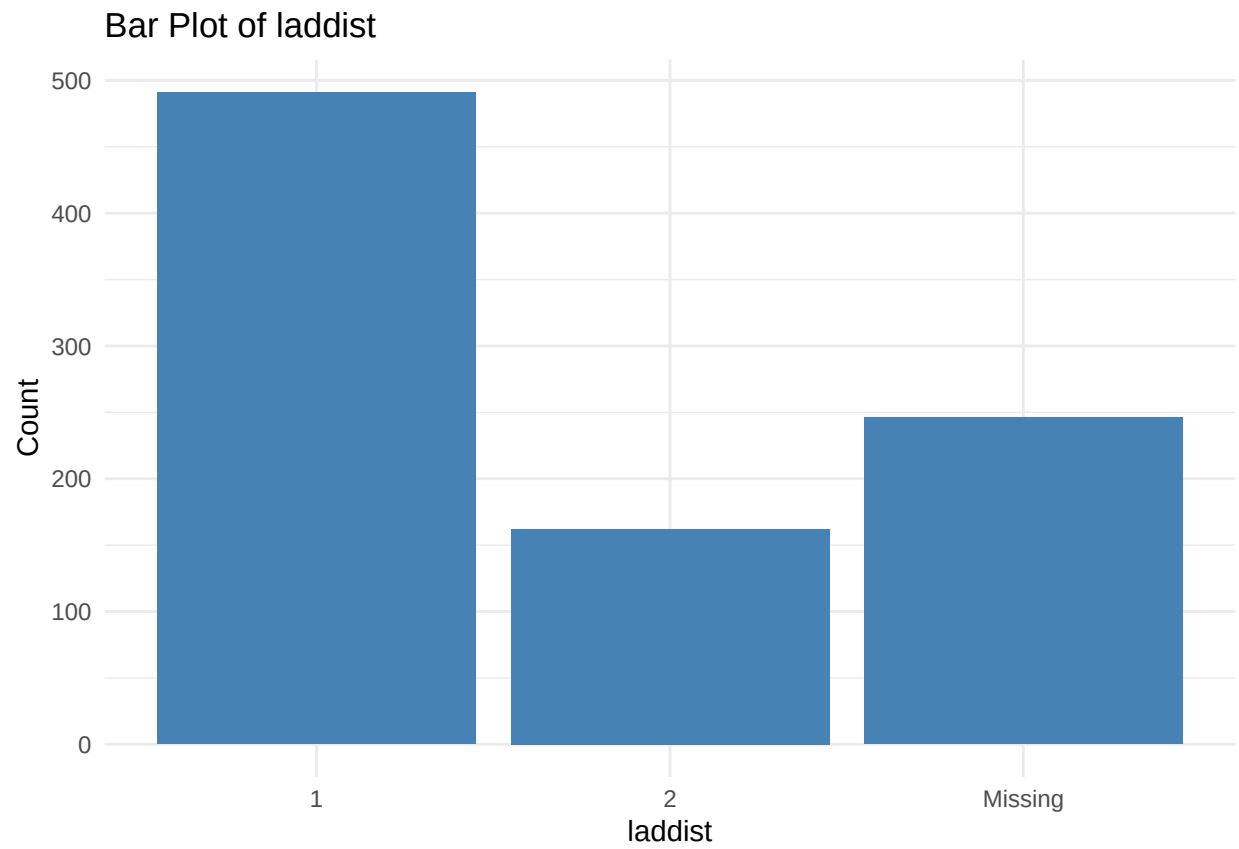


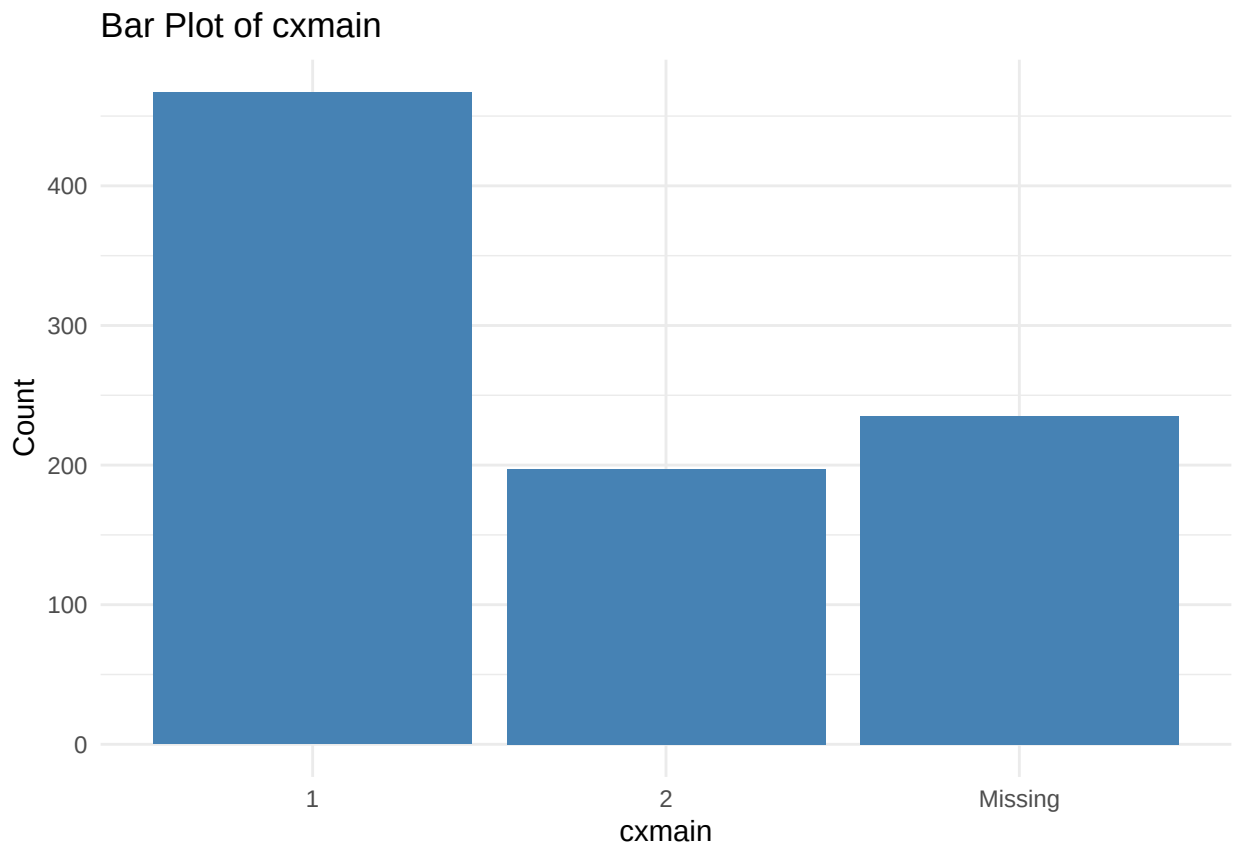


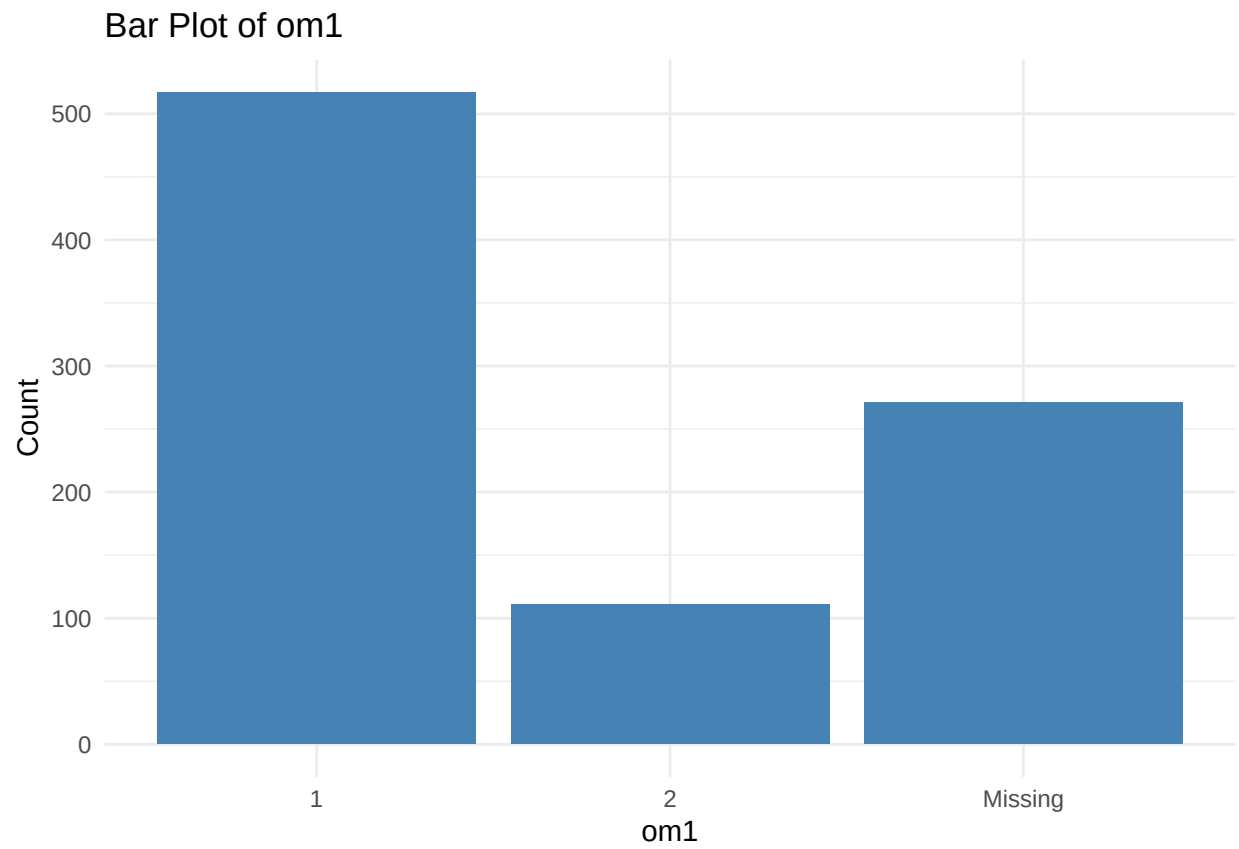


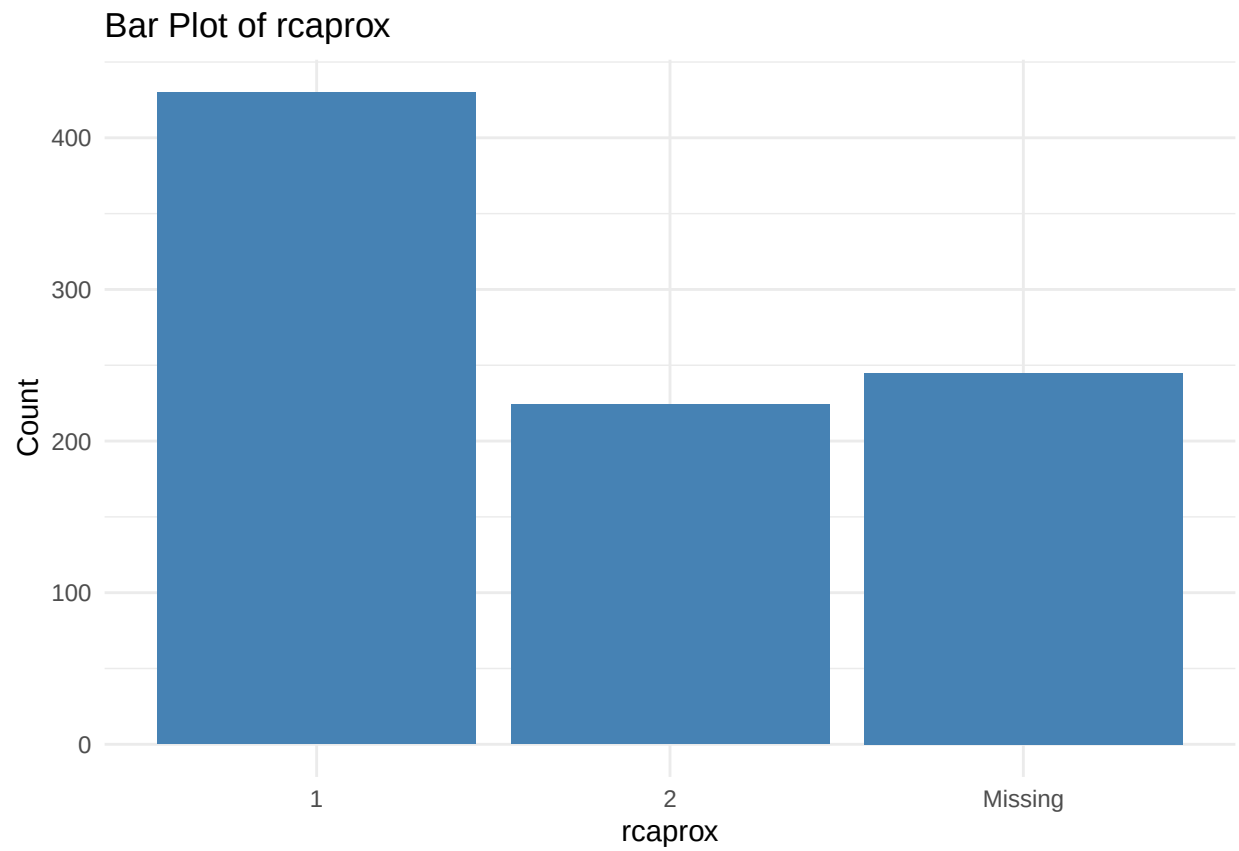


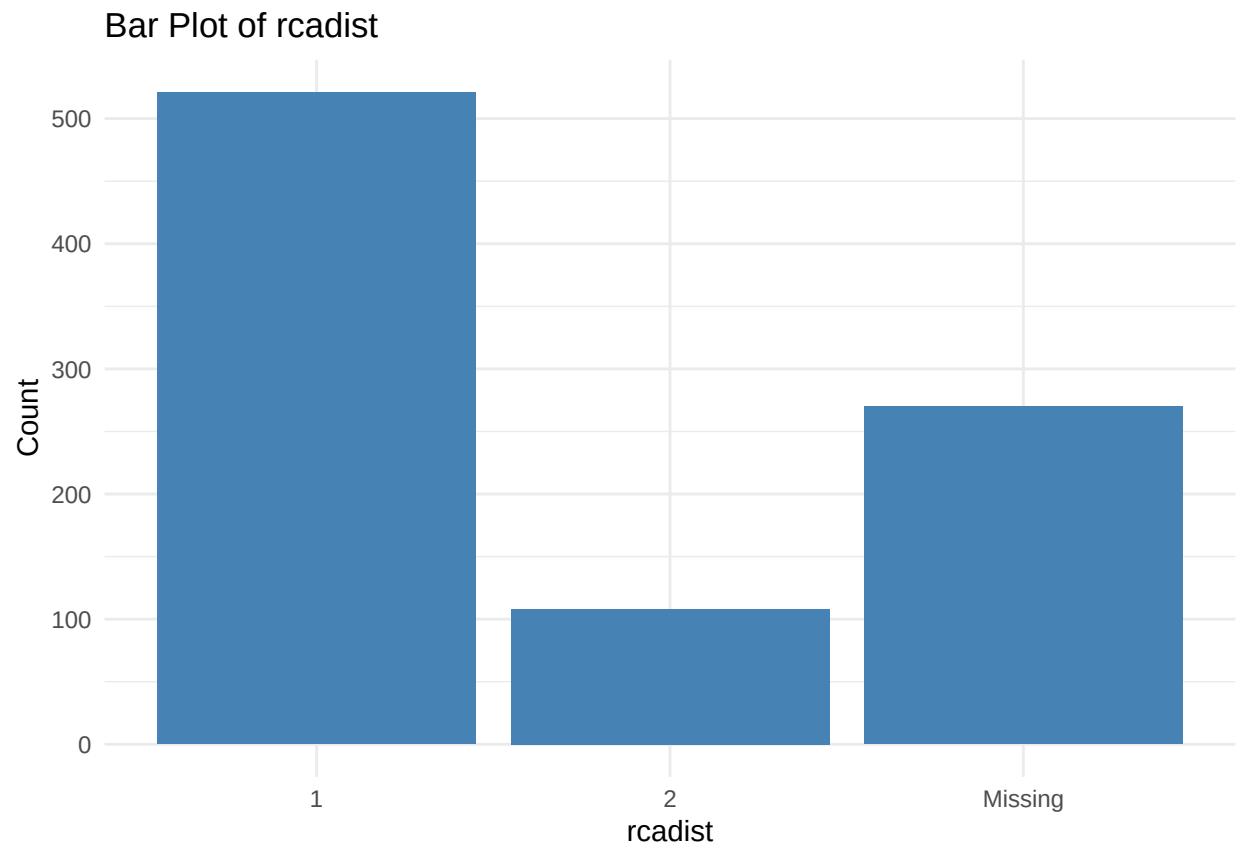


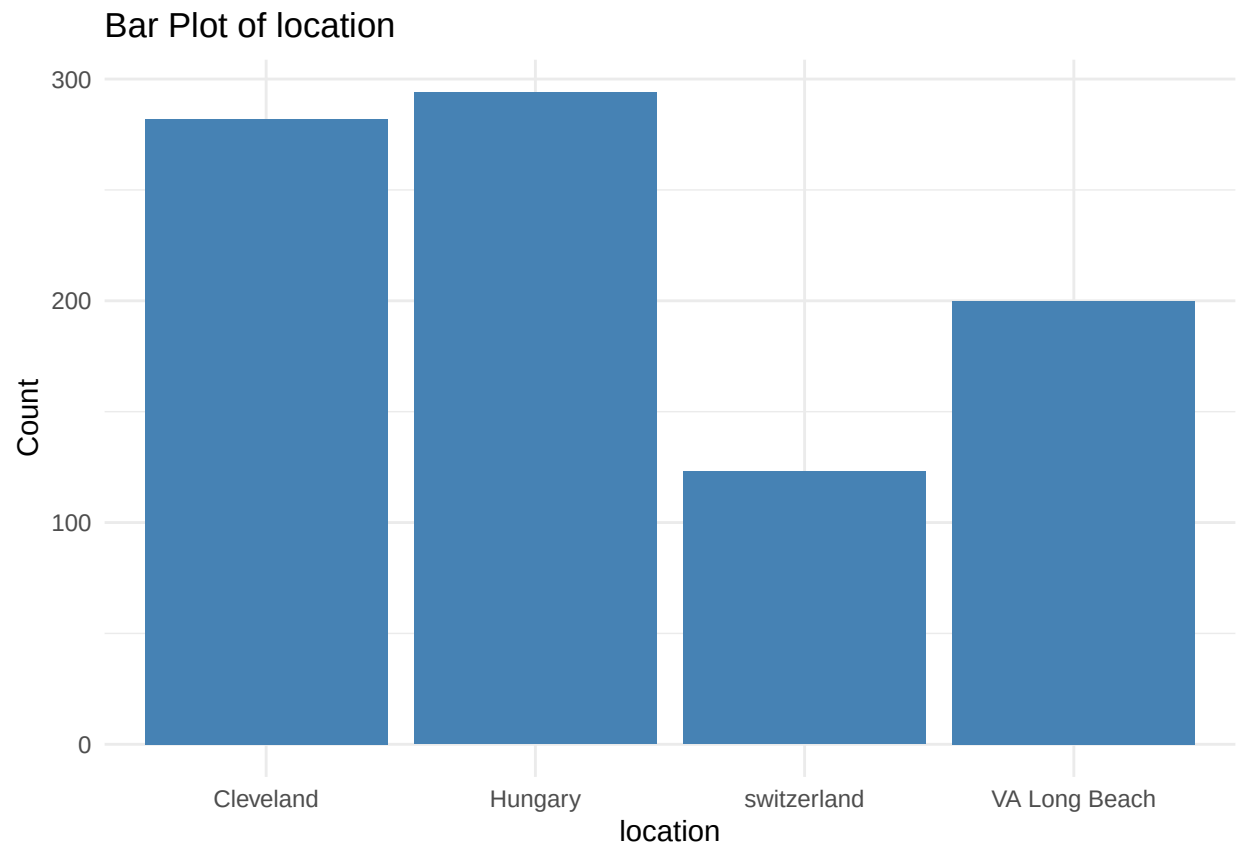


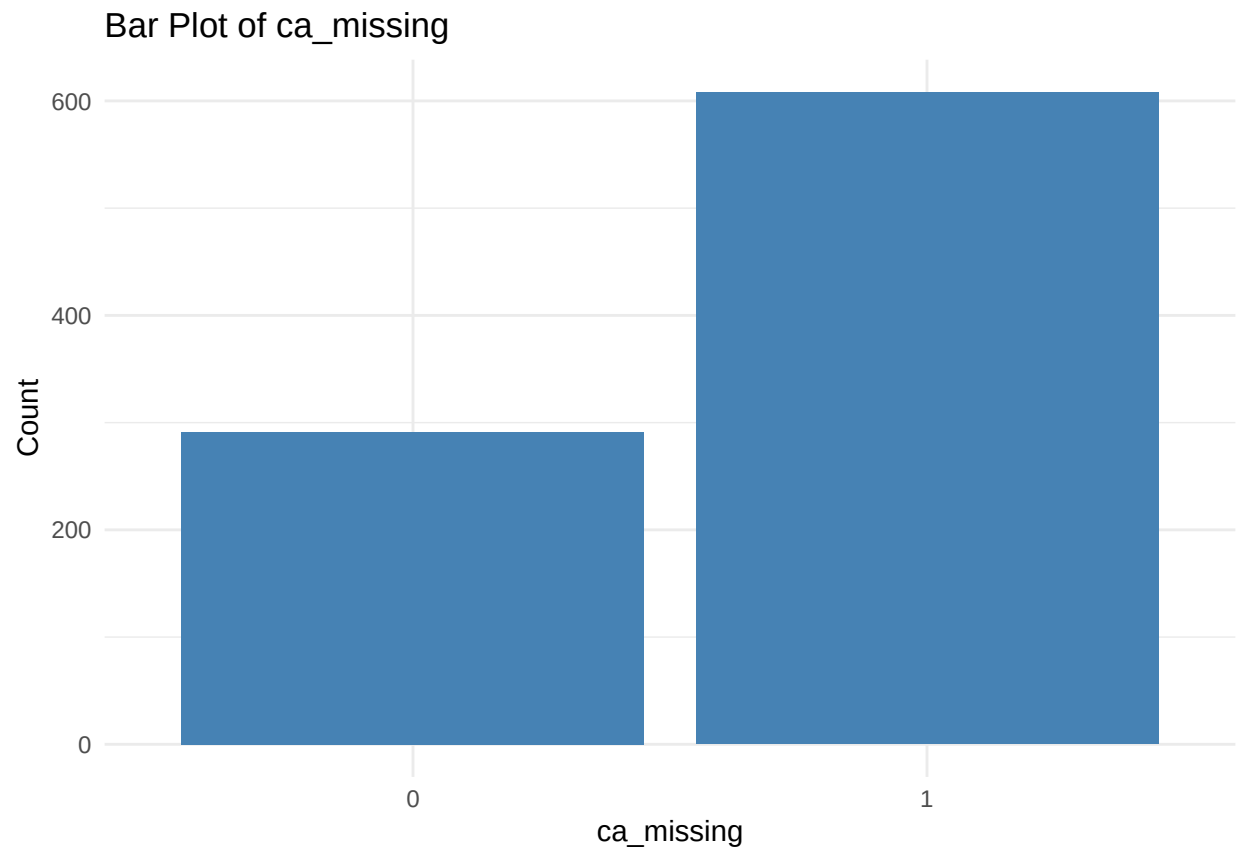


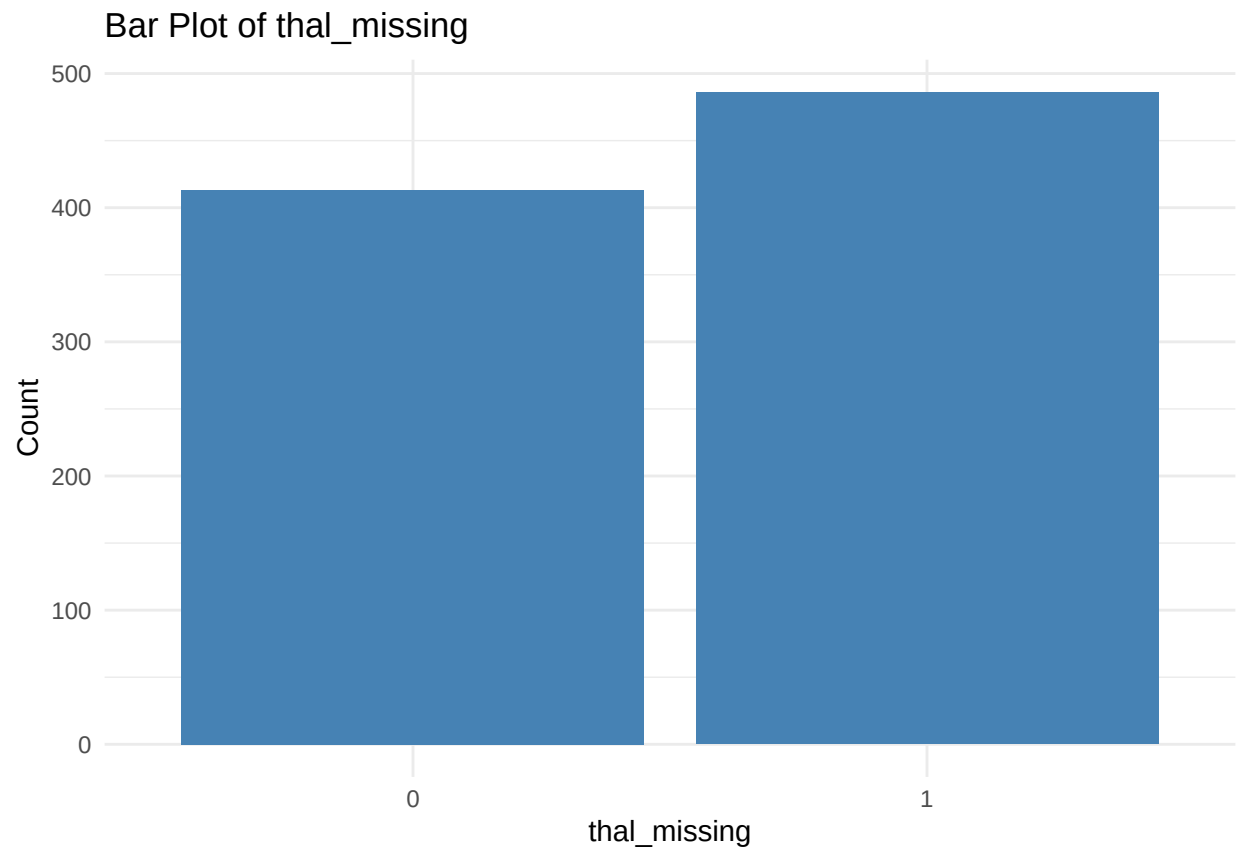


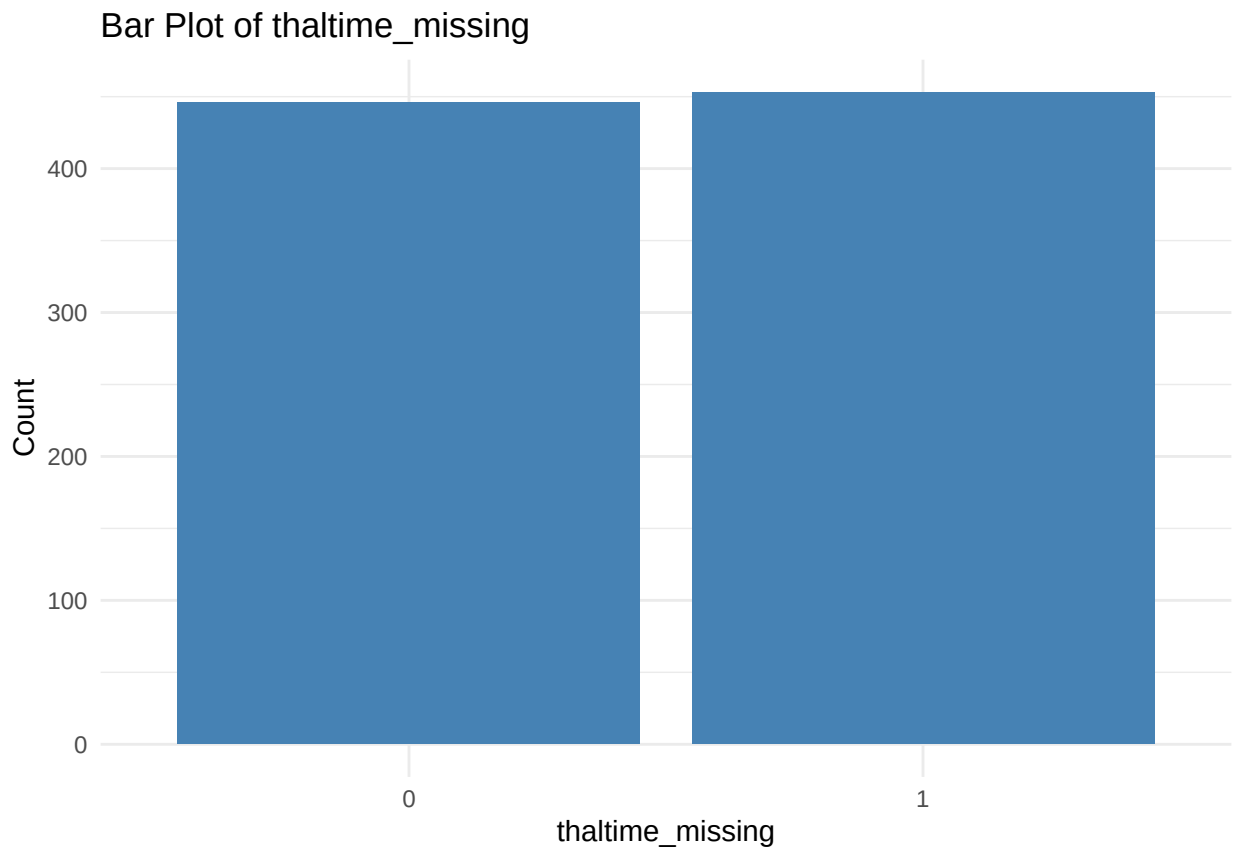


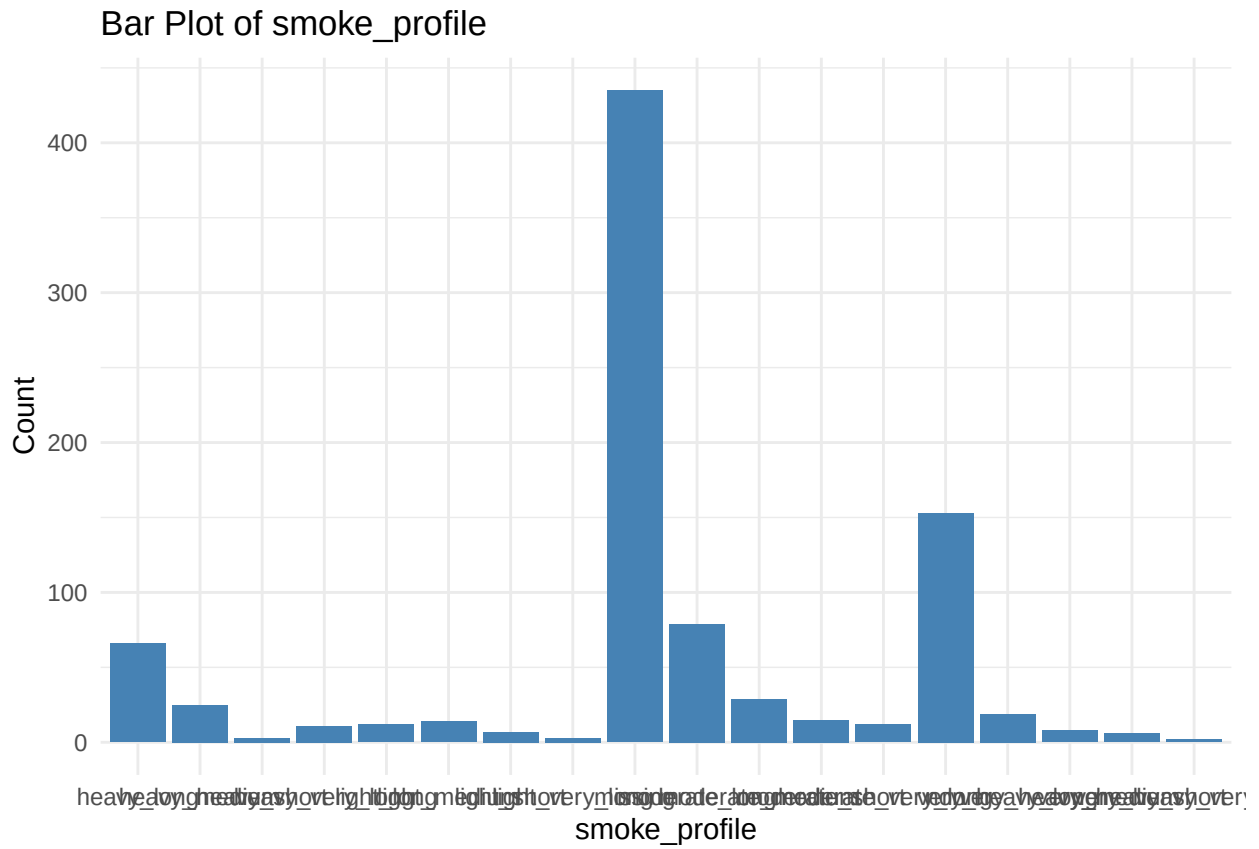




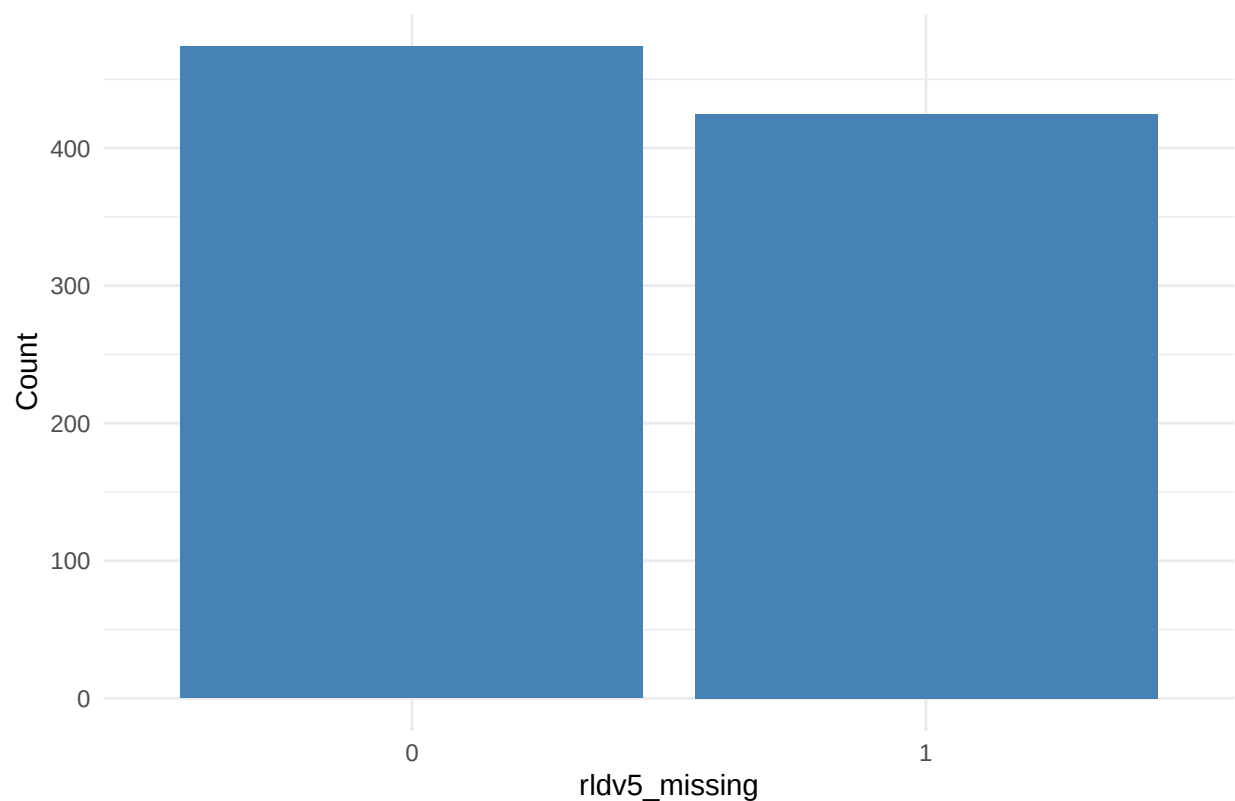








Bar Plot of rldv5_missing



Feature Types

```
X_df %>%  
  summarise(  
    numeric = sum(sapply(., is.numeric)),  
    categorical = sum(sapply(., is.factor))  
  )
```

```
## # A tibble: 1 x 2  
##   numeric categorical  
##   <int>      <int>  
## 1      29         26
```

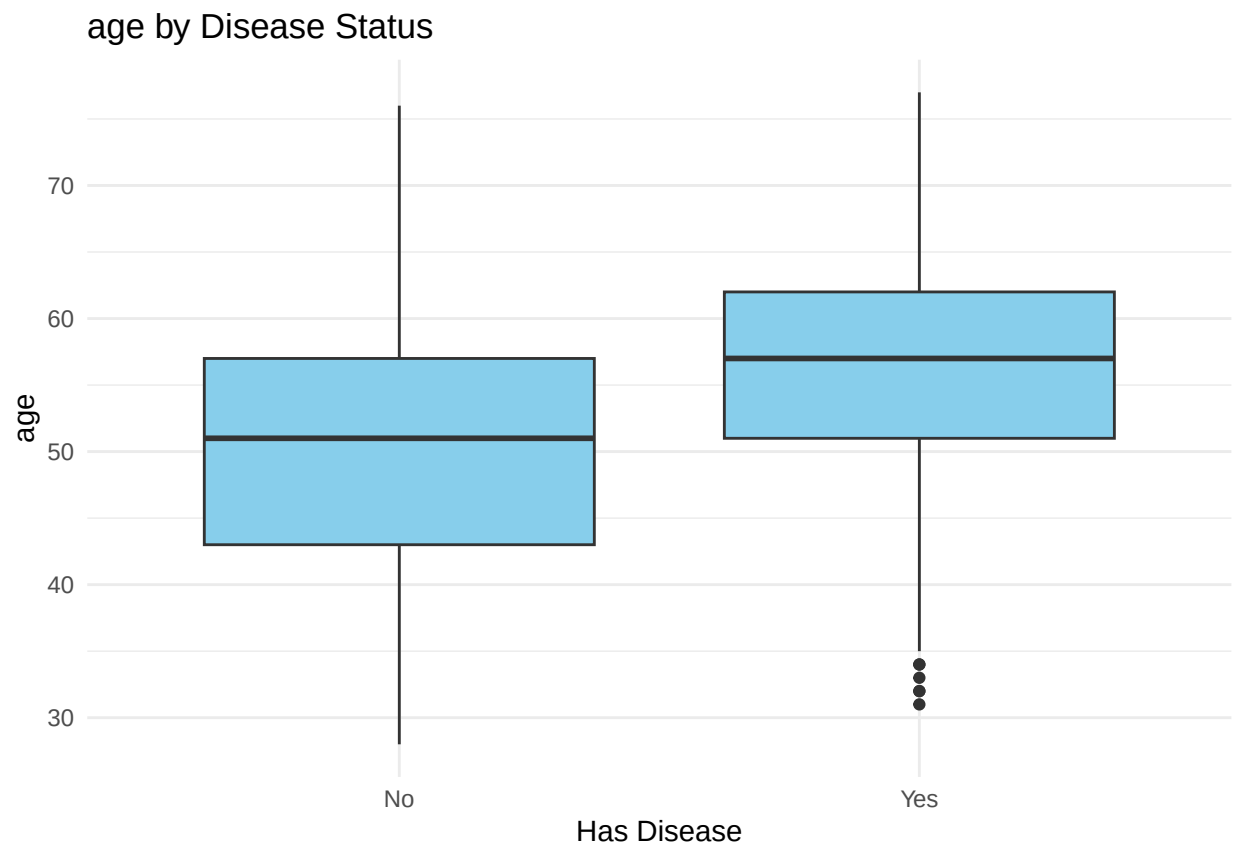
Bivariate Analysis

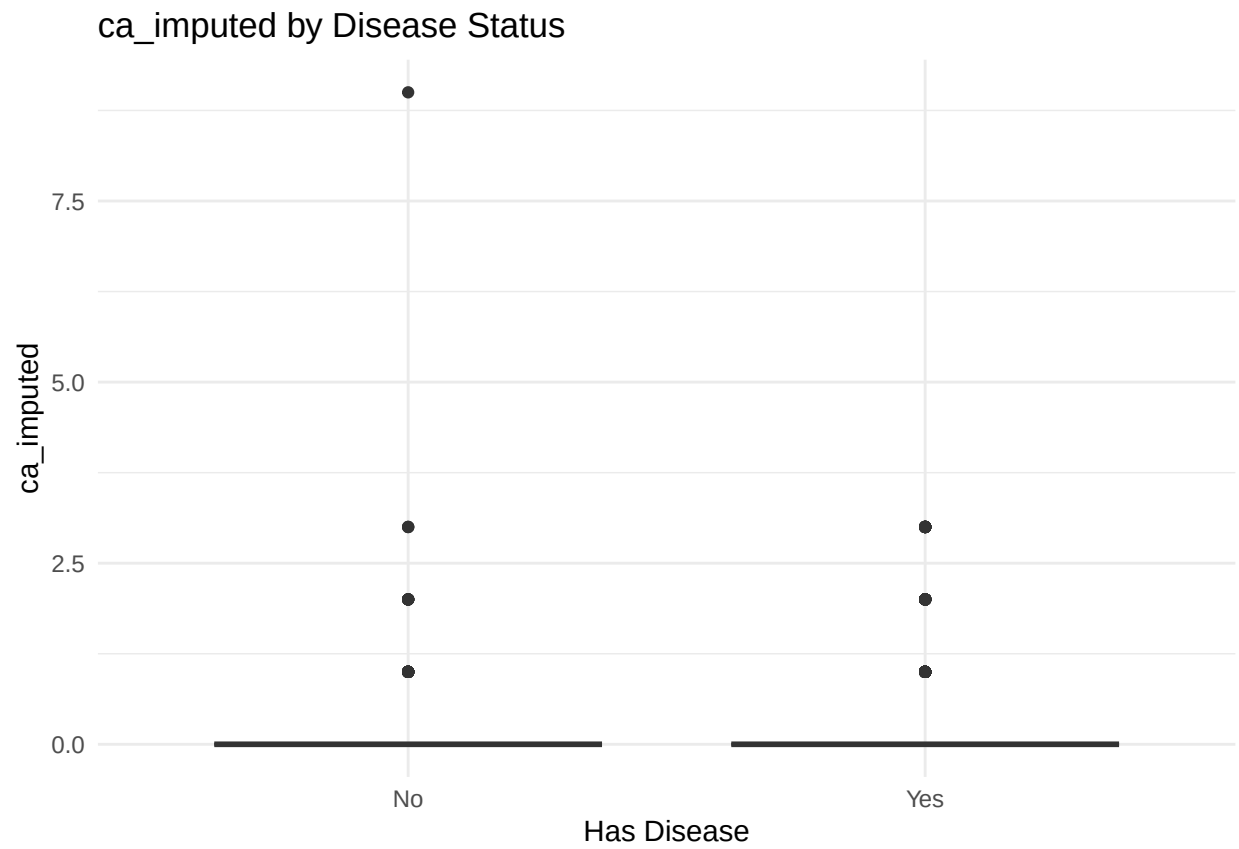
Numerical Vars

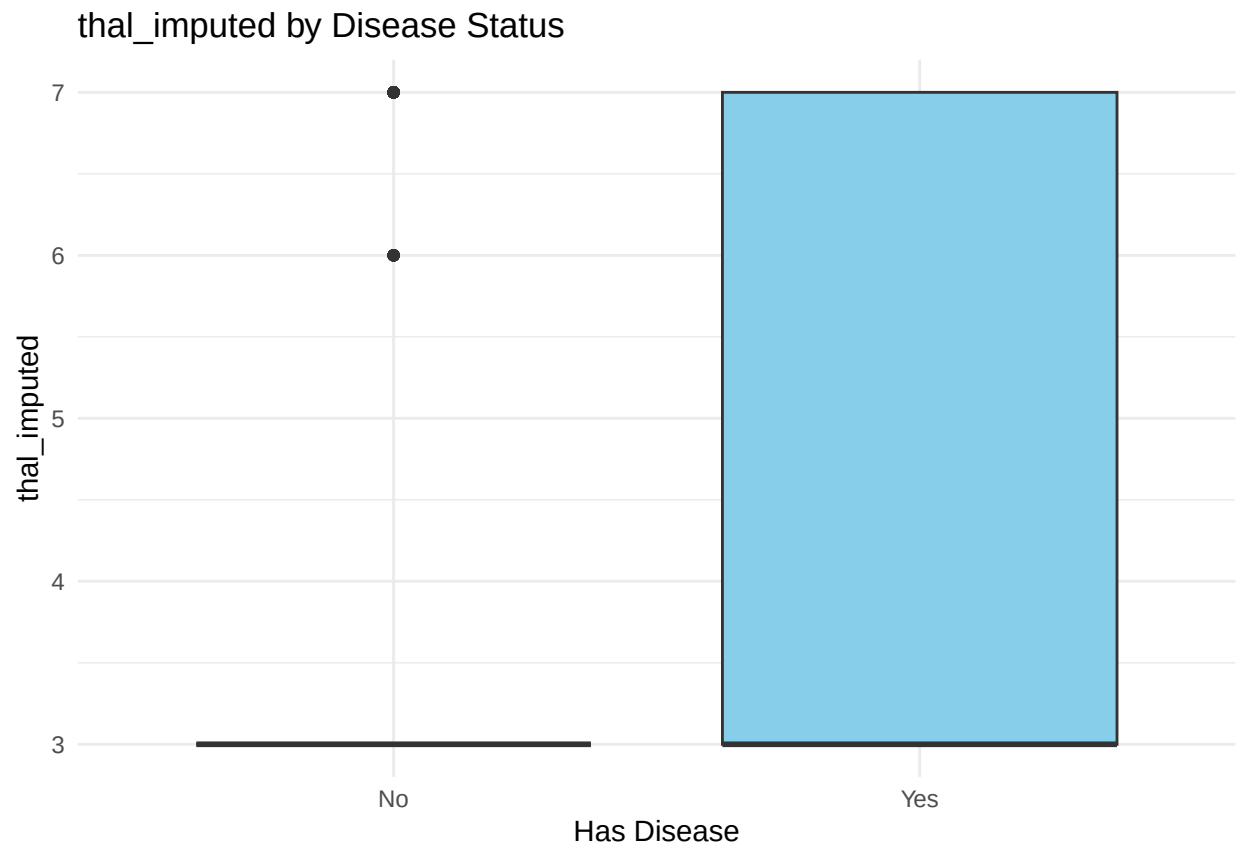
Box Plots

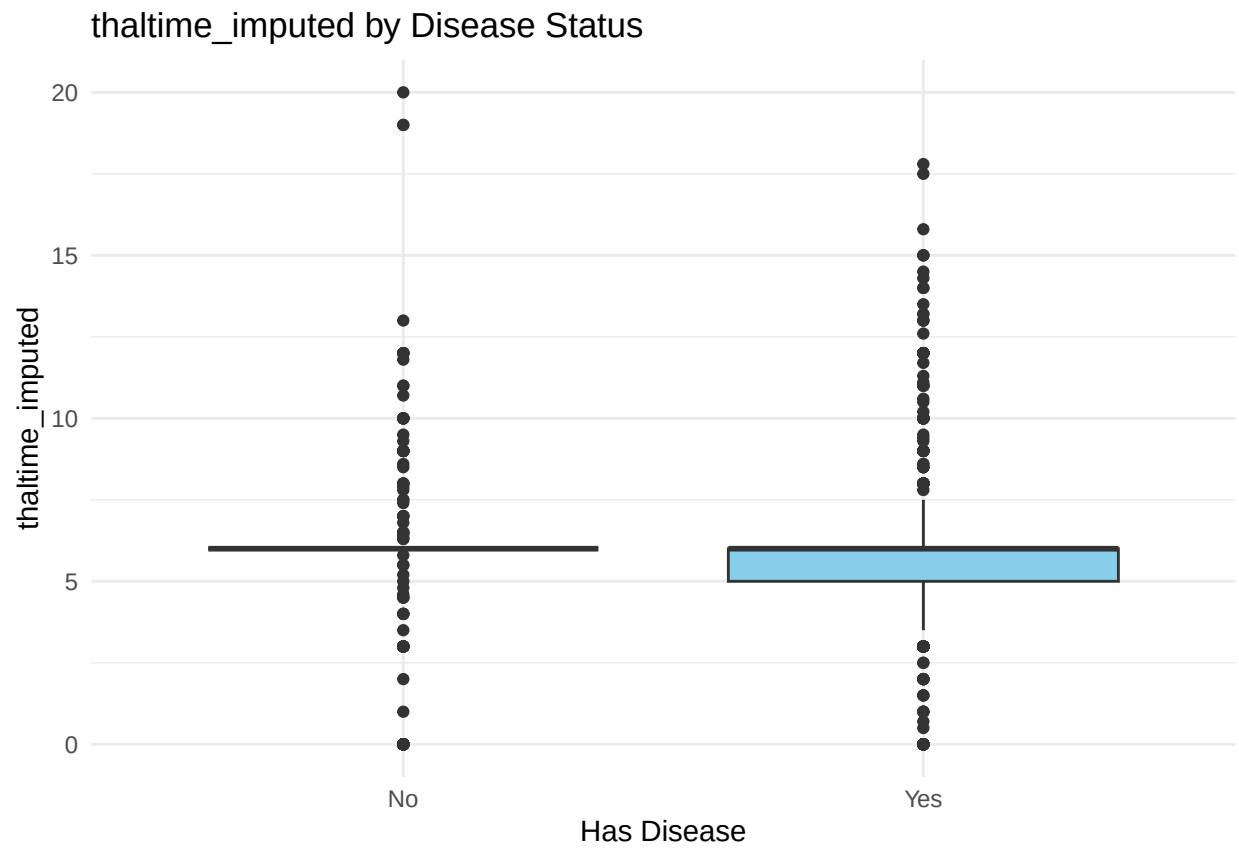
```
plot_bivariate_num <- function(var) {  
  ggplot(df, aes(x = factor(has_disease), y = .data[[var]])) +  
    geom_boxplot(fill = "skyblue") +  
    labs(title = paste(var, "by Disease Status"), x = "Has Disease", y = var) +  
    theme_minimal()  
}
```

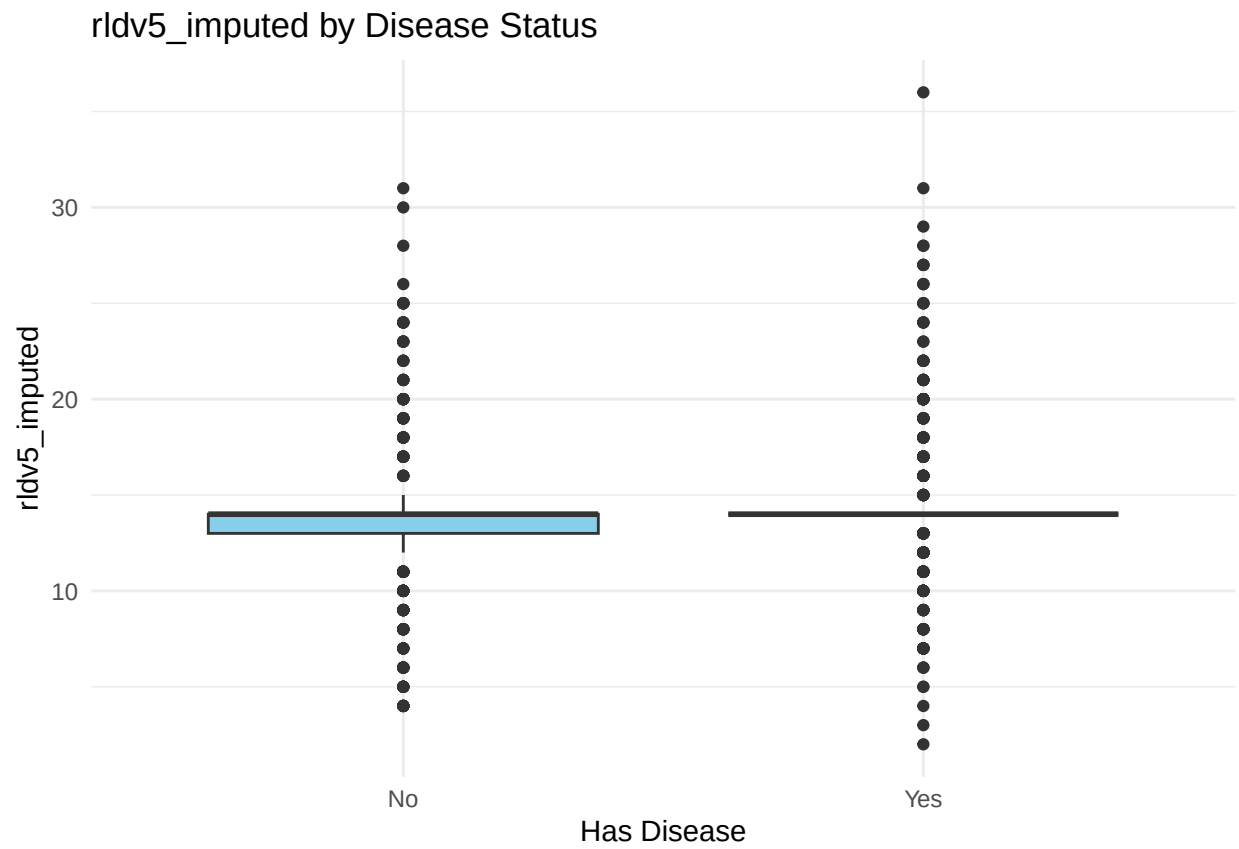
```
for (var in num_vars){  
  print(plot_bivariate_num(var))  
}
```

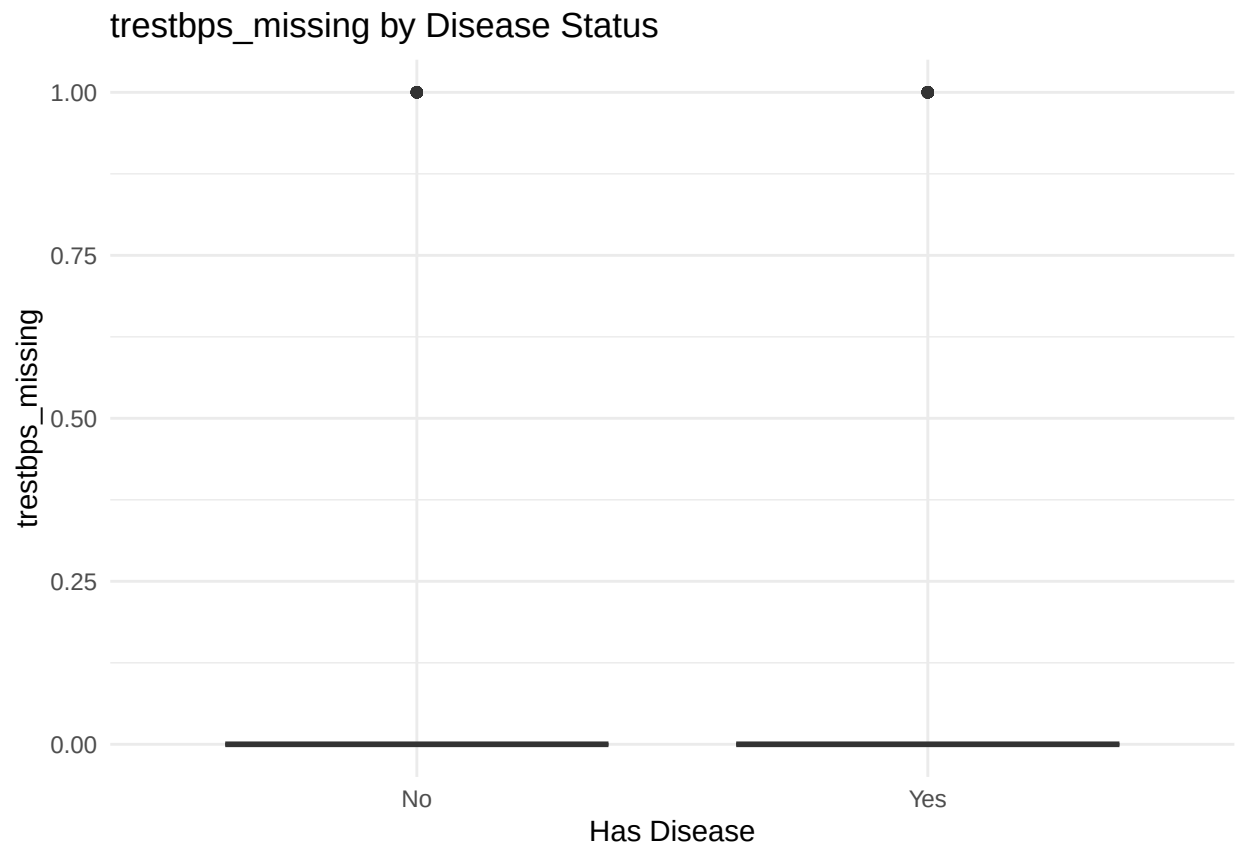


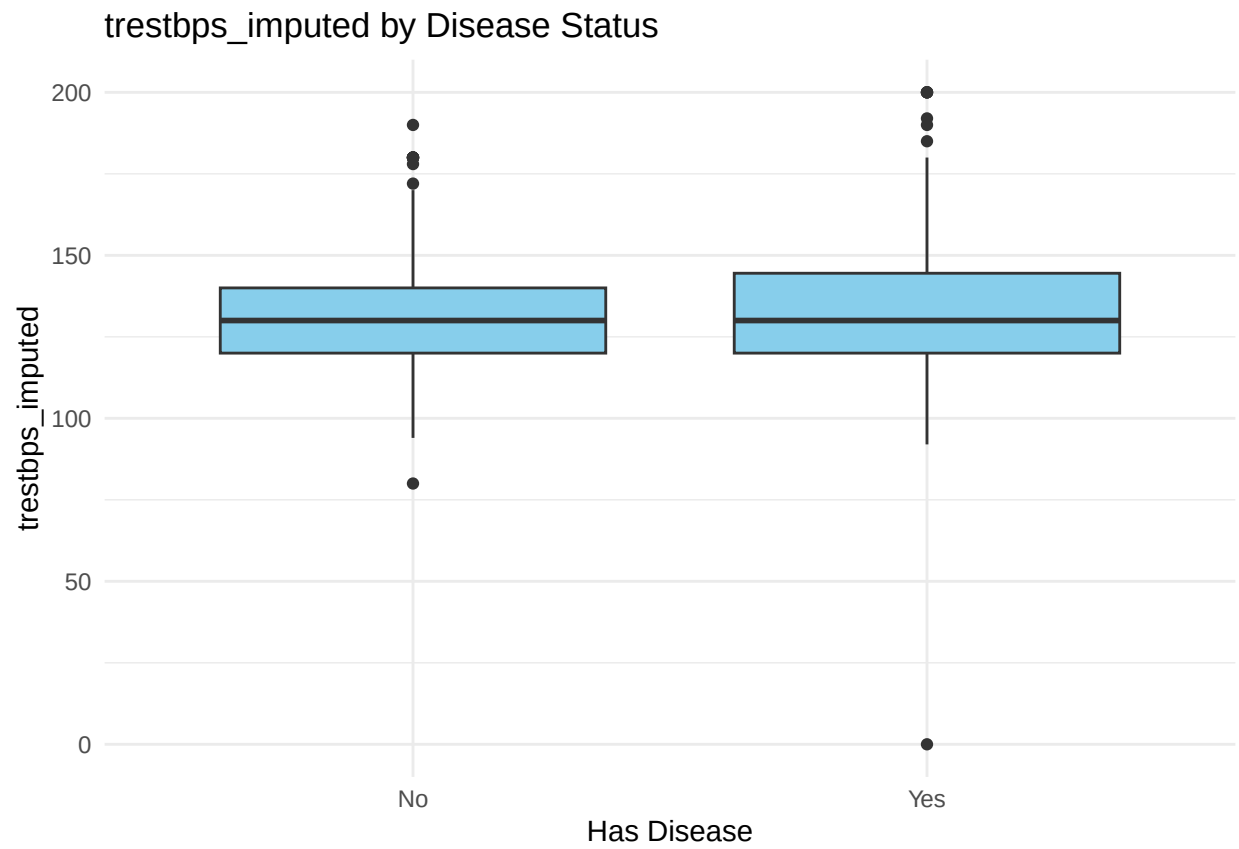


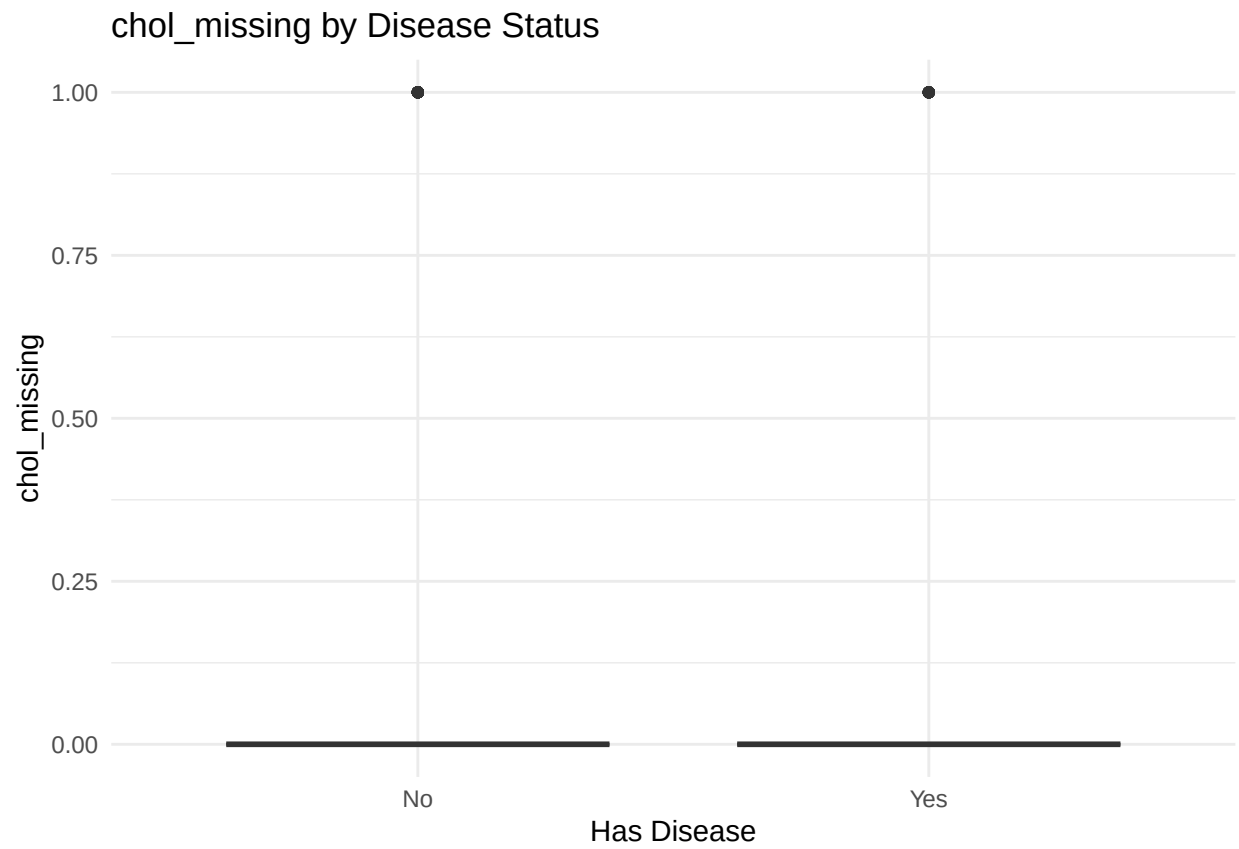


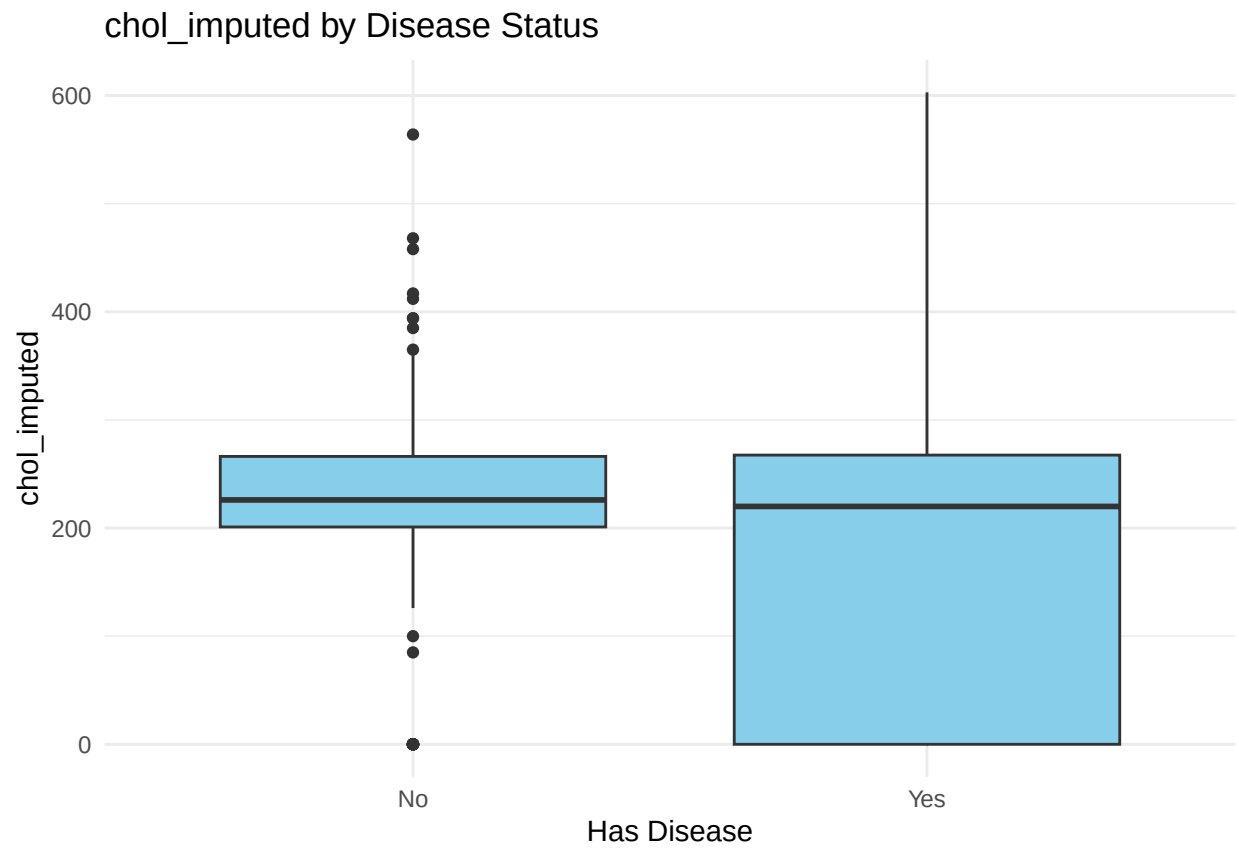


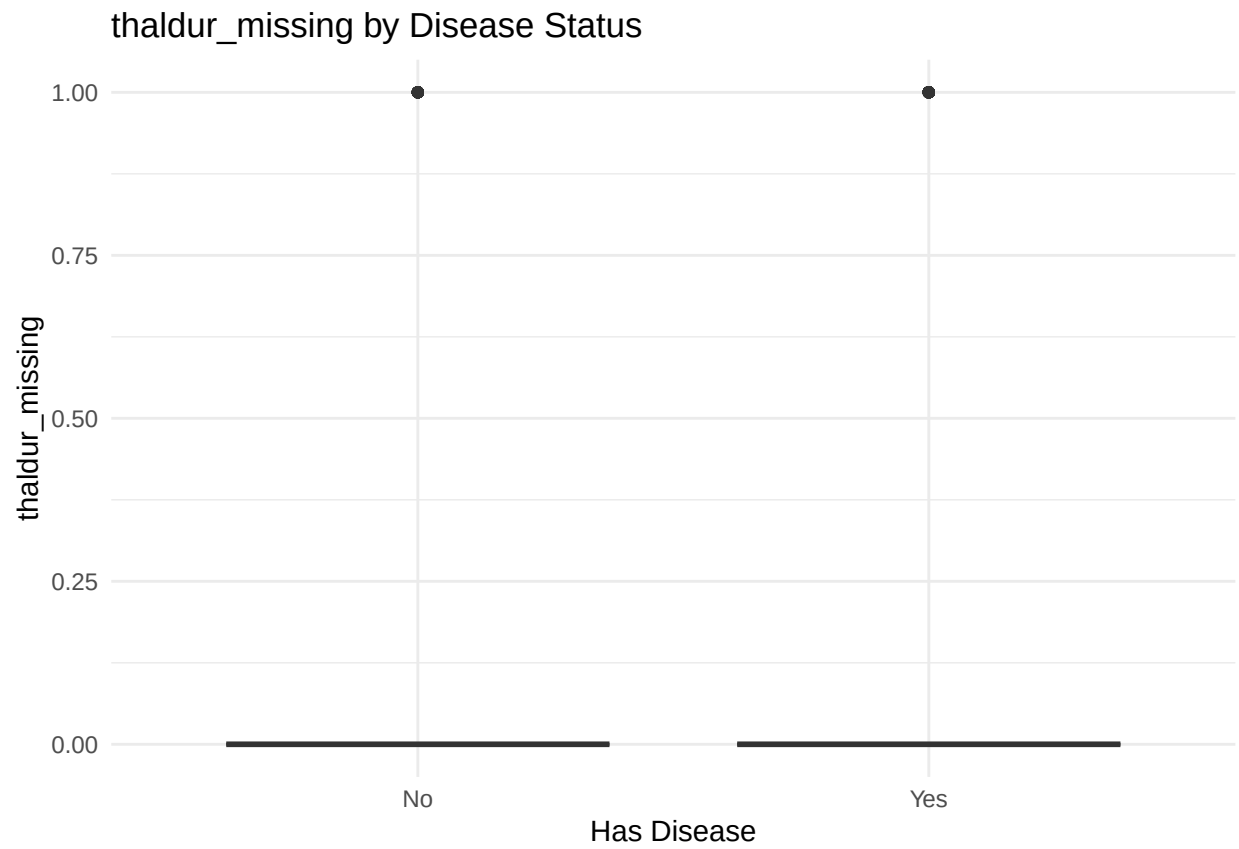


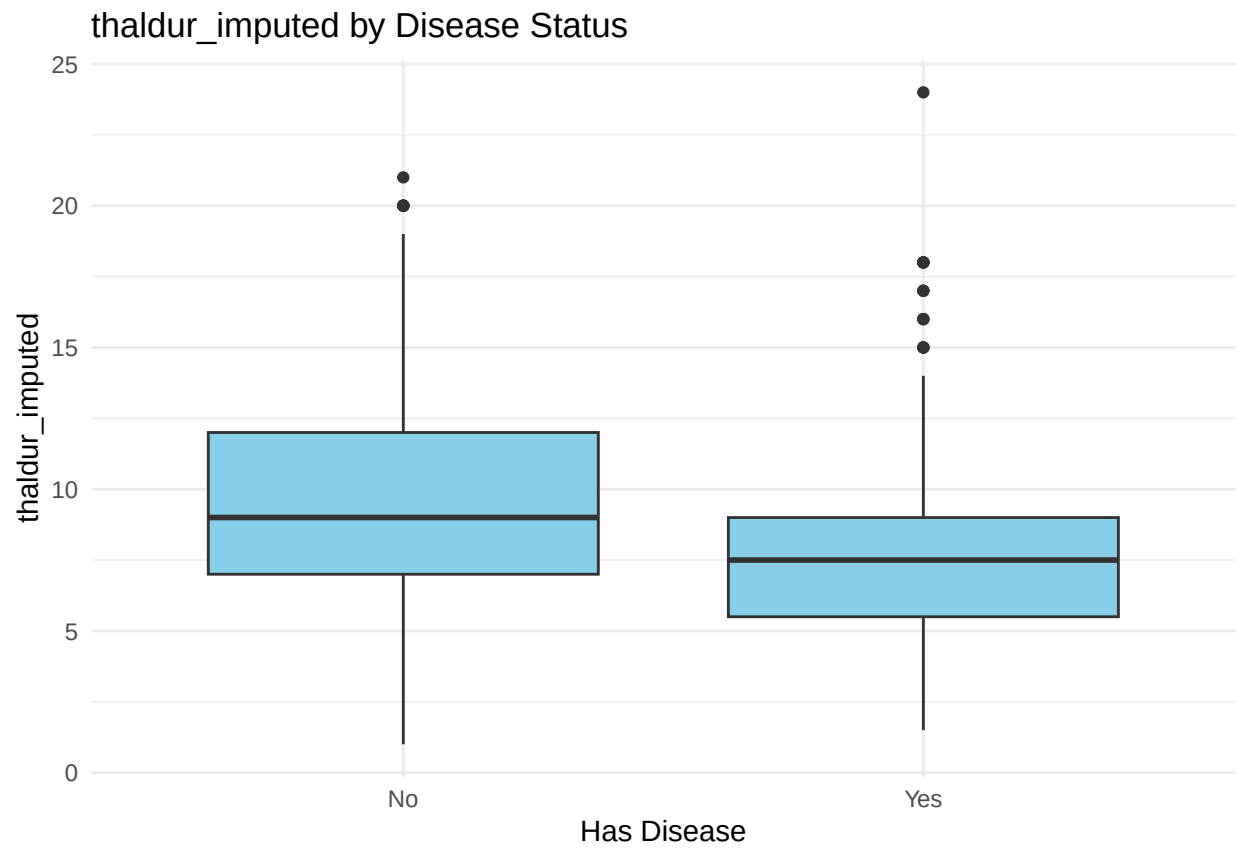


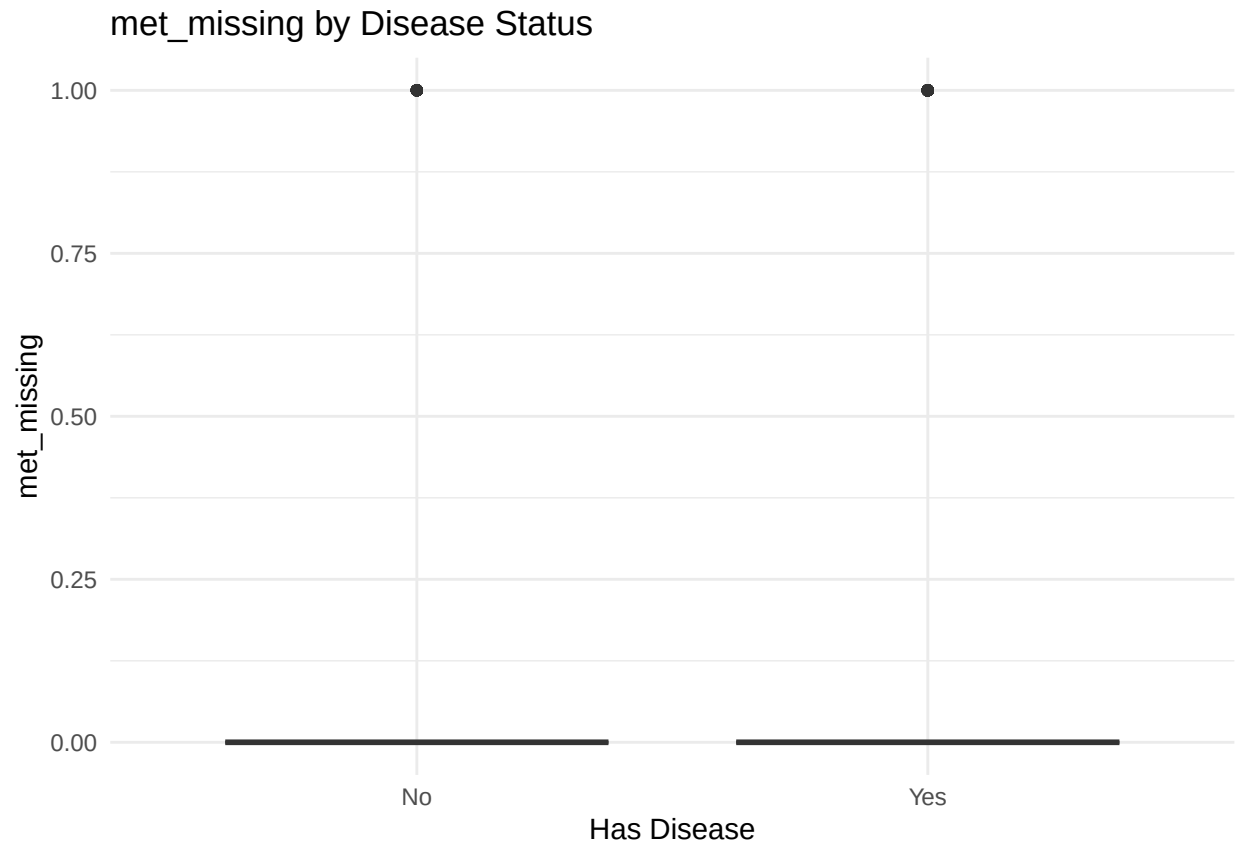


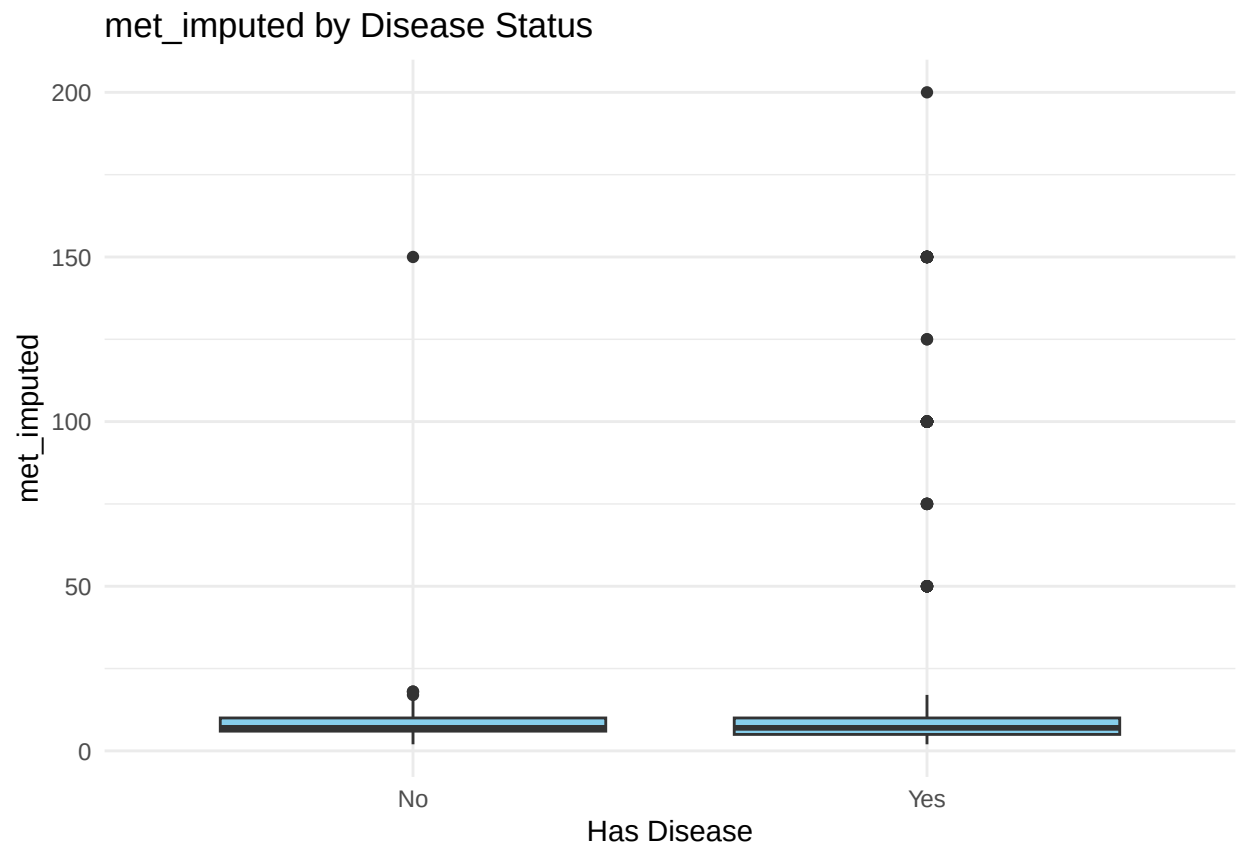


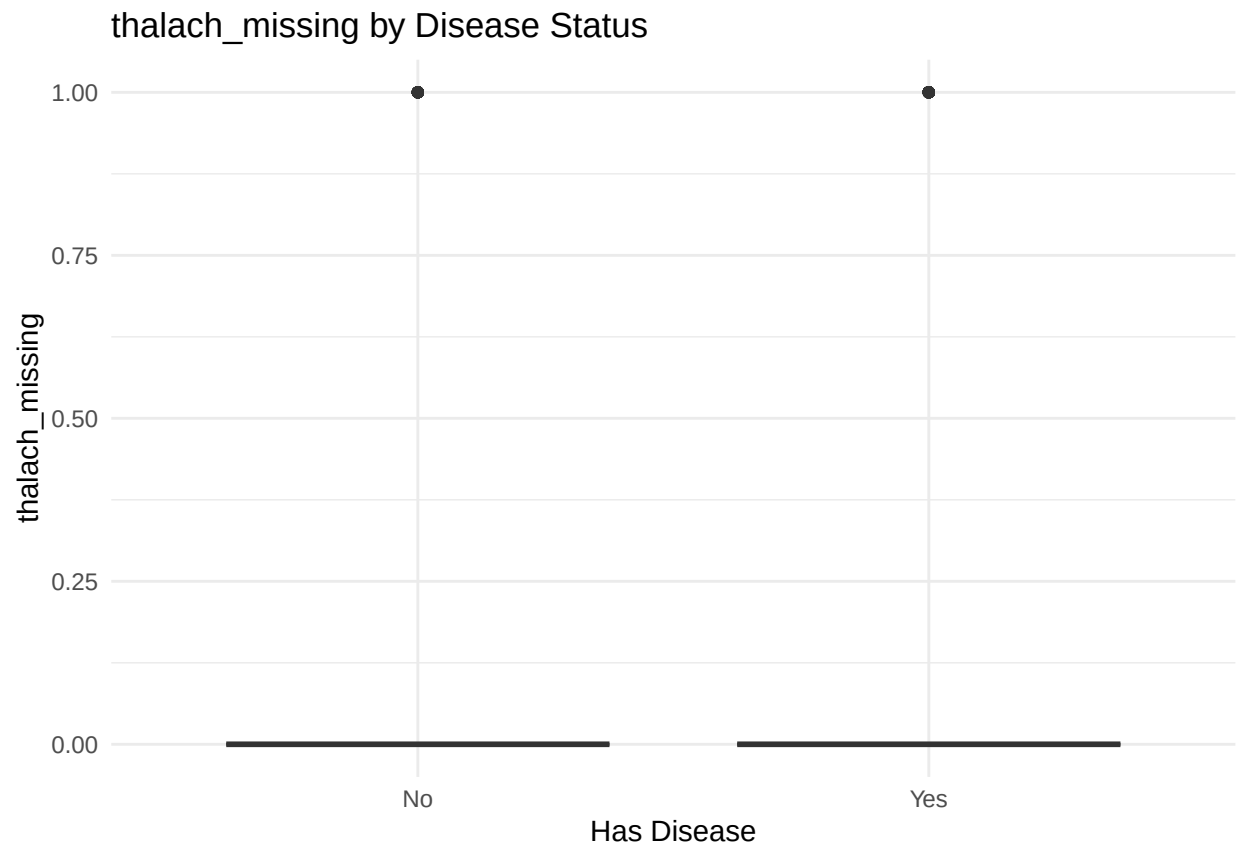


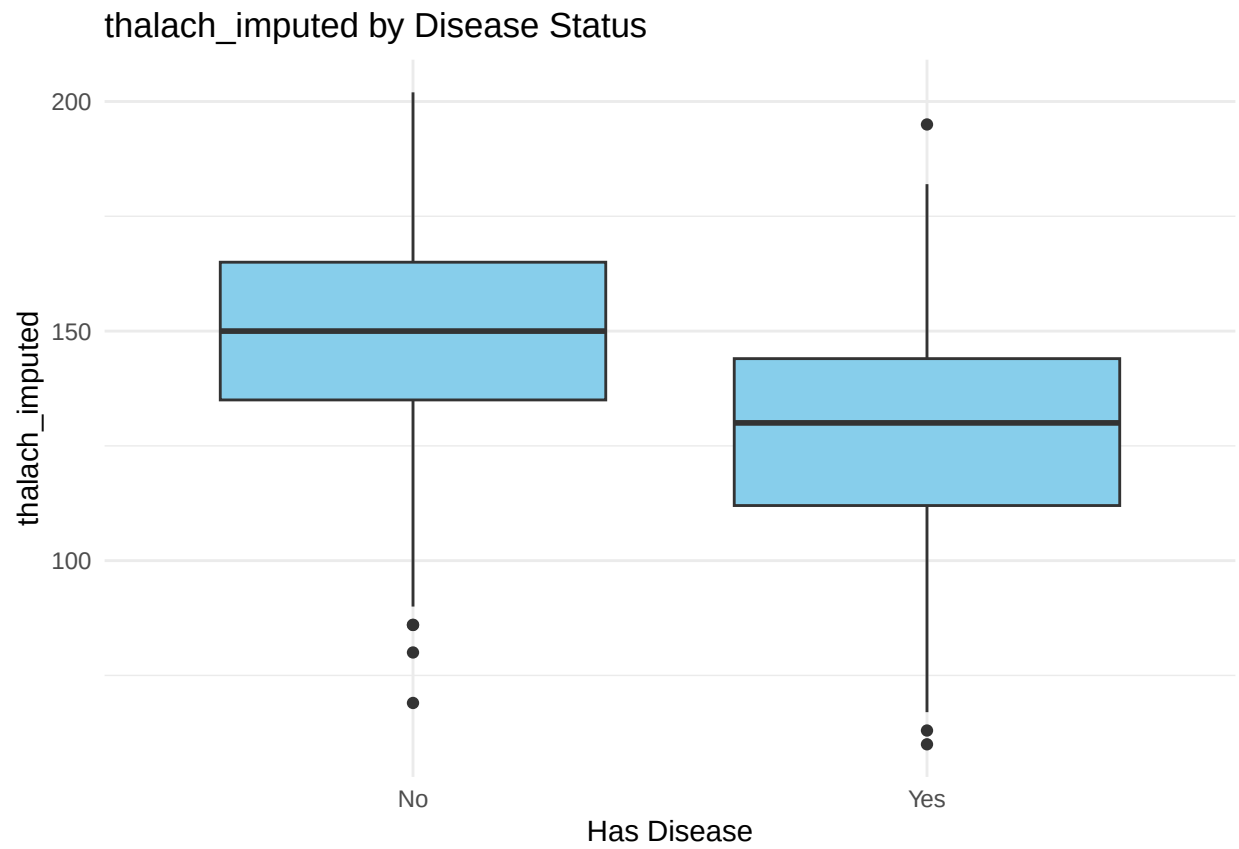


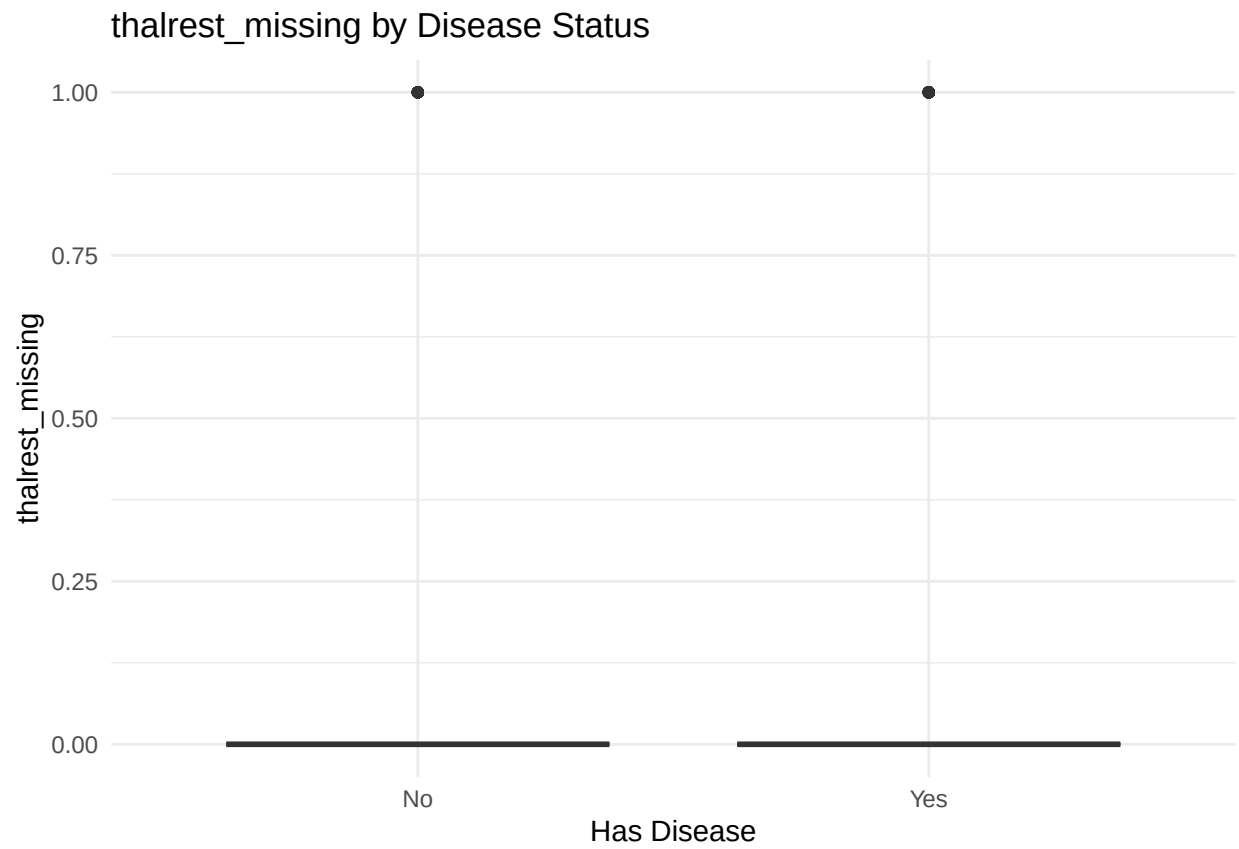


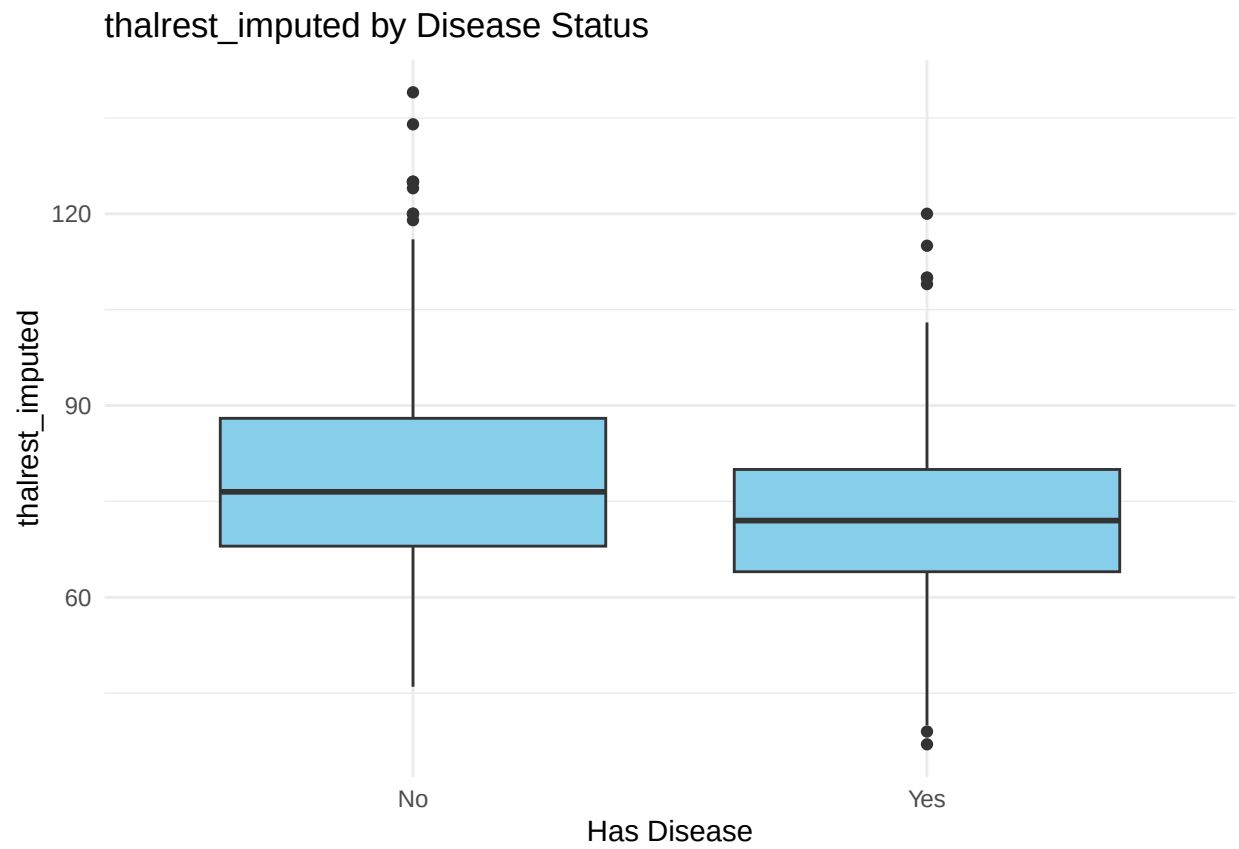


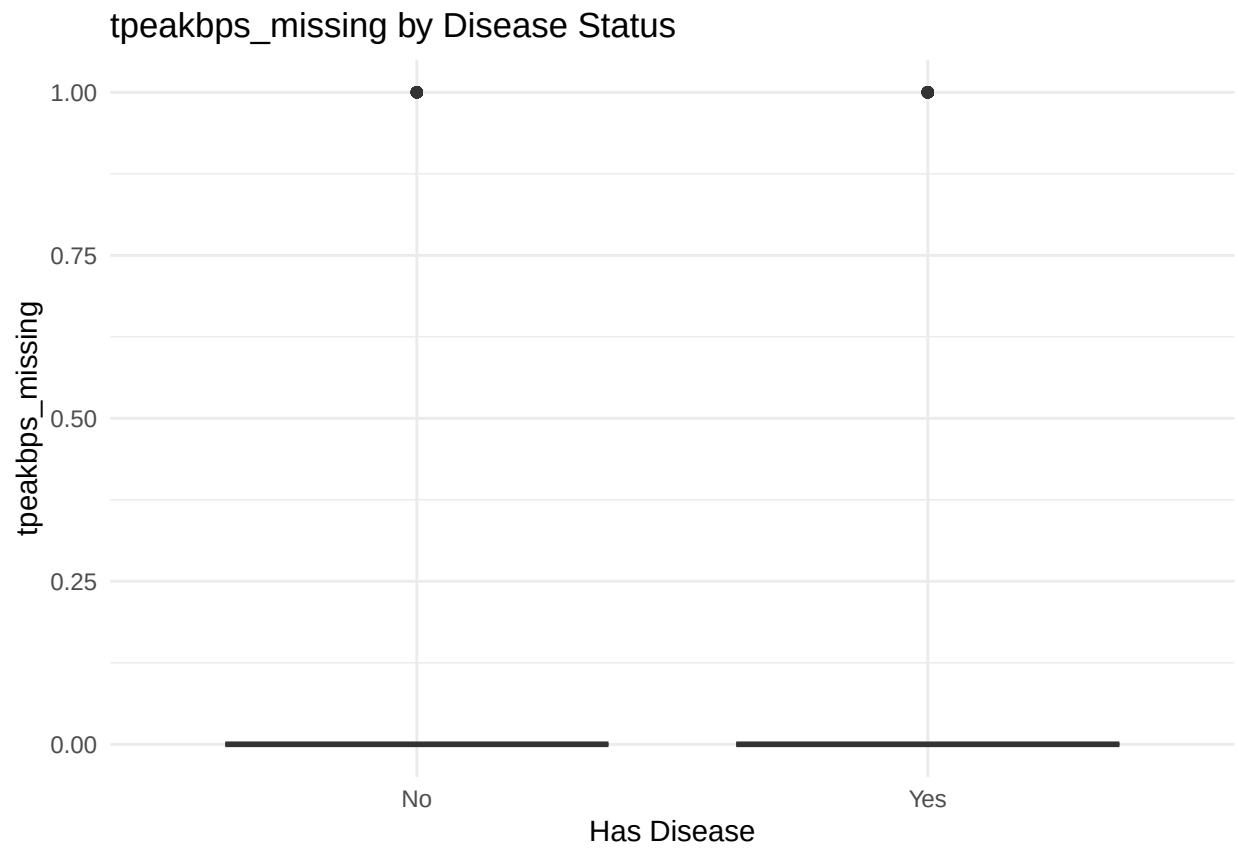


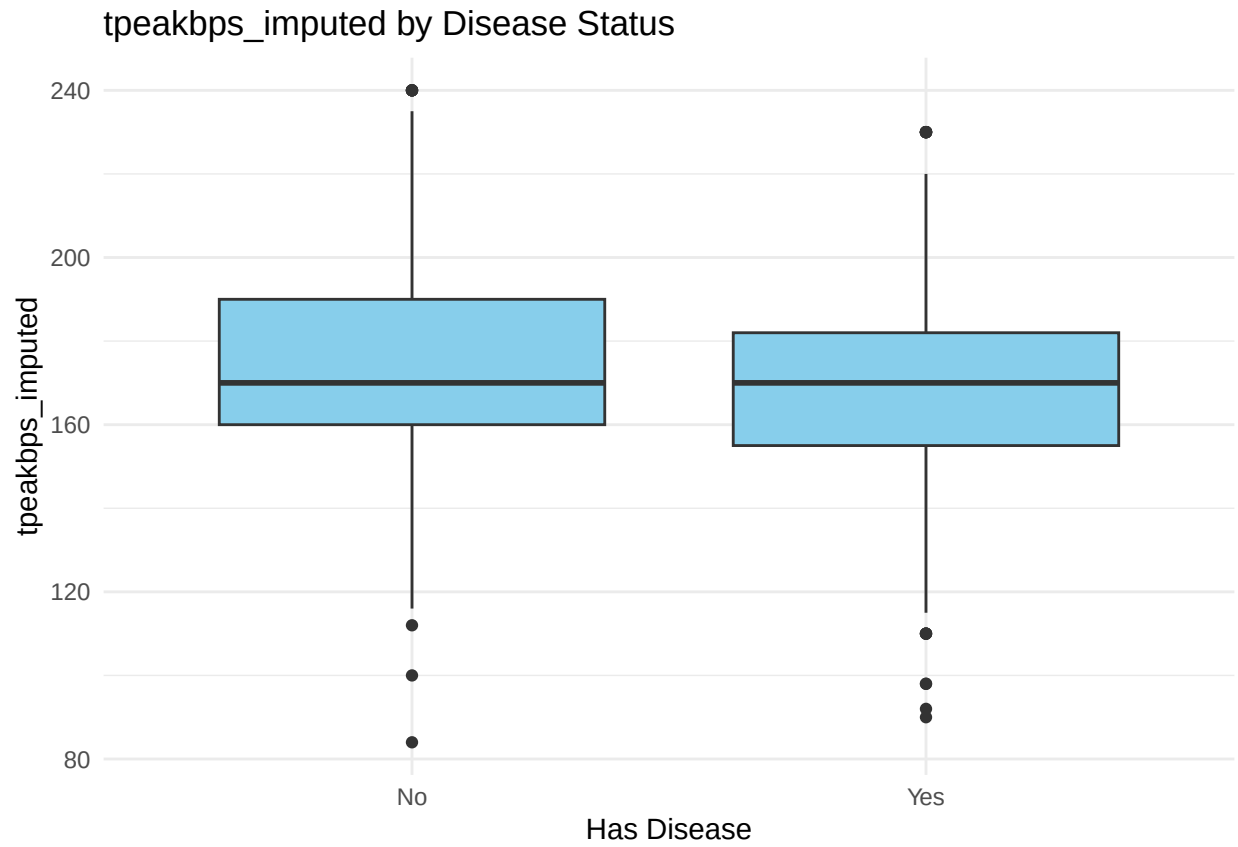


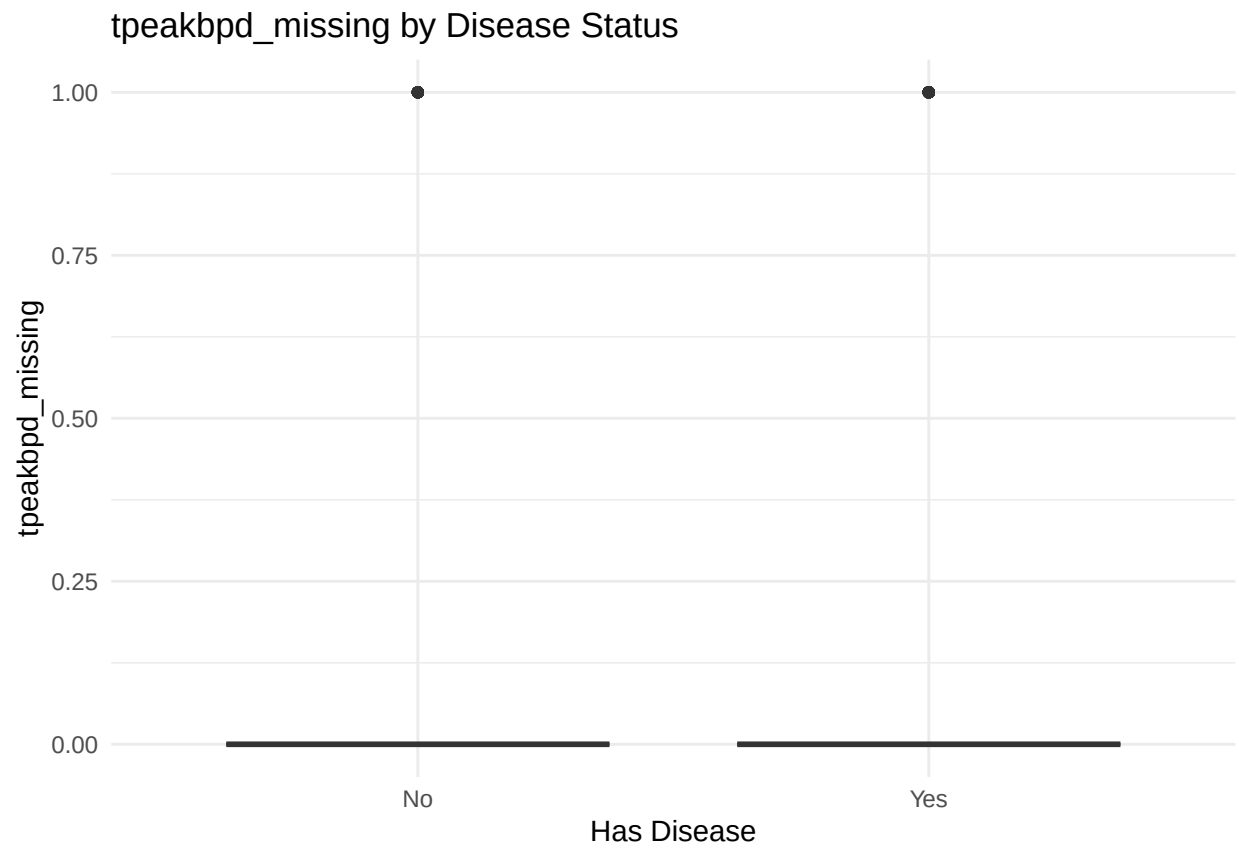


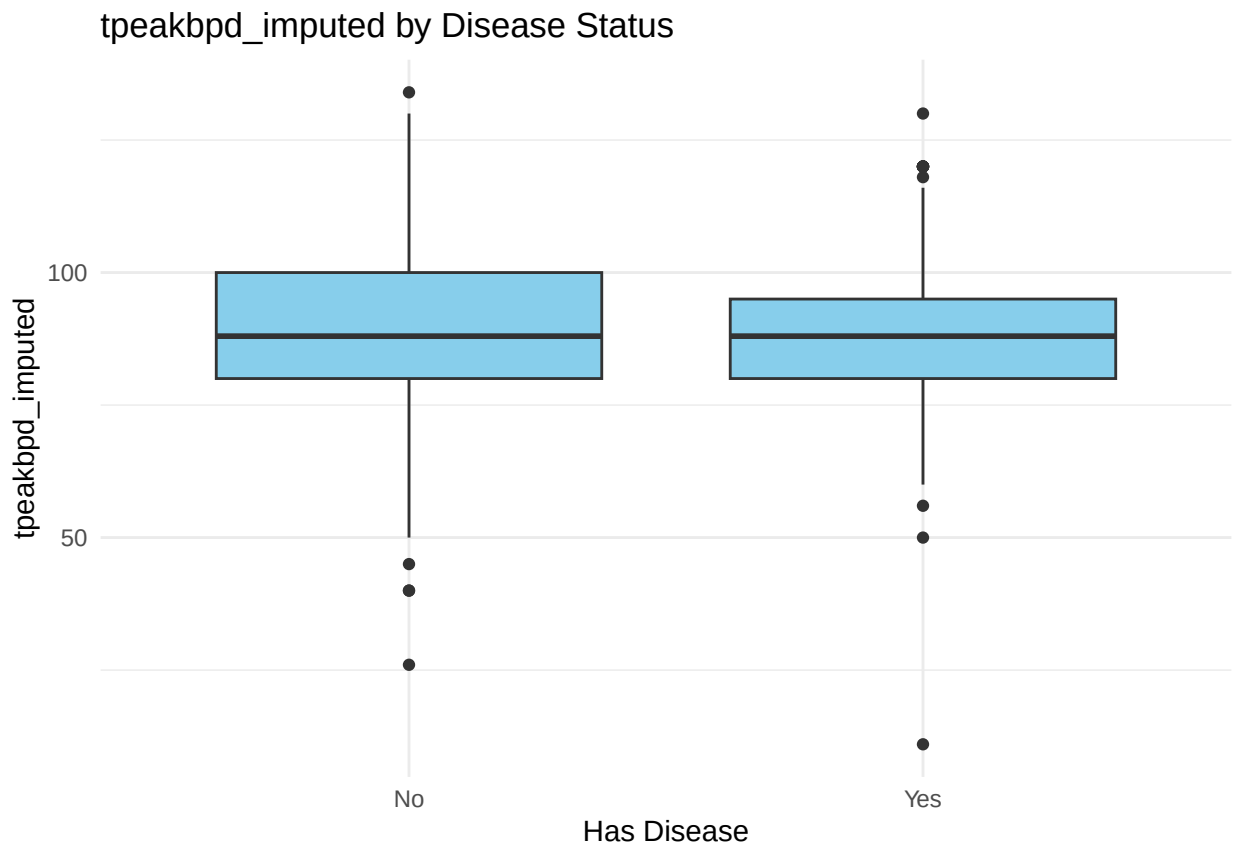


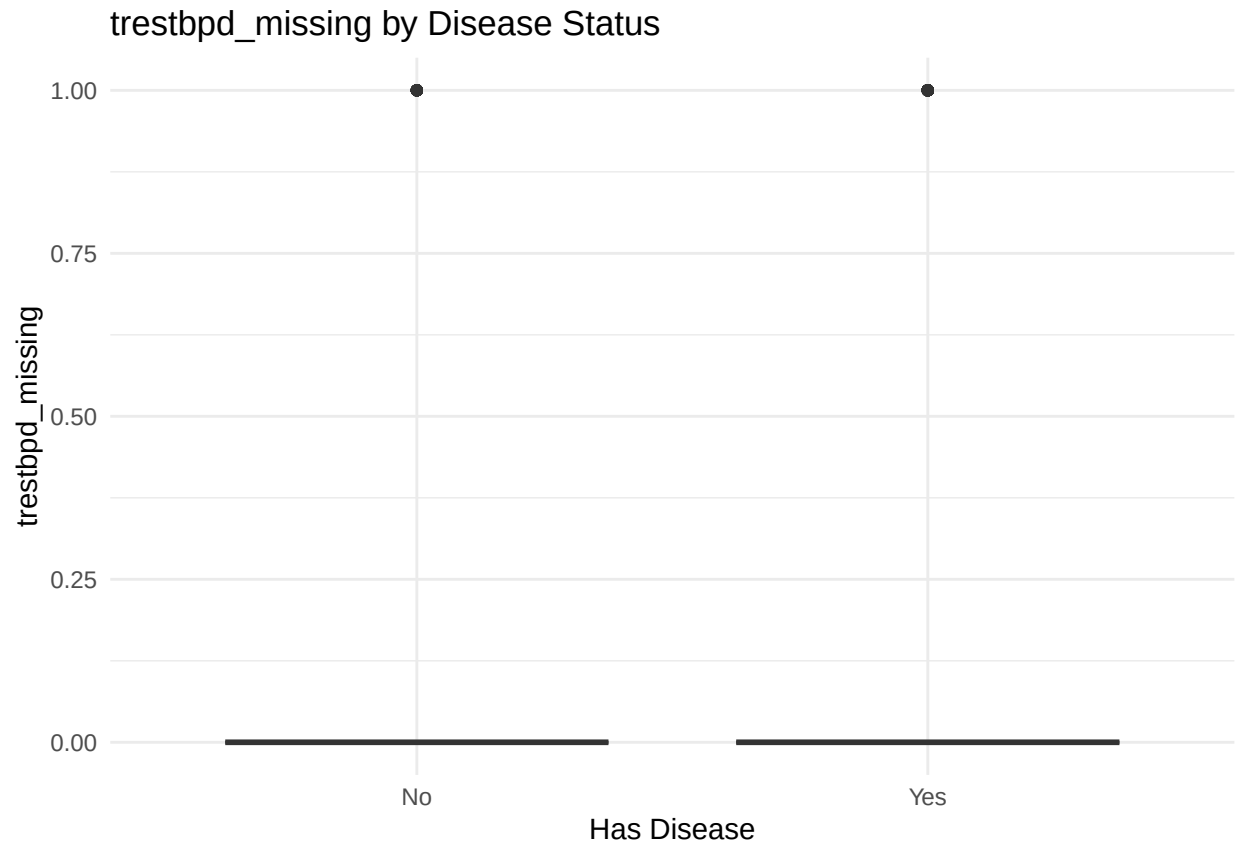


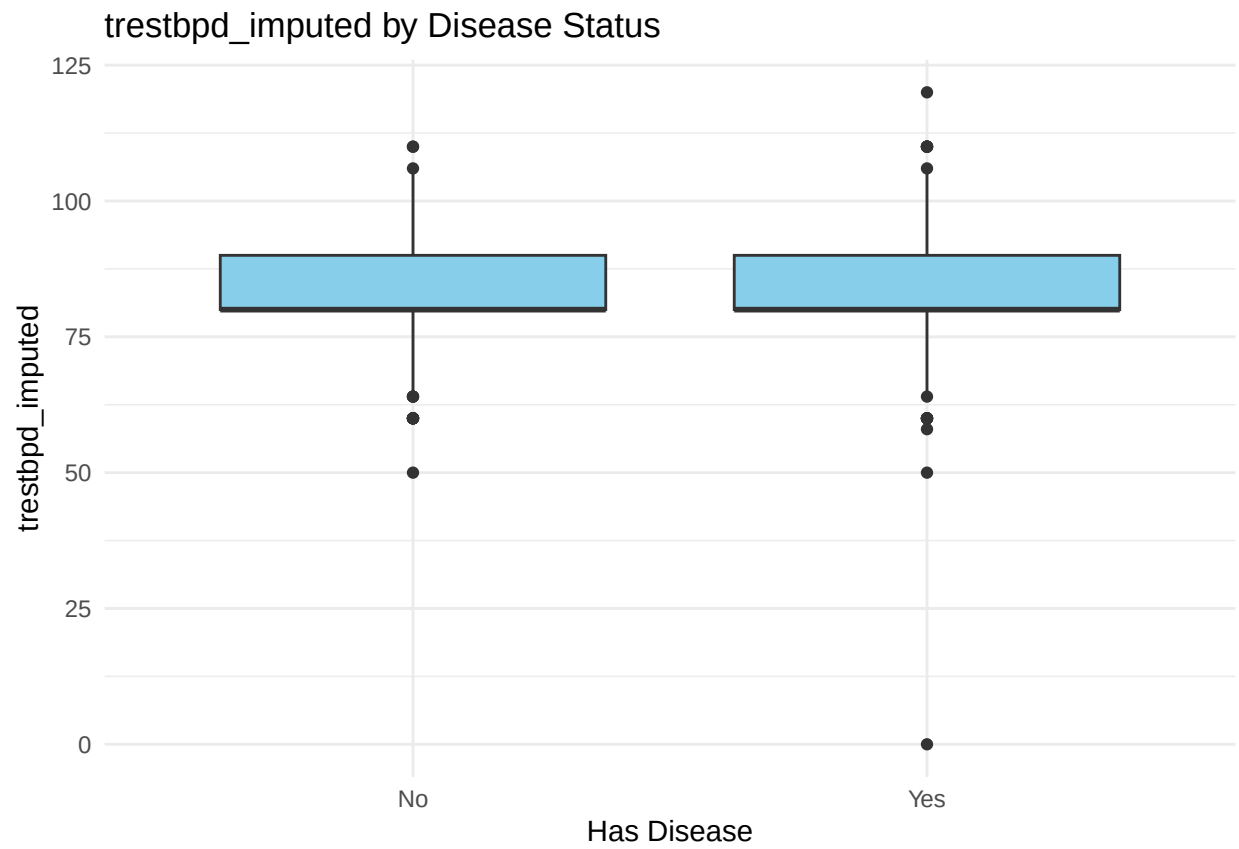


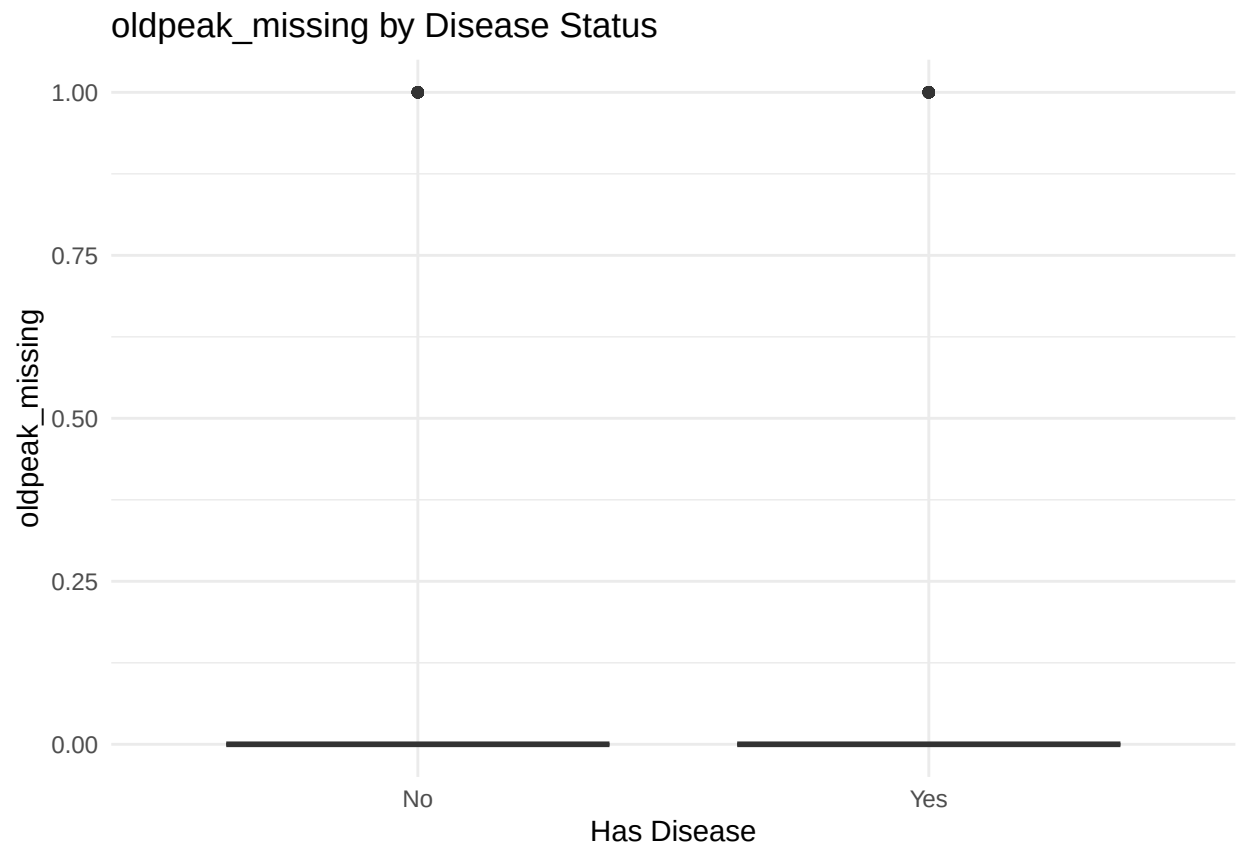


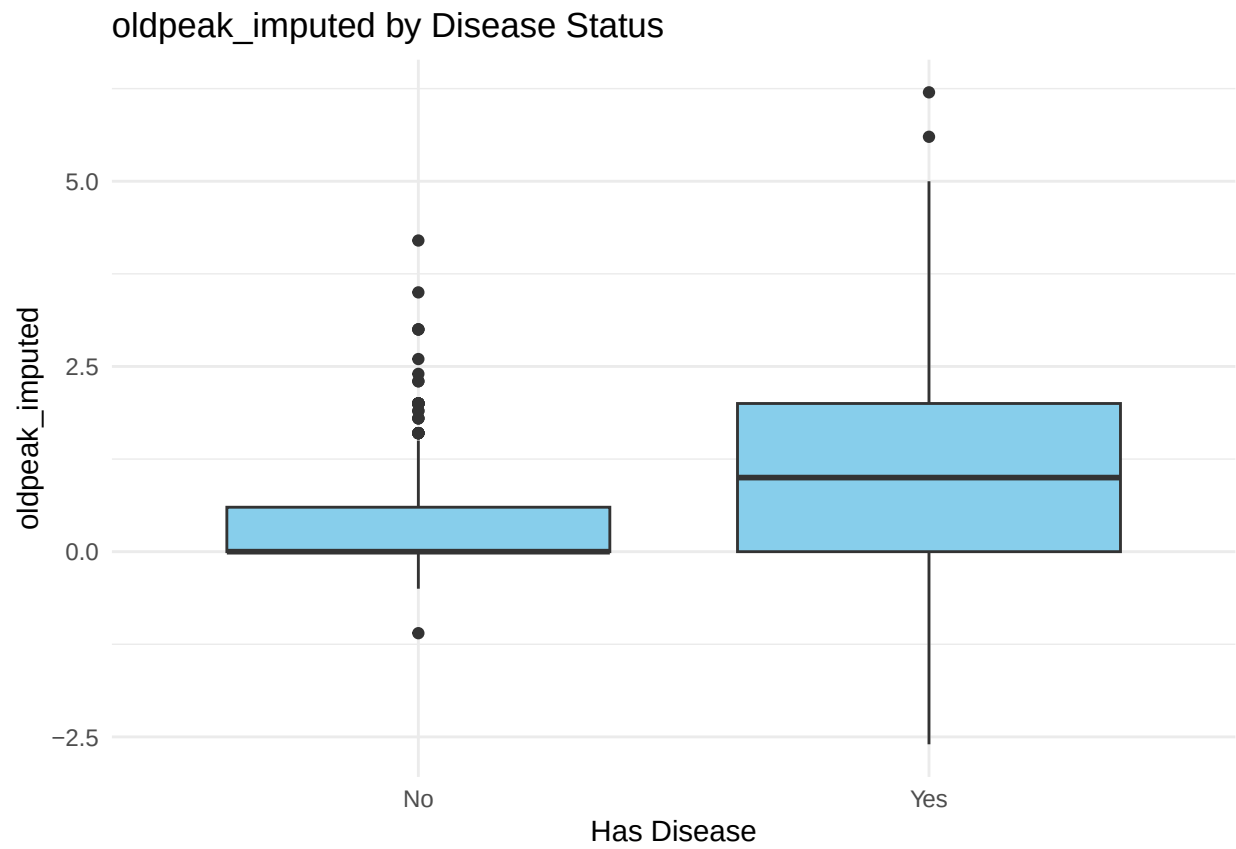


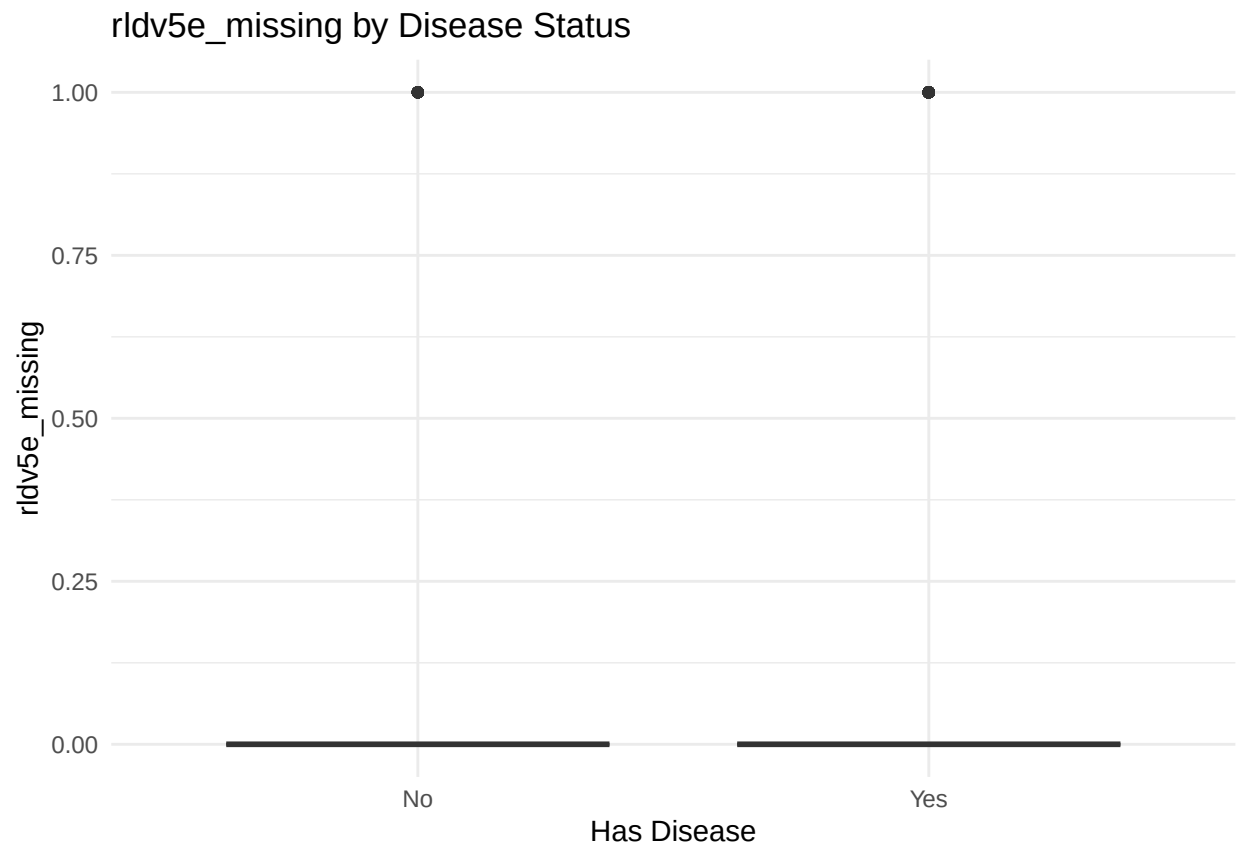


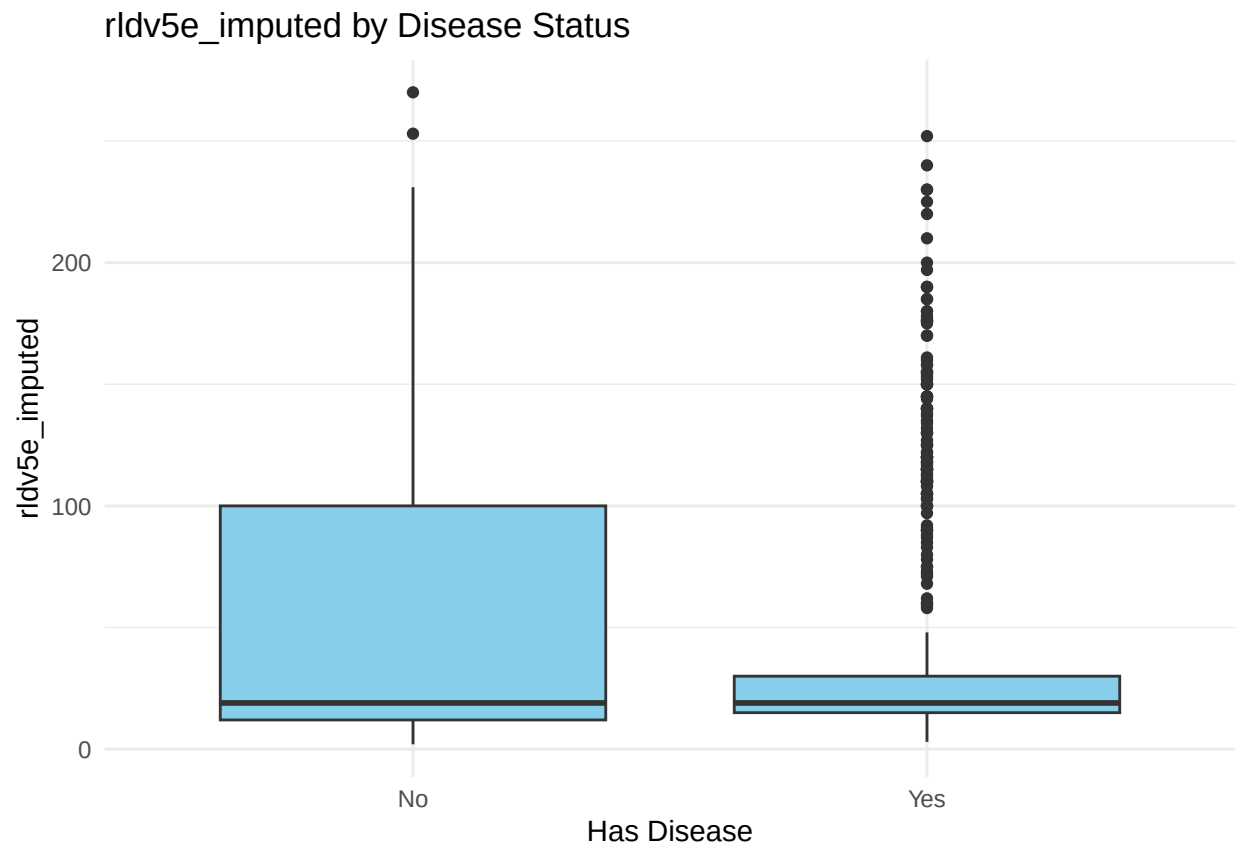


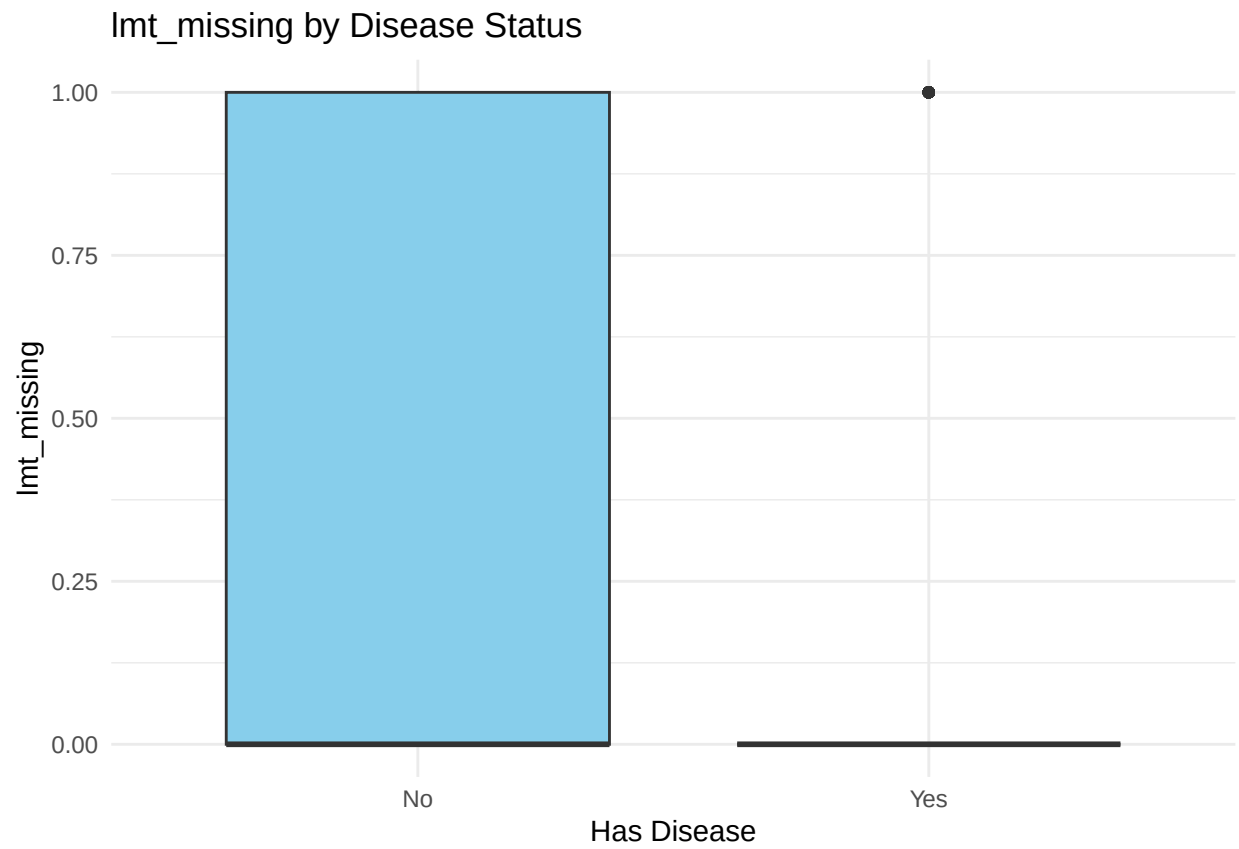


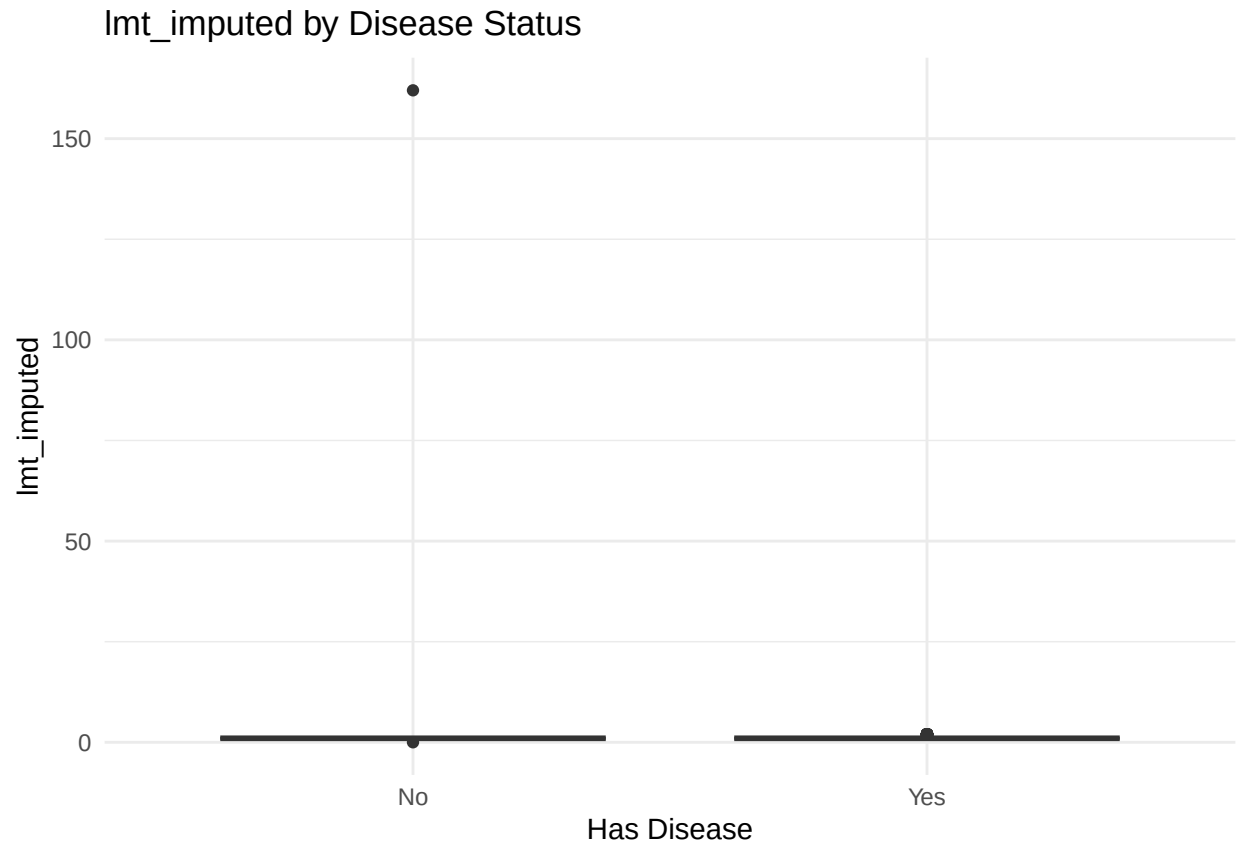












Welch's two-sample t-test

Statistical testing to determine if there are significant differences between the means of two features.

```
for (var in num_vars) {
  formula <- reformulate("has_disease", response = var)
  test_result <- t.test(formula, data = df)
  cat("=== T-test for", var, "===\n")
  print(test_result)
  cat("\n\n")
}
```

```
## === T-test for age ===
##
##  Welch Two Sample t-test
##
## data:  age by has_disease
## t = -8.5853, df = 833.36, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##   -6.462916 -4.057656
## sample estimates:
##  mean in group No mean in group Yes
##       50.58416      55.84444
##
##
##
```

```

## === T-test for ca_imputed ===
##
## Welch Two Sample t-test
##
## data: ca_imputed by has_disease
## t = -3.6443, df = 895.1, p-value = 0.0002836
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -0.25055510 -0.07514748
## sample estimates:
## mean in group No mean in group Yes
## 0.1361386 0.2989899
##
##
##
## === T-test for thal_imputed ===
##
## Welch Two Sample t-test
##
## data: thal_imputed by has_disease
## t = -9.0385, df = 860.57, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -1.1314140 -0.7277019
## sample estimates:
## mean in group No mean in group Yes
## 3.438119 4.367677
##
##
##
## === T-test for thaltime_imputed ===
##
## Welch Two Sample t-test
##
## data: thaltime_imputed by has_disease
## t = -0.82499, df = 890.94, p-value = 0.4096
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -0.5200975 0.2122537
## sample estimates:
## mean in group No mean in group Yes
## 5.761634 5.915556
##
##
##
## === T-test for rldv5_imputed ===
##
## Welch Two Sample t-test
##
## data: rldv5_imputed by has_disease
## t = -1.7981, df = 828.98, p-value = 0.07252
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -1.05248435 0.04609371

```



```

## sample estimates:
## mean in group No mean in group Yes
##      13.93317      14.43636
##
##
##
## === T-test for trestbps_missing ===
##
## Welch Two Sample t-test
##
## data:  trestbps_missing by has_disease
## t = -1.8033, df = 896.85, p-value = 0.07167
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -0.061152495  0.002586638
## sample estimates:
## mean in group No mean in group Yes
##      0.04950495      0.07878788
##
##
##
## === T-test for trestbps_imputed ===
##
## Welch Two Sample t-test
##
## data:  trestbps_imputed by has_disease
## t = -2.977, df = 896.52, p-value = 0.002989
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -5.998366 -1.231847
## sample estimates:
## mean in group No mean in group Yes
##      129.9728      133.5879
##
##
##
## === T-test for chol_missing ===
##
## Welch Two Sample t-test
##
## data:  chol_missing by has_disease
## t = 1.9913, df = 696, p-value = 0.04684
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  0.0003482239 0.0492667376
## sample estimates:
## mean in group No mean in group Yes
##      0.04702970      0.02222222
##
##
##
## === T-test for chol_imputed ===
##
## Welch Two Sample t-test

```

```

##
## data: chol_imputed by has_disease
## t = 7.5093, df = 816.55, p-value = 1.559e-13
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## 37.88176 64.69433
## sample estimates:
## mean in group No mean in group Yes
## 227.8416 176.5535
##
##
##
## === T-test for thaldur_missing ===
##
## Welch Two Sample t-test
##
## data: thaldur_missing by has_disease
## t = -1.1721, df = 893.81, p-value = 0.2415
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -0.05008523 0.01263149
## sample estimates:
## mean in group No mean in group Yes
## 0.05198020 0.07070707
##
##
##
## === T-test for thaldur_imputed ===
##
## Welch Two Sample t-test
##
## data: thaldur_imputed by has_disease
## t = 8.7171, df = 775.94, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## 1.608559 2.543589
## sample estimates:
## mean in group No mean in group Yes
## 9.764356 7.688283
##
##
##
## === T-test for met_missing ===
##
## Welch Two Sample t-test
##
## data: met_missing by has_disease
## t = -4.1789, df = 877.82, p-value = 3.222e-05
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -0.12675602 -0.04574123
## sample estimates:
## mean in group No mean in group Yes
## 0.06930693 0.15555556

```

```

##
##
##
## === T-test for met_imputed ===
##
## Welch Two Sample t-test
##
## data: met_imputed by has_disease
## t = -7.6036, df = 544.71, p-value = 1.27e-13
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -16.563228 -9.762245
## sample estimates:
## mean in group No mean in group Yes
##      8.12797      21.29071
##
##
##
## === T-test for thalach_missing ===
##
## Welch Two Sample t-test
##
## data: thalach_missing by has_disease
## t = -1.3416, df = 895.8, p-value = 0.1801
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -0.052219591 0.009815351
## sample estimates:
## mean in group No mean in group Yes
##      0.04950495      0.07070707
##
##
##
## === T-test for thalach_imputed ===
##
## Welch Two Sample t-test
##
## data: thalach_imputed by has_disease
## t = 12.322, df = 865.57, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## 16.15533 22.27721
## sample estimates:
## mean in group No mean in group Yes
##      148.0446      128.8283
##
##
##
## === T-test for thalrest_missing ===
##
## Welch Two Sample t-test
##
## data: thalrest_missing by has_disease
## t = -1.4592, df = 896.5, p-value = 0.1449

```

```

## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -0.05445650 0.00801186
## sample estimates:
## mean in group No mean in group Yes
## 0.04950495 0.07272727
##
##
##
## === T-test for thalrest_imputed ===
##
## Welch Two Sample t-test
##
## data: thalrest_imputed by has_disease
## t = 6.3454, df = 773.7, p-value = 3.774e-10
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## 4.183520 7.931421
## sample estimates:
## mean in group No mean in group Yes
## 78.73020 72.67273
##
##
##
## === T-test for tpeakbps_missing ===
##
## Welch Two Sample t-test
##
## data: tpeakbps_missing by has_disease
## t = -2.2436, df = 894.11, p-value = 0.0251
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -0.070047808 -0.004679664
## sample estimates:
## mean in group No mean in group Yes
## 0.04950495 0.08686869
##
##
##
## === T-test for tpeakbps_imputed ===
##
## Welch Two Sample t-test
##
## data: tpeakbps_imputed by has_disease
## t = 3.1634, df = 860.4, p-value = 0.001614
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## 1.988506 8.489841
## sample estimates:
## mean in group No mean in group Yes
## 174.4109 169.1717
##
##
##

```

```

## === T-test for tpeakbpd_missing ===
##
## Welch Two Sample t-test
##
## data: tpeakbpd_missing by has_disease
## t = -2.2436, df = 894.11, p-value = 0.0251
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -0.070047808 -0.004679664
## sample estimates:
## mean in group No mean in group Yes
## 0.04950495 0.08686869
##
##
##
## === T-test for tpeakbpd_imputed ===
##
## Welch Two Sample t-test
##
## data: tpeakbpd_imputed by has_disease
## t = 0.38958, df = 814.22, p-value = 0.6969
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -1.517519 2.269054
## sample estimates:
## mean in group No mean in group Yes
## 87.54950 87.17374
##
##
##
## === T-test for trestbpd_missing ===
##
## Welch Two Sample t-test
##
## data: trestbpd_missing by has_disease
## t = -1.8033, df = 896.85, p-value = 0.07167
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -0.061152495 0.002586638
## sample estimates:
## mean in group No mean in group Yes
## 0.04950495 0.07878788
##
##
##
## === T-test for trestbpd_imputed ===
##
## Welch Two Sample t-test
##
## data: trestbpd_imputed by has_disease
## t = 0.20883, df = 879.12, p-value = 0.8346
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -1.163256 1.440273

```

```

## sample estimates:
## mean in group No mean in group Yes
##      83.36881      83.23030
##
##
##
## === T-test for oldpeak_missing ===
##
## Welch Two Sample t-test
##
## data: oldpeak_missing by has_disease
## t = -1.8566, df = 896.85, p-value = 0.06369
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -0.063456965 0.001760795
## sample estimates:
## mean in group No mean in group Yes
##      0.05198020      0.08282828
##
##
##
## === T-test for oldpeak_imputed ===
##
## Welch Two Sample t-test
##
## data: oldpeak_imputed by has_disease
## t = -12.186, df = 832.96, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -0.8839527 -0.6387035
## sample estimates:
## mean in group No mean in group Yes
##      0.4257426      1.1870707
##
##
##
## === T-test for rldv5e_missing ===
##
## Welch Two Sample t-test
##
## data: rldv5e_missing by has_disease
## t = -8.4101, df = 779.1, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -0.2318428 -0.1440948
## sample estimates:
## mean in group No mean in group Yes
##      0.05445545      0.24242424
##
##
##
## === T-test for rldv5e_imputed ===
##
## Welch Two Sample t-test

```

```
##
## data:  rldv5e_imputed by has_disease
## t = 2.8953, df = 816.16, p-value = 0.003889
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##    3.581025 18.657808
## sample estimates:
## mean in group No mean in group Yes
##      55.36386      44.24444
##
##
##
## === T-test for lmt_missing ===
##
## Welch Two Sample t-test
##
## data:  lmt_missing by has_disease
## t = 9.4361, df = 742.88, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##    0.2257806 0.3444065
## sample estimates:
## mean in group No mean in group Yes
##      0.4628713      0.1777778
##
##
##
## === T-test for lmt_imputed ===
##
## Welch Two Sample t-test
##
## data:  lmt_imputed by has_disease
## t = 0.78046, df = 403.8, p-value = 0.4356
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##   -0.4726466  1.0950288
## sample estimates:
## mean in group No mean in group Yes
##      1.396040      1.084848
```

Summary for numerical variables

```
missing_flags <- colnames(df)[str_detect(colnames(df), "_missing$")]

# recommendation code generated with the help of ChatGPT.
# Create an empty data frame to store results
bivariate_summary <- data.frame(
  Variable = character(),
  p_value = numeric(),
  Missingness_Flag = logical(),
  Recommendation = character(),
  stringsAsFactors = FALSE
)
```

```

# Loop through and compute t-tests
for (var in num_vars) {
  formula <- reformulate("has_disease", response = var)
  p_val <- tryCatch(t.test(formula, data = df)$p.value, error = function(e) NA)

  is_missing_flag <- paste0(var, "_missing") %in% colnames(df) | var %in% missing_flags

  recommendation <- if (!is.na(p_val) && p_val < 0.05) {
    "Include (predictive)"
  } else if (is_missing_flag) {
    "Include (missingness informative)"
  } else {
    "Optional (evaluate with model)"
  }

  bivariate_summary <- rbind(
    bivariate_summary,
    data.frame(
      Variable = var,
      p_value = round(p_val, 5),
      Missingness_Flag = is_missing_flag,
      Recommendation = recommendation
    )
  )
}

# View result
bivariate_summary

```

##	Variable	p_value	Missingness_Flag	Recommendation
## 1	age	0.00000	FALSE	Include (predictive)
## 2	ca_imputed	0.00028	FALSE	Include (predictive)
## 3	thal_imputed	0.00000	FALSE	Include (predictive)
## 4	thaltime_imputed	0.40960	FALSE	Optional (evaluate with model)
## 5	rldv5_imputed	0.07252	FALSE	Optional (evaluate with model)
## 6	trestbps_missing	0.07167	TRUE	Include (missingness informative)
## 7	trestbps_imputed	0.00299	FALSE	Include (predictive)
## 8	chol_missing	0.04684	TRUE	Include (predictive)
## 9	chol_imputed	0.00000	FALSE	Include (predictive)
## 10	thaldur_missing	0.24149	TRUE	Include (missingness informative)
## 11	thaldur_imputed	0.00000	FALSE	Include (predictive)
## 12	met_missing	0.00003	TRUE	Include (predictive)
## 13	met_imputed	0.00000	FALSE	Include (predictive)
## 14	thalach_missing	0.18008	TRUE	Include (missingness informative)
## 15	thalach_imputed	0.00000	FALSE	Include (predictive)
## 16	thalrest_missing	0.14486	TRUE	Include (missingness informative)
## 17	thalrest_imputed	0.00000	FALSE	Include (predictive)
## 18	tpeakbps_missing	0.02510	TRUE	Include (predictive)
## 19	tpeakbps_imputed	0.00161	FALSE	Include (predictive)
## 20	tpeakbpd_missing	0.02510	TRUE	Include (predictive)
## 21	tpeakbpd_imputed	0.69695	FALSE	Optional (evaluate with model)
## 22	trestbpd_missing	0.07167	TRUE	Include (missingness informative)
## 23	trestbpd_imputed	0.83463	FALSE	Optional (evaluate with model)


```
## 24 oldpeak_missing 0.06369 TRUE Include (missingness informative)
## 25 oldpeak_imputed 0.00000 FALSE Include (predictive)
## 26 rldv5e_missing 0.00000 TRUE Include (predictive)
## 27 rldv5e_imputed 0.00389 FALSE Include (predictive)
## 28 lmt_missing 0.00000 TRUE Include (predictive)
## 29 lmt_imputed 0.43558 FALSE Optional (evaluate with model)

# create list of variables recommended as Include
selected_num_vars <- bivariate_summary %>%
  filter(str_detect(Recommendation, "Include")) %>%
  pull(Variable)
```

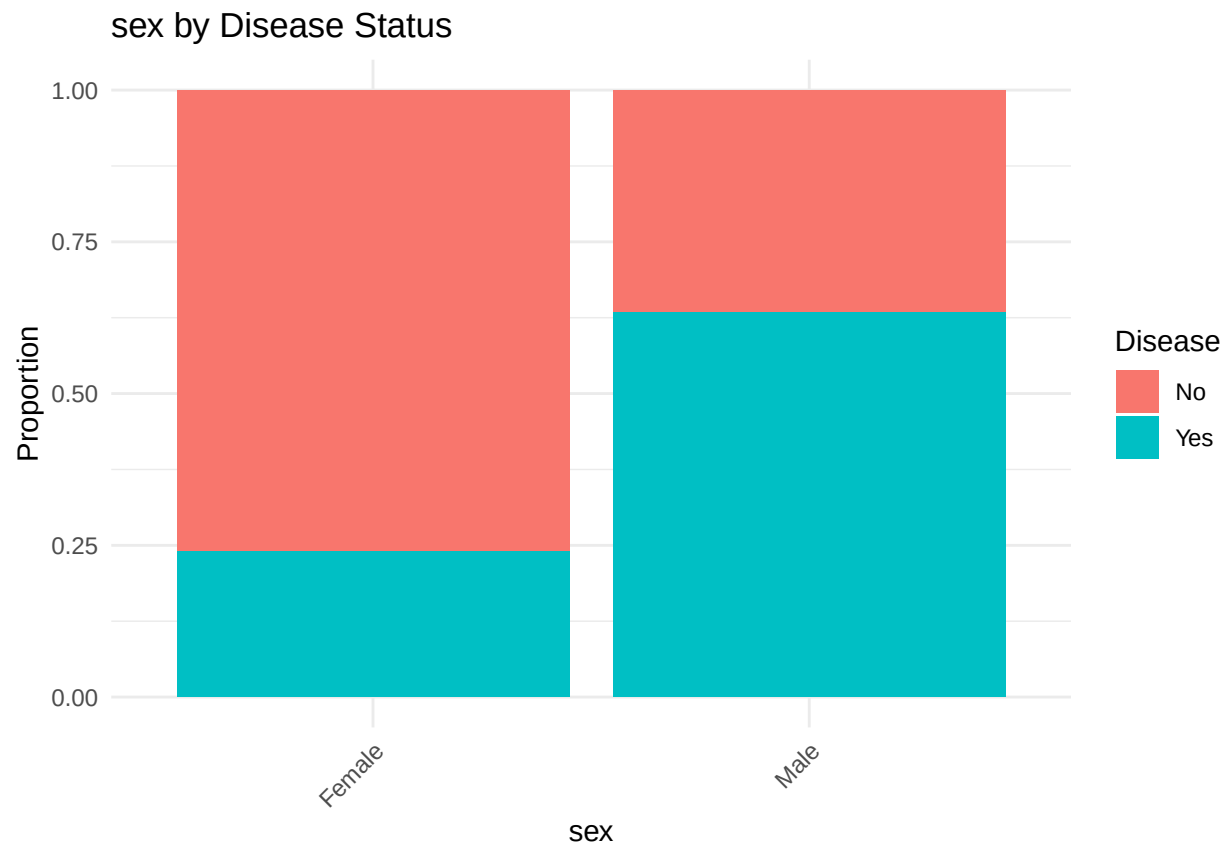
Based on recommendation, all variables recommended as 'Include' will be kept.

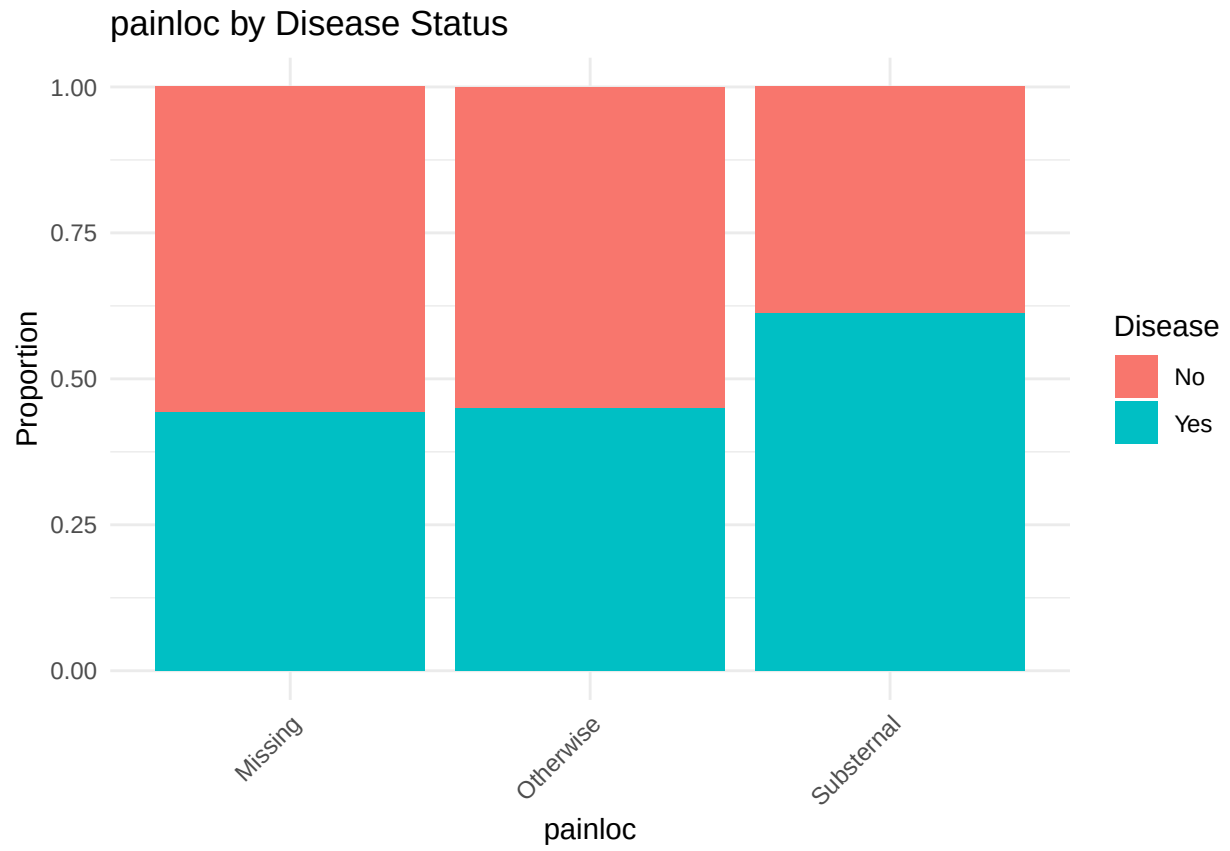
Categorical Vars

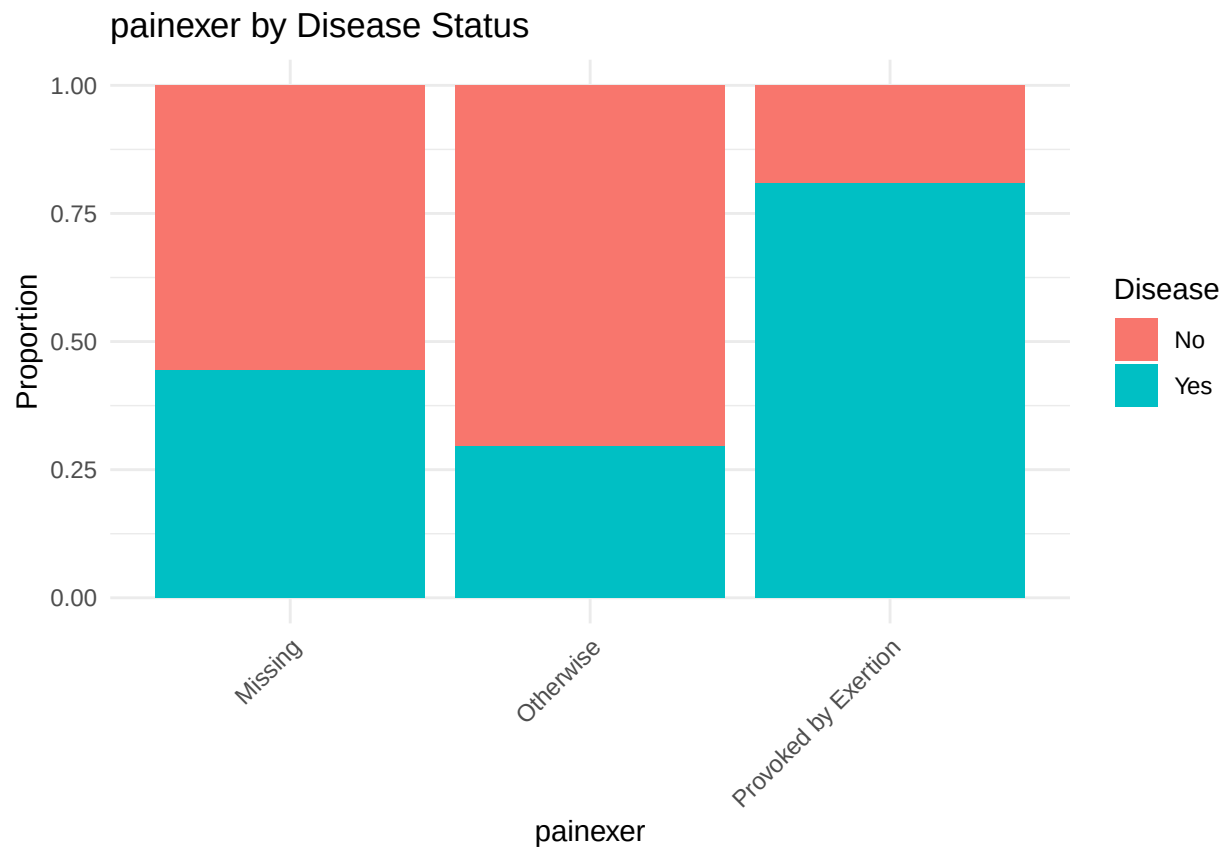
Bar Plots

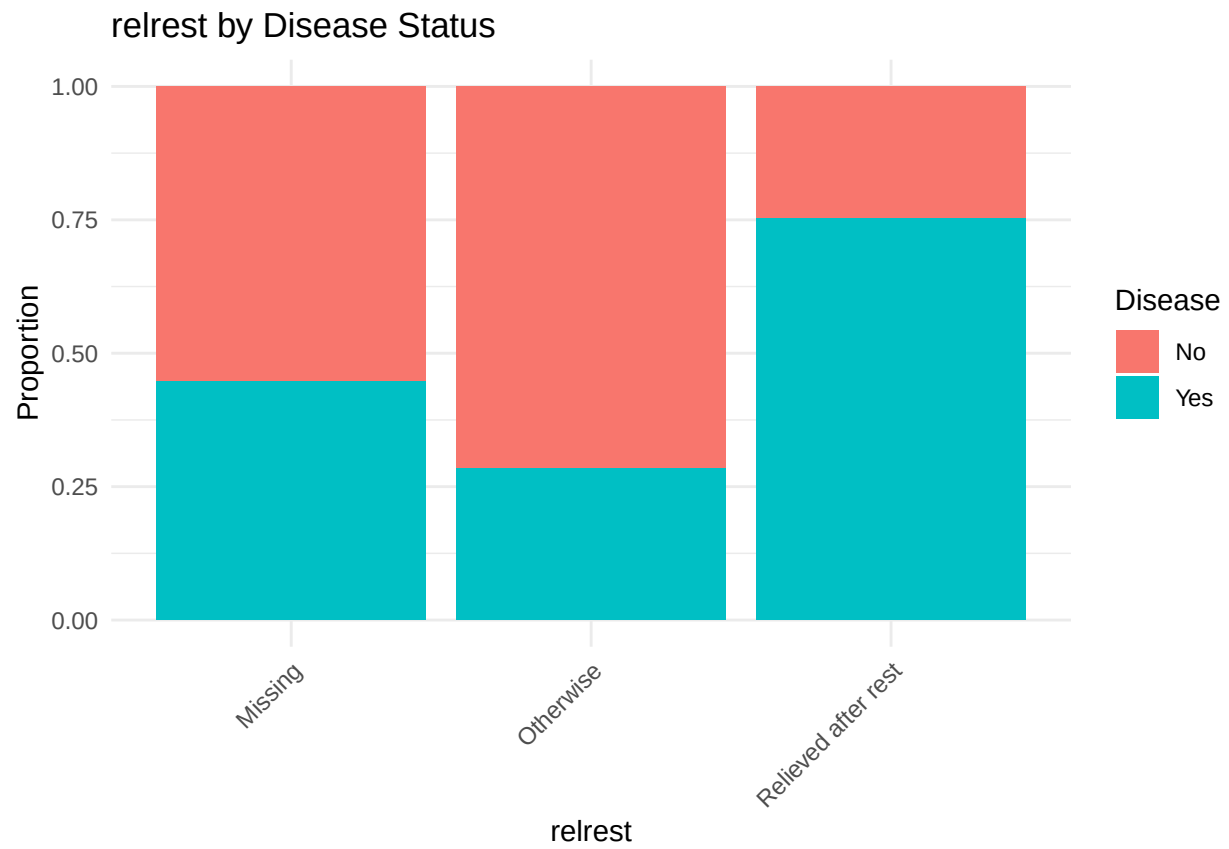
```
plot_bivariate_cat <- function(var) {
  ggplot(df, aes(x = .data[[var]], fill = factor(has_disease))) +
    geom_bar(position = "fill") + # proportion stacked
    labs(title = paste(var, "by Disease Status"), y = "Proportion", fill = "Disease") +
    theme_minimal() +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))
}

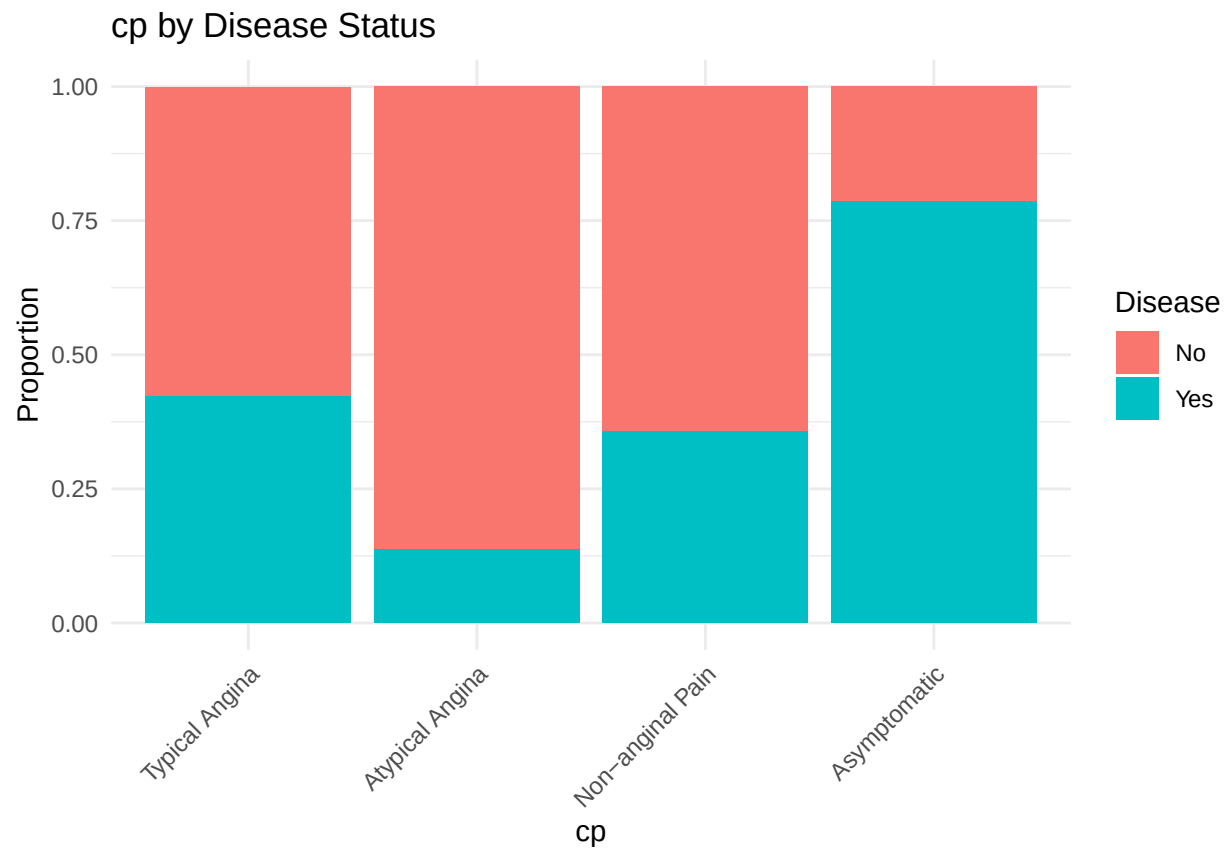
for (var in cat_vars) {
  print(plot_bivariate_cat(var))
}
```

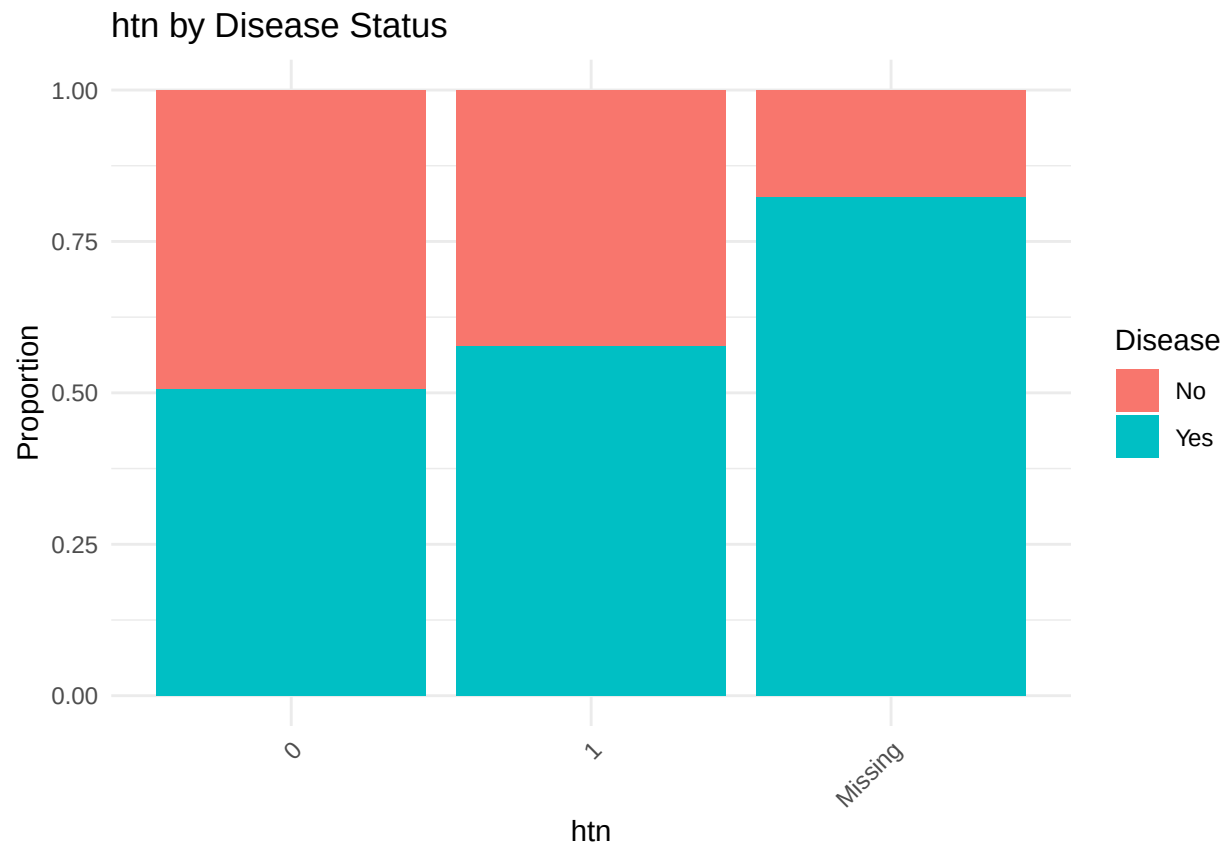


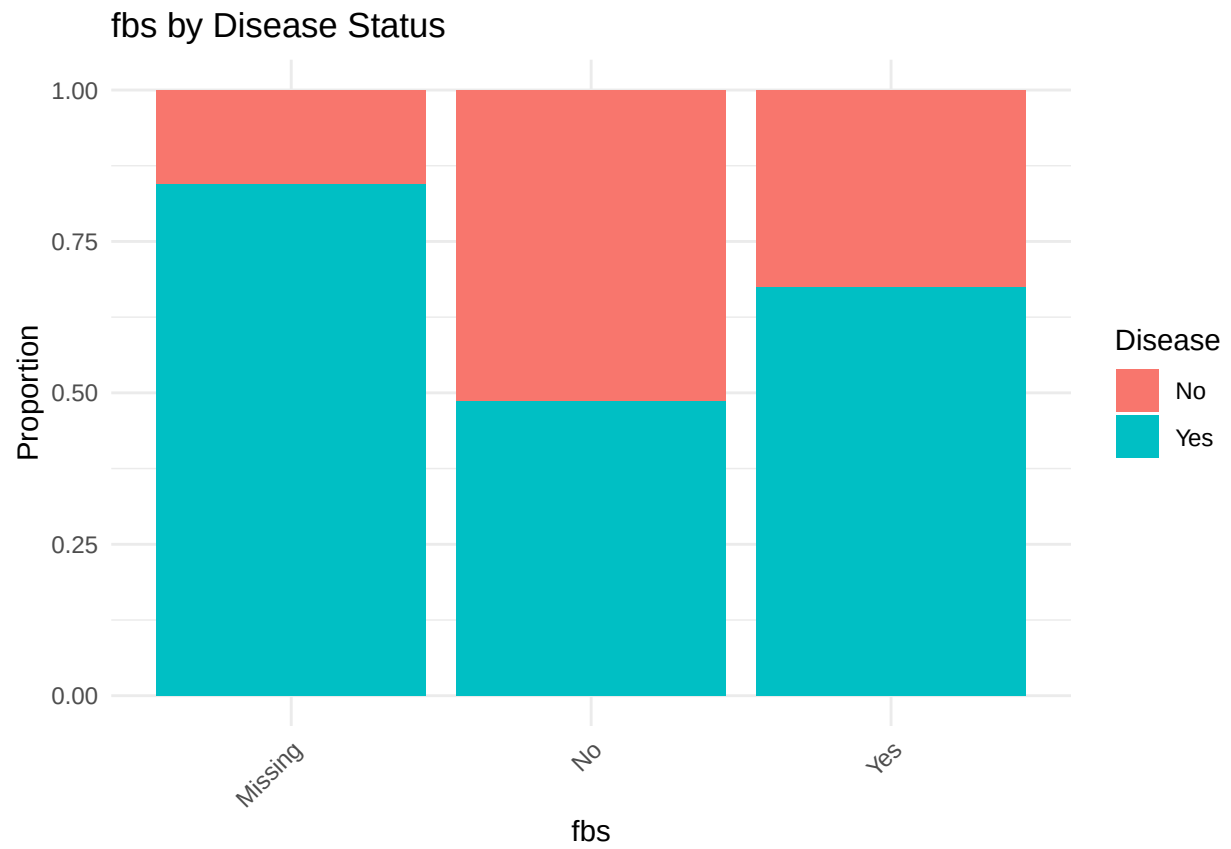


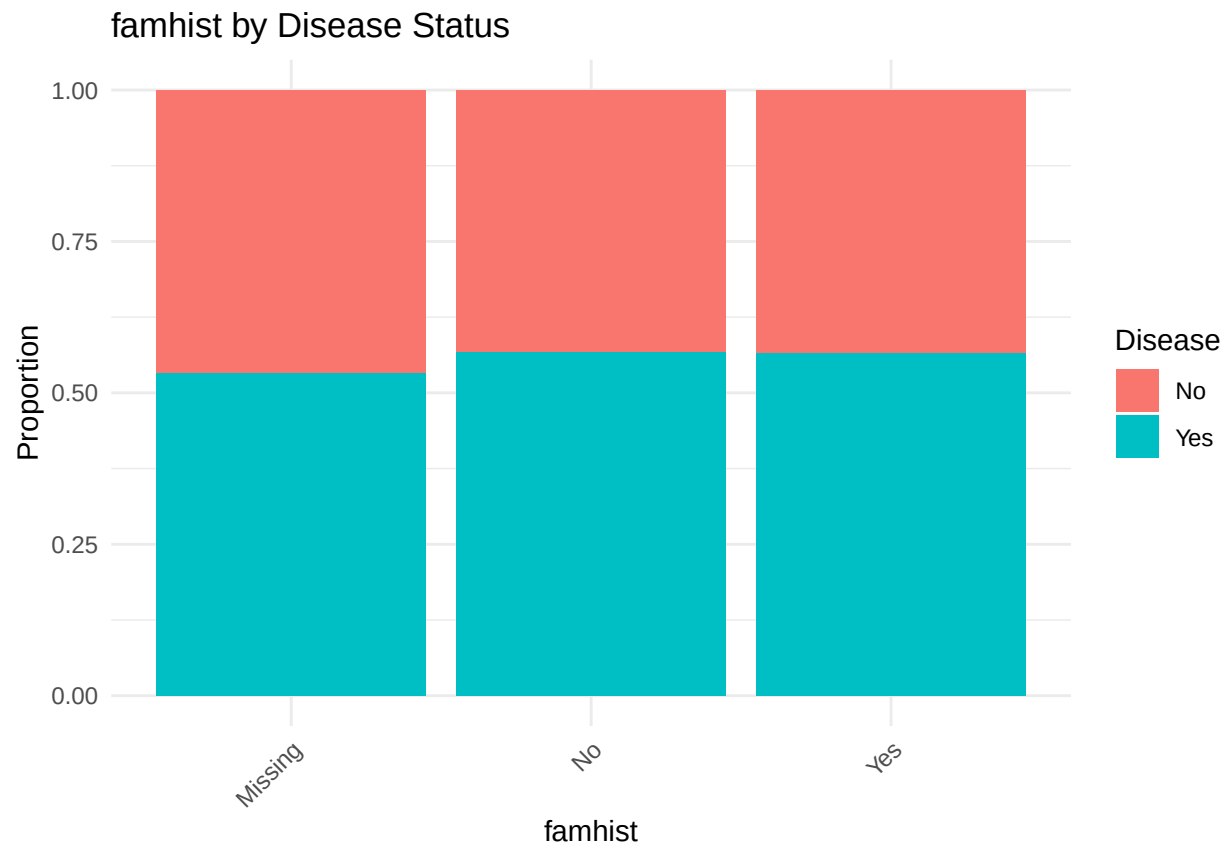


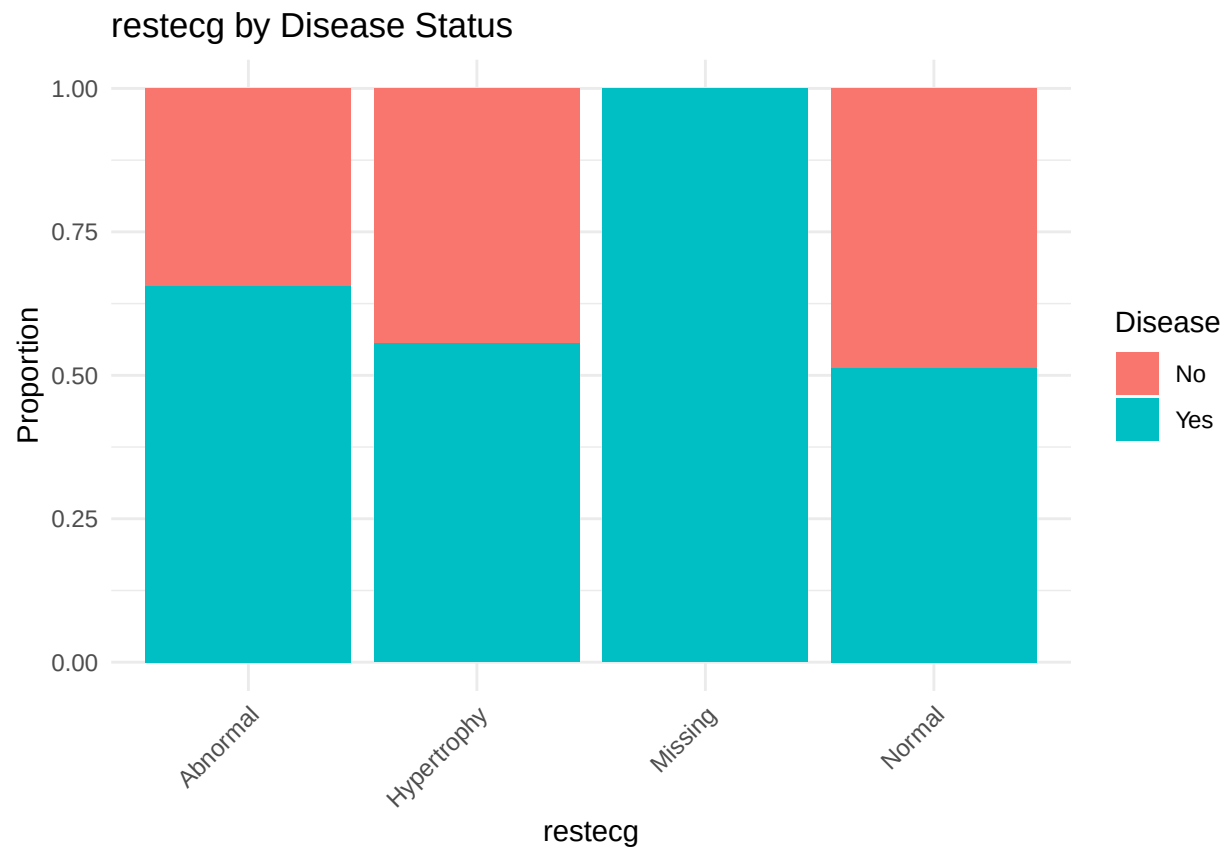


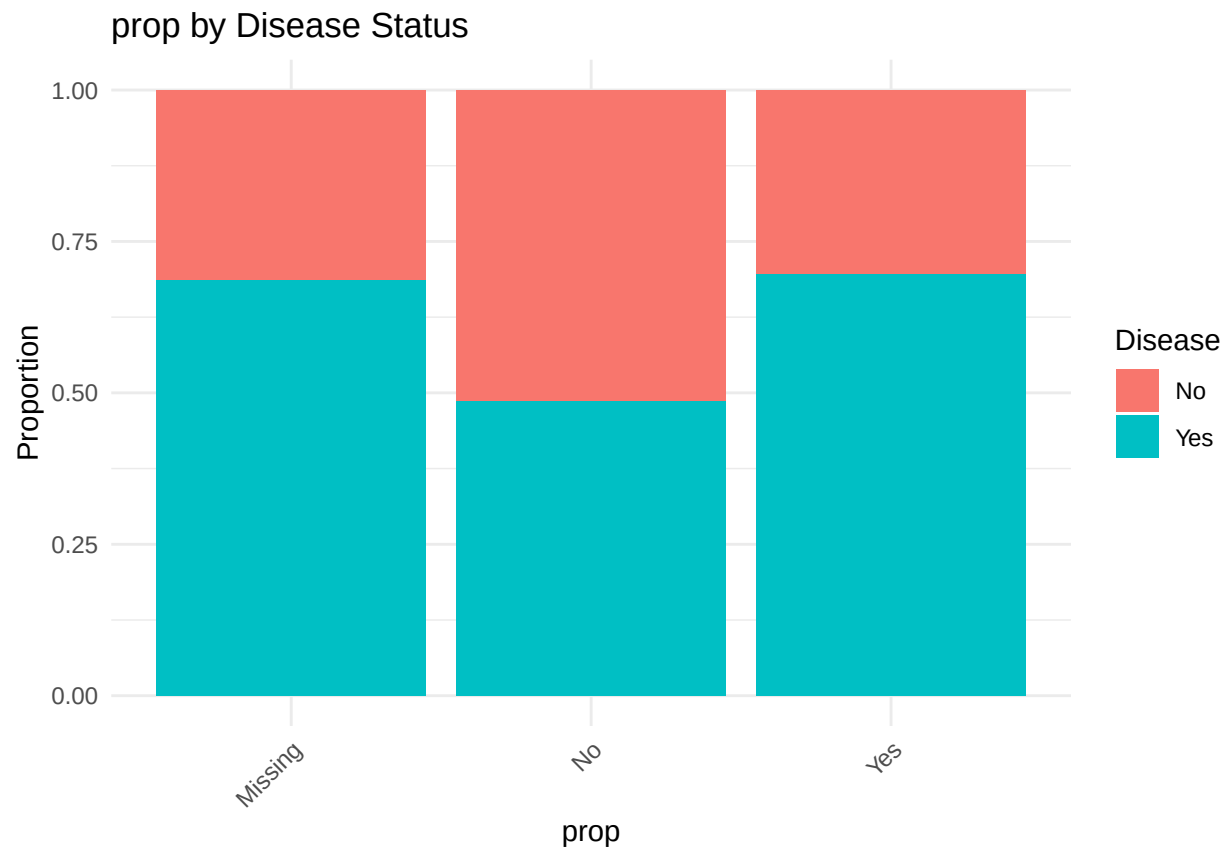


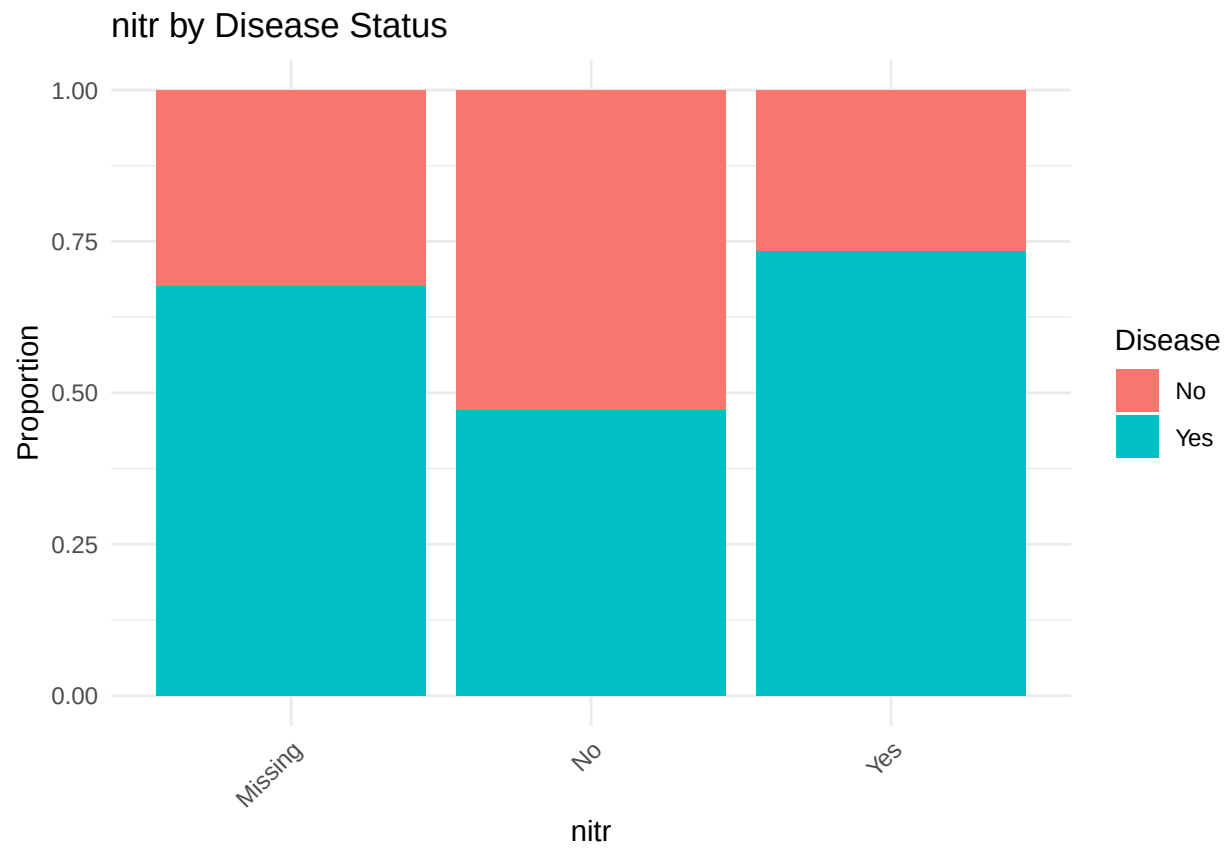


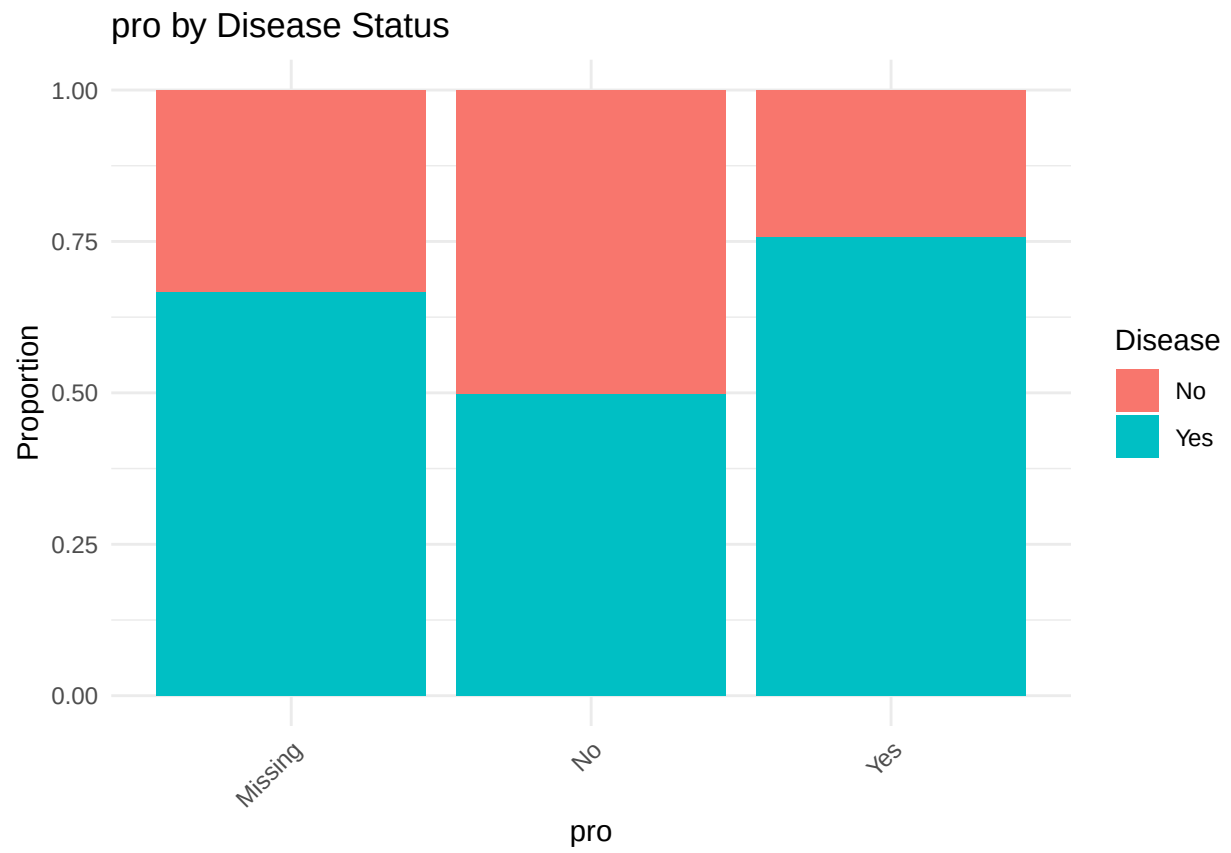


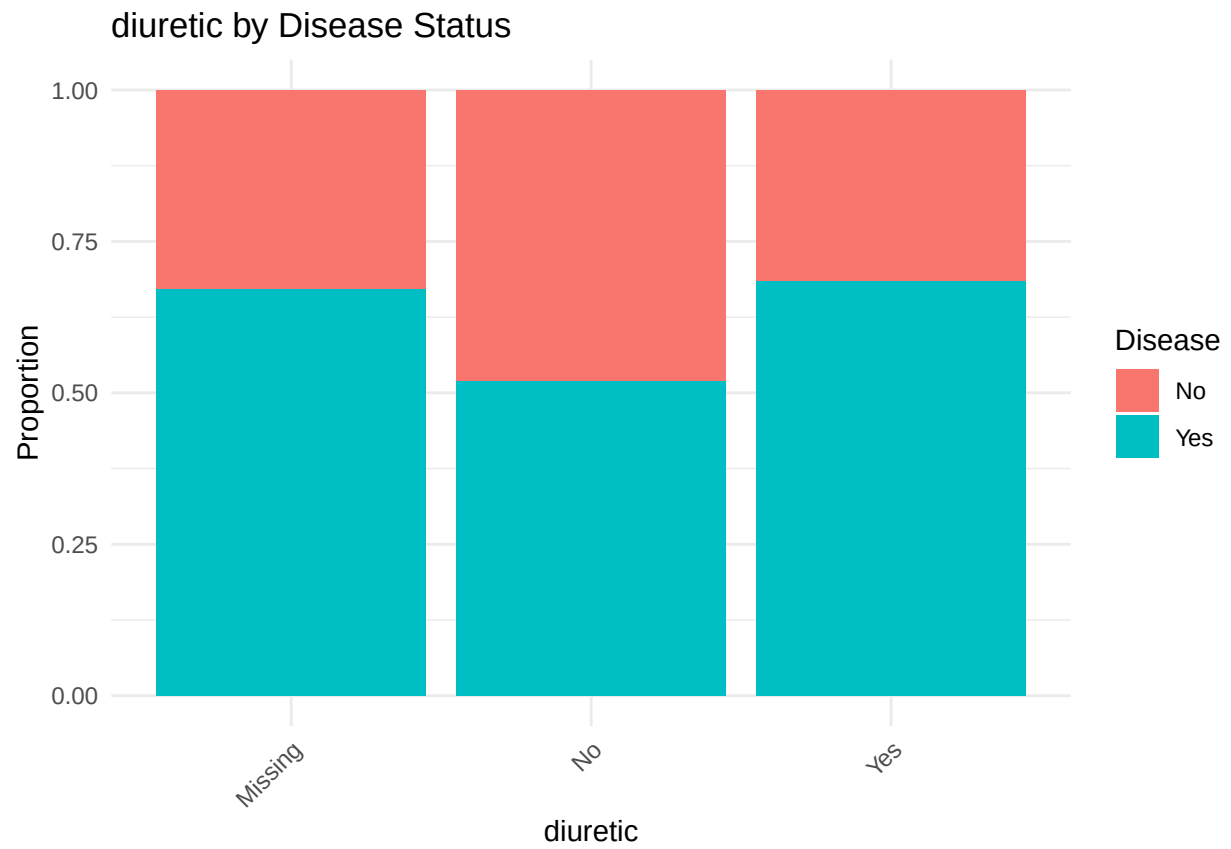


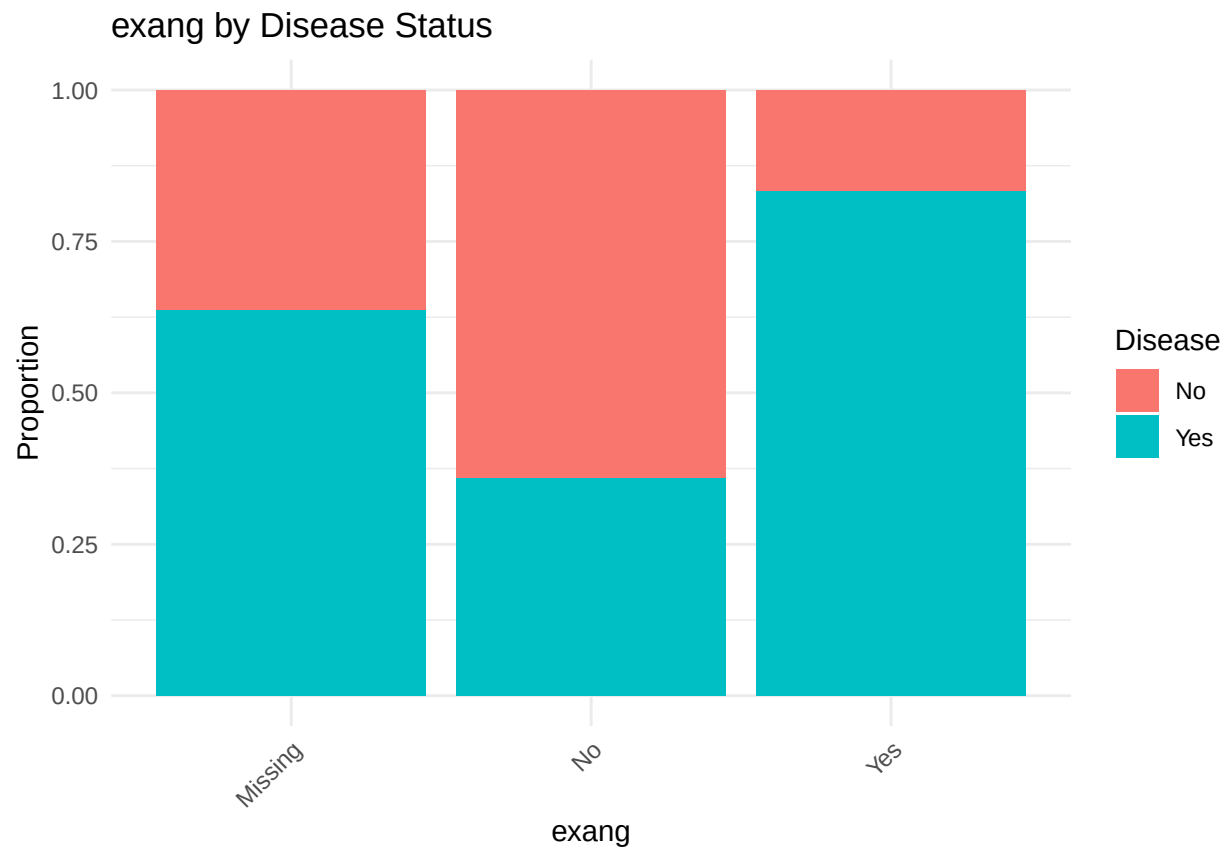


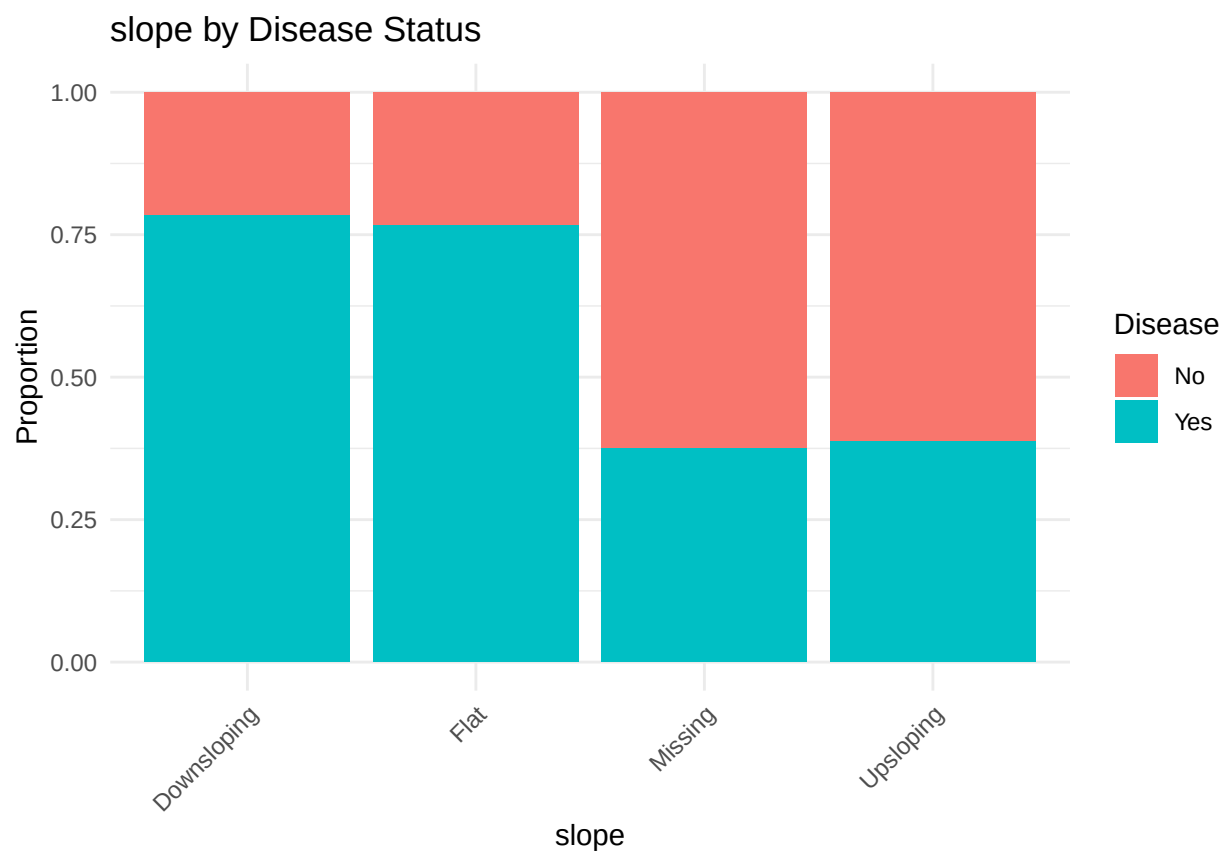


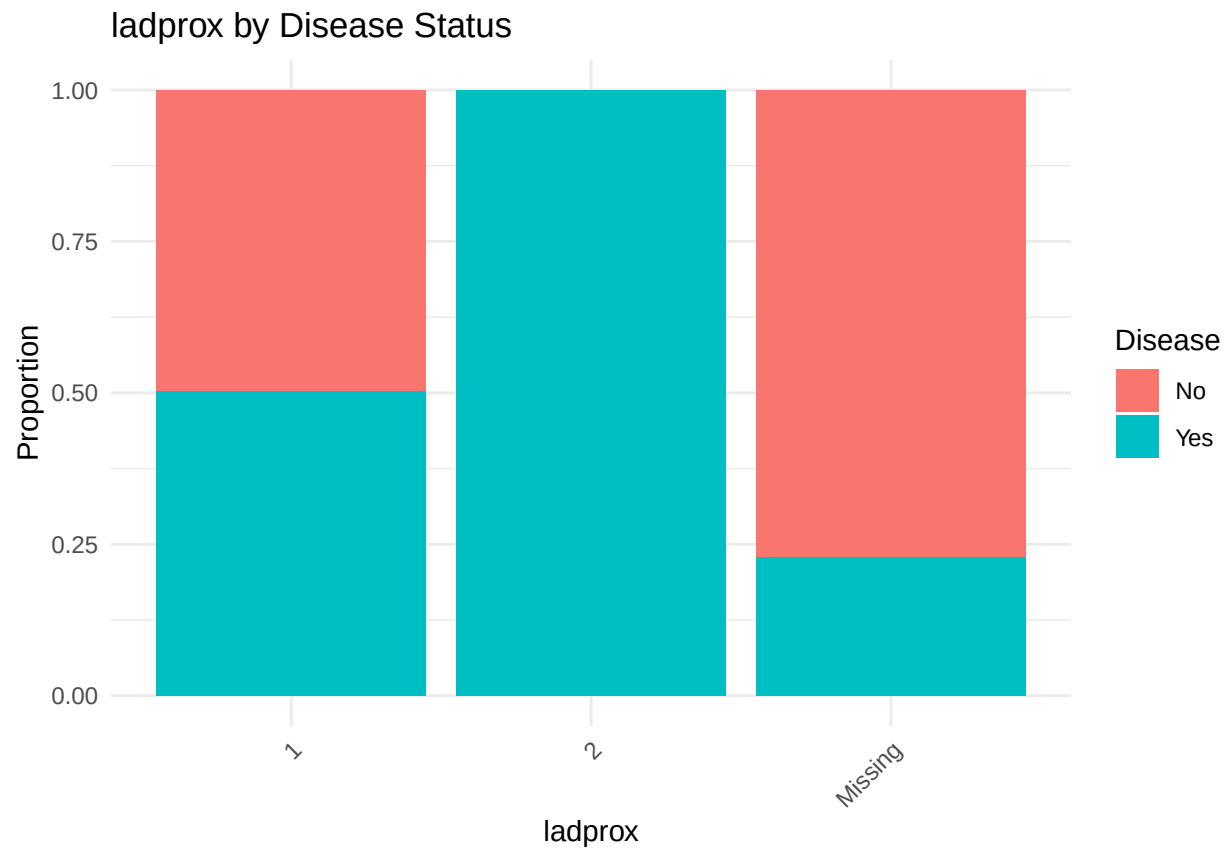


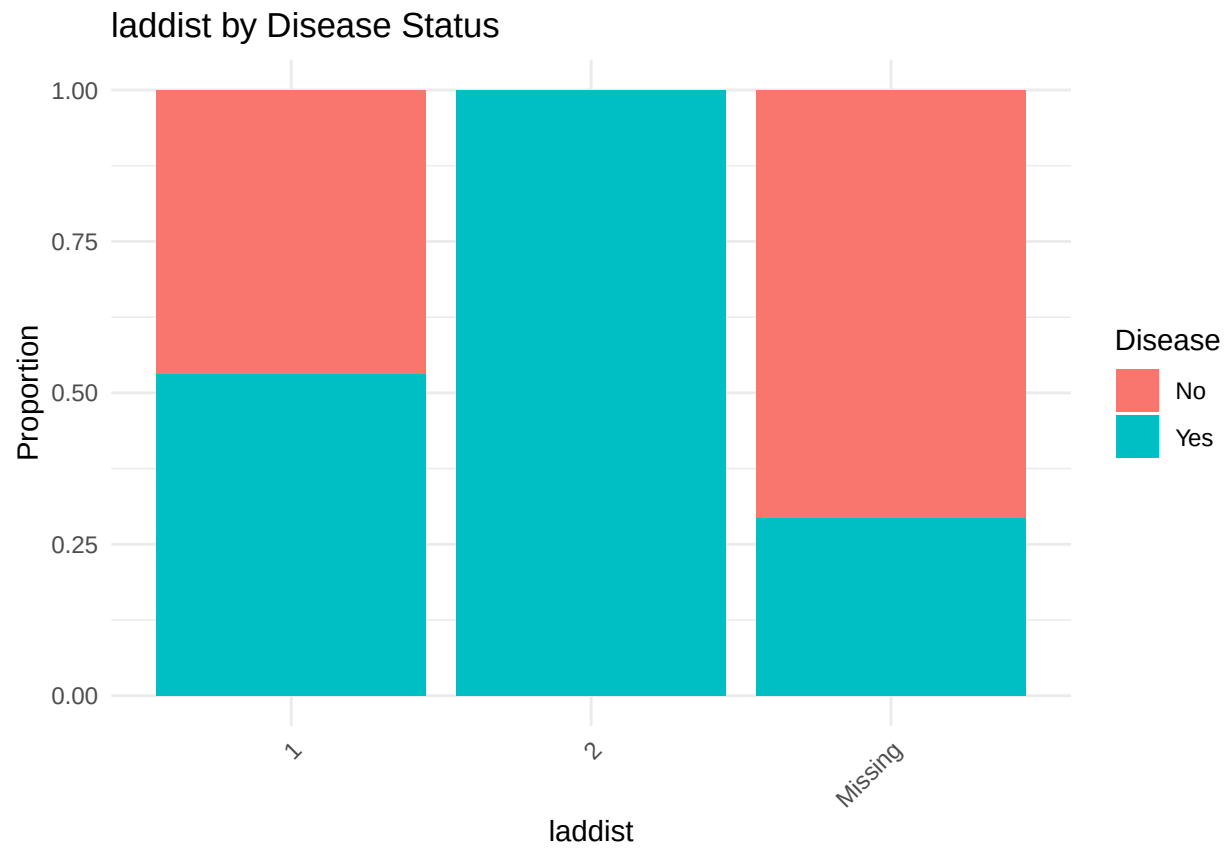


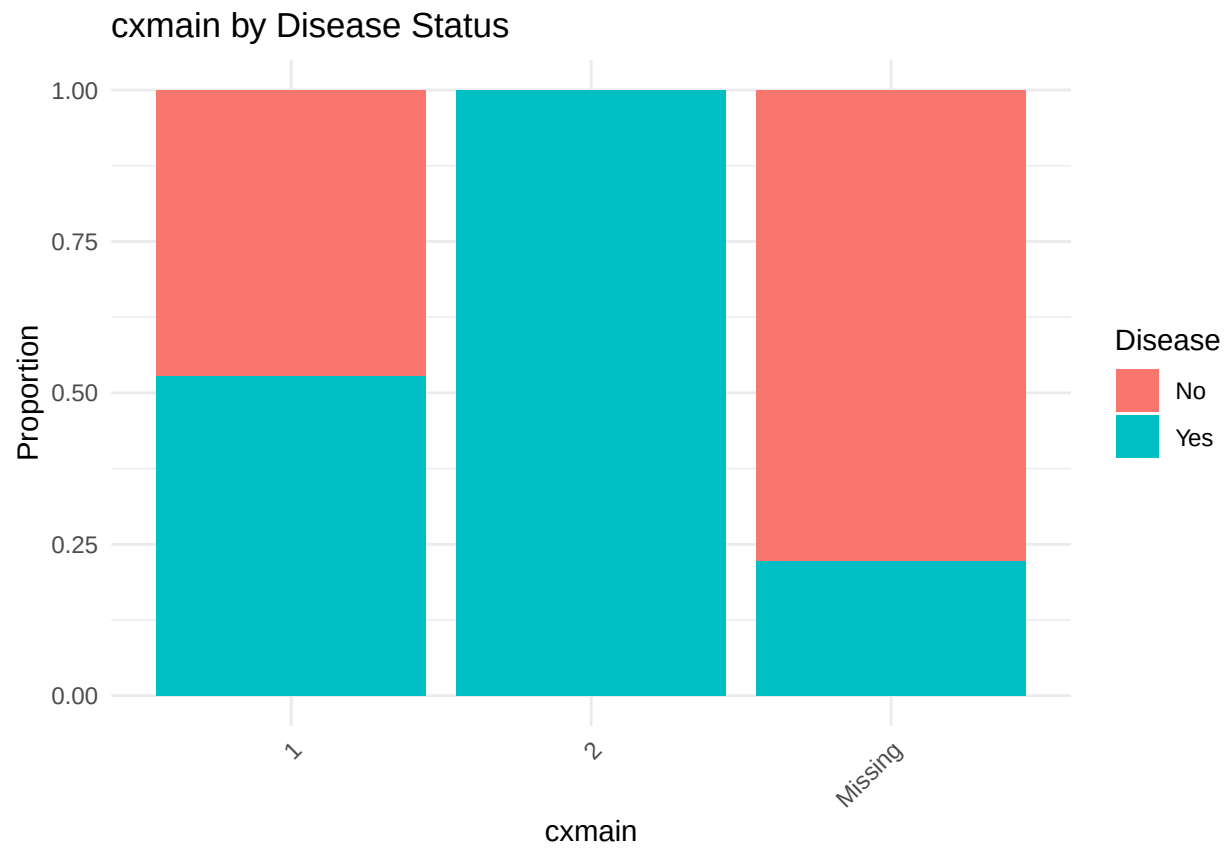


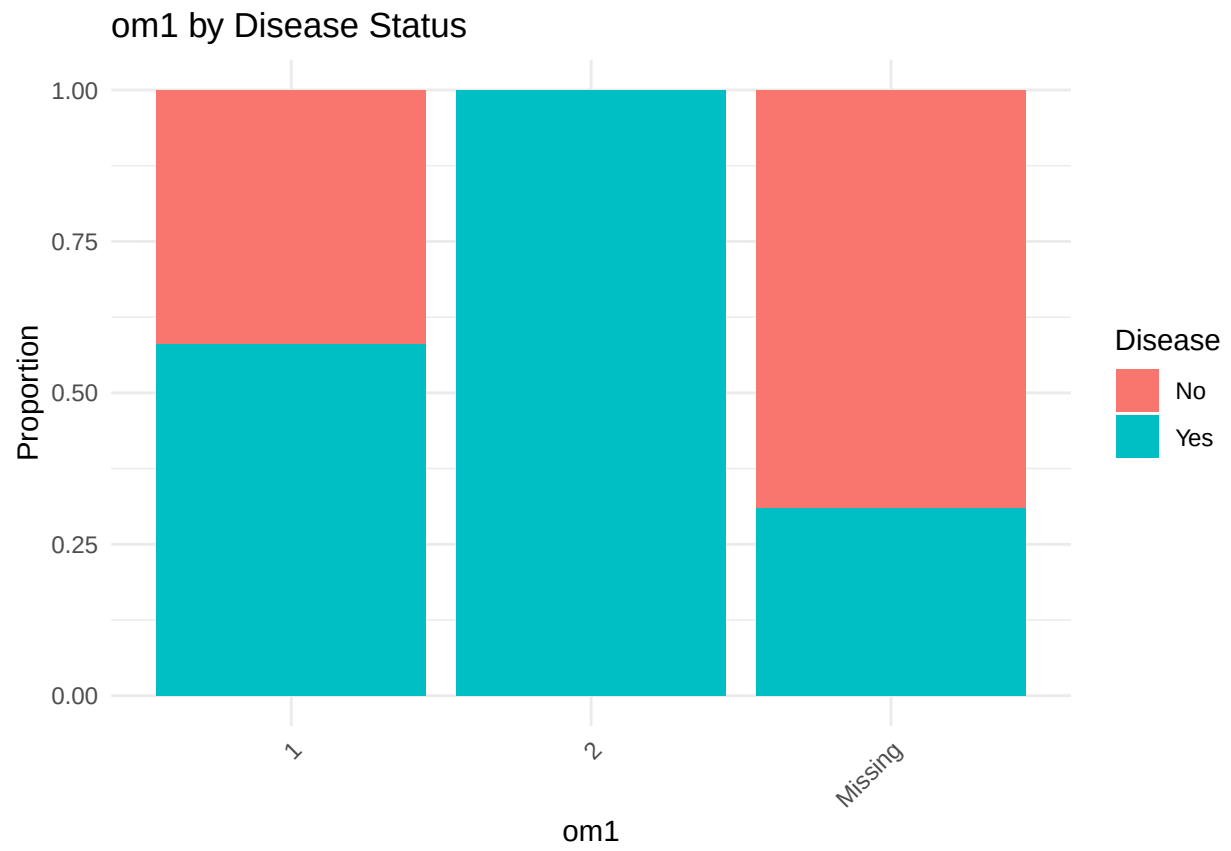


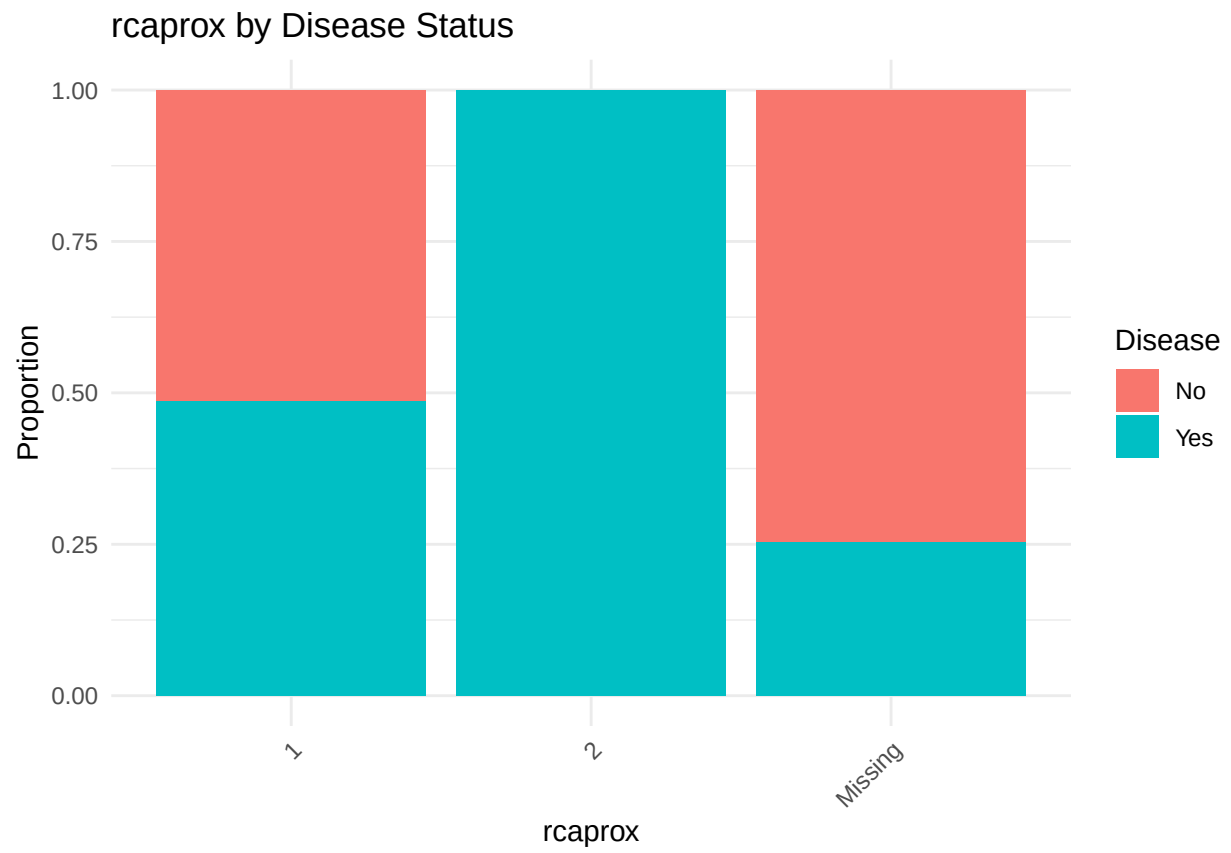


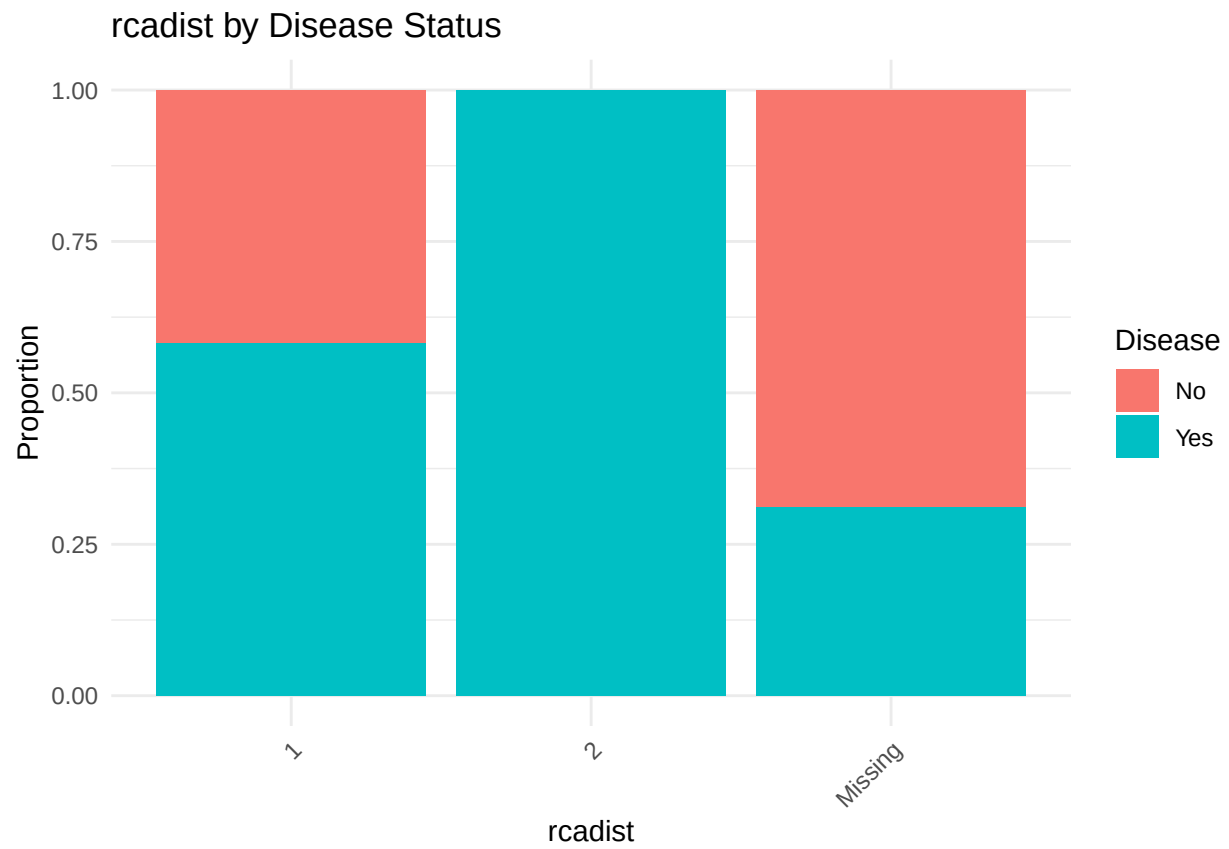


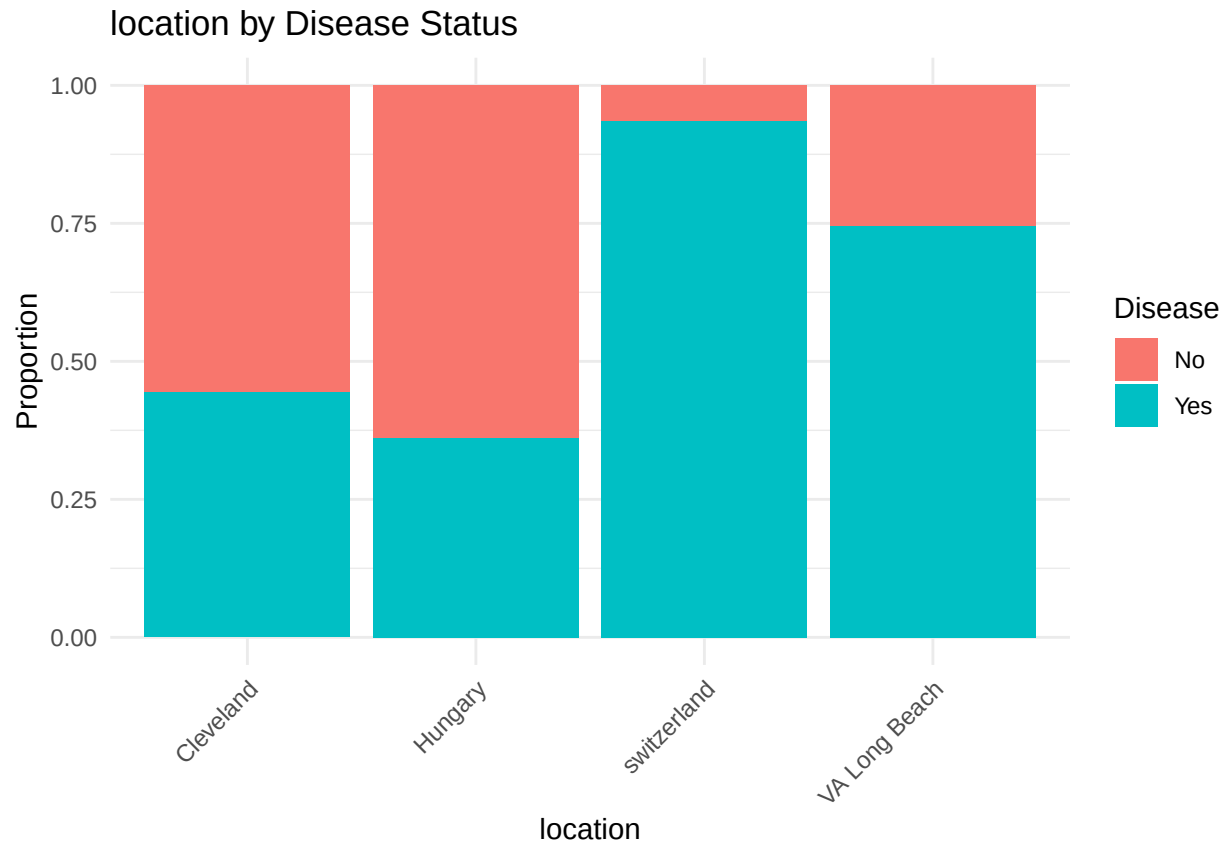


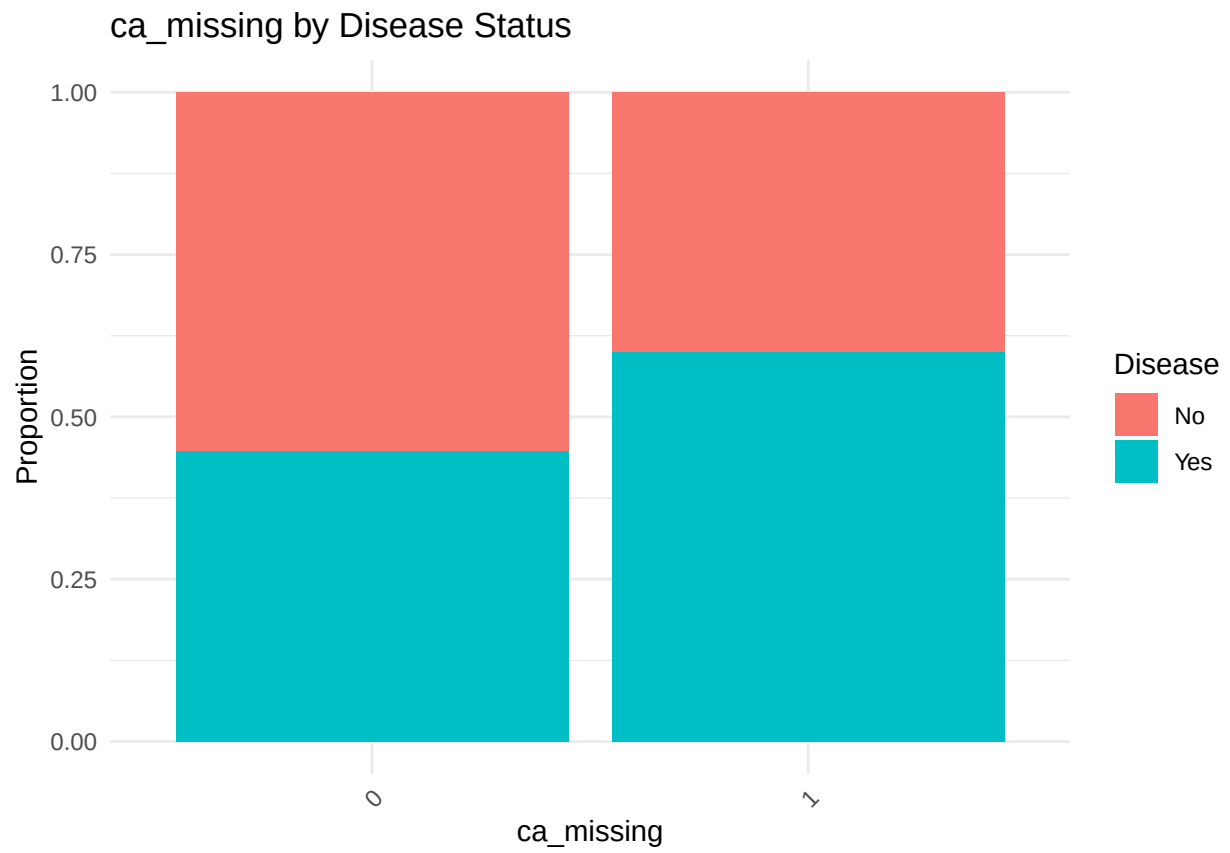


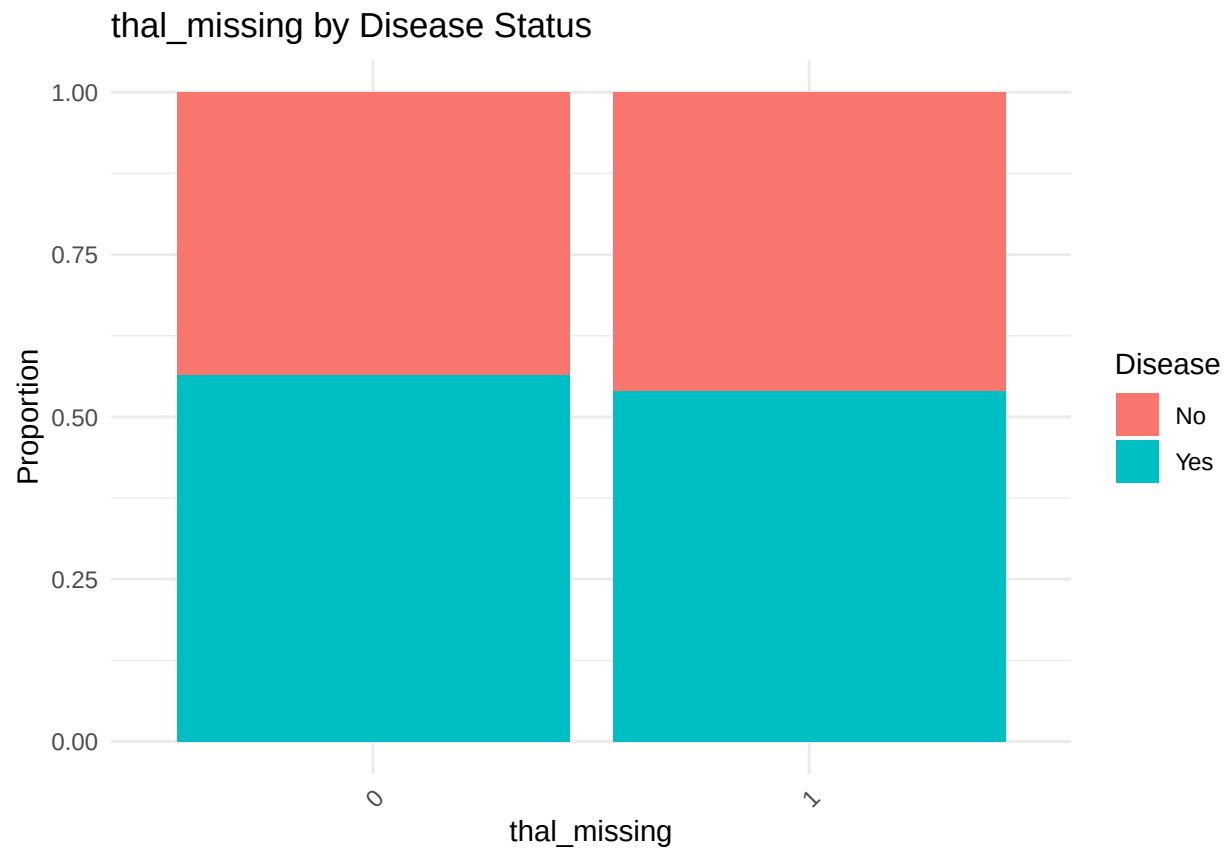




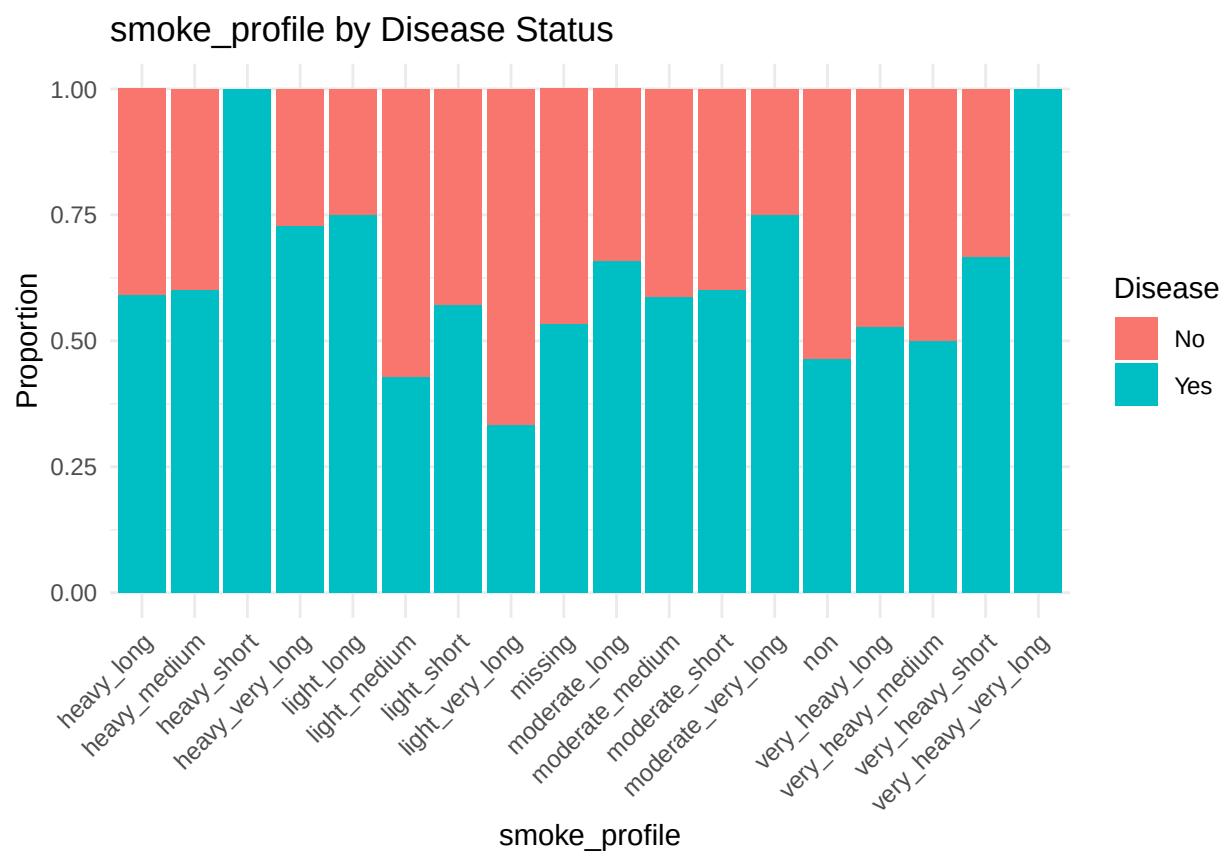


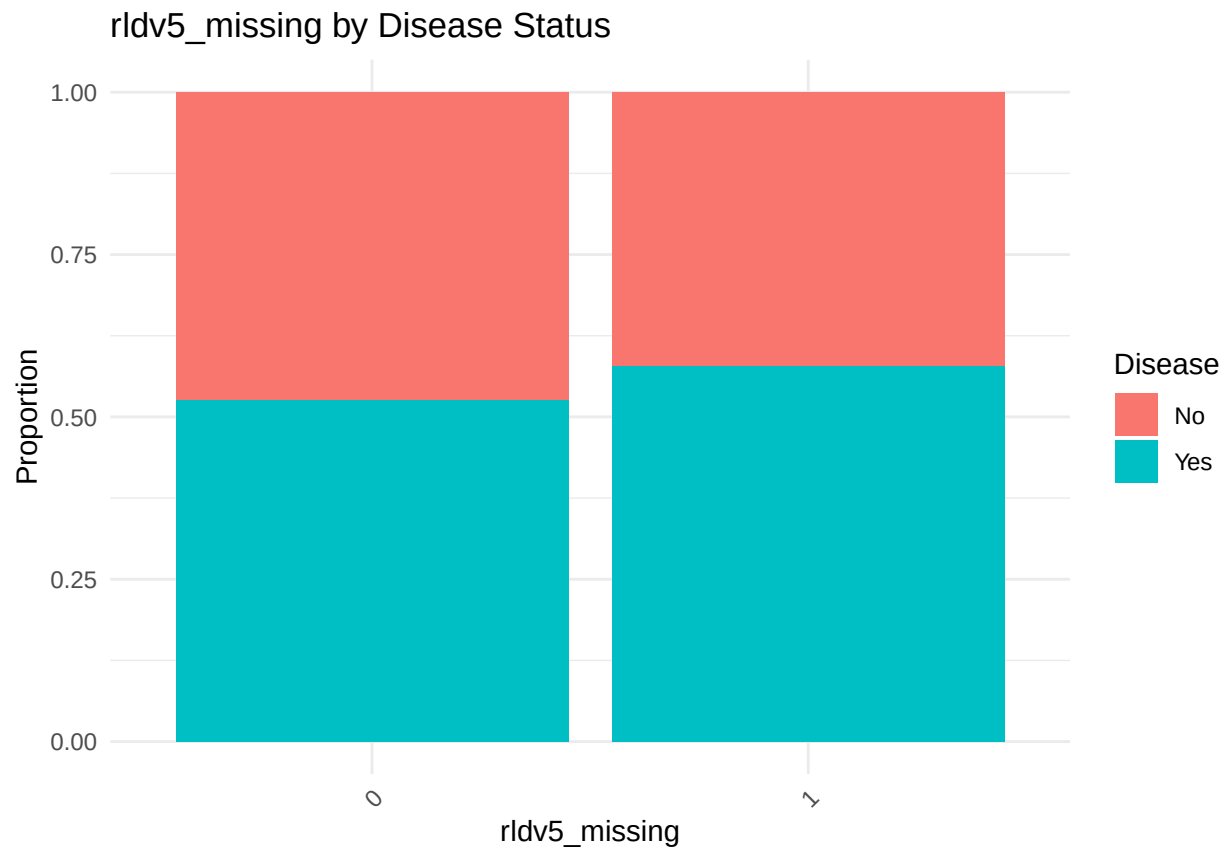












The following groups of variables have similar distributions which means they are possibly correlated:

Group 1: “prop”, “nitr”, “pro”, “diuretic”, “exang” Group 2: “ladprox”, “laddist”, “cxmain”, “om1”, “rcaprox”, “readist”

```
cat_vars_to_keep <- c("sex", "painexer", "relrest", "cp", "htn", "fbs", "restecg", "exang", "slope", "l
```

```
# cramer v
cramerv <- function(dataframe, var_list){
  cramerV_matrix <- matrix(NA, nrow = length(var_list), ncol = length(var_list),
    dimnames = list(var_list, var_list))

  for (i in seq_along(var_list)) {
    for (j in seq_along(var_list)) {
      tbl <- table(dataframe[[var_list[i]]], dataframe[[var_list[j]]])
      cramerV_matrix[i, j] <- suppressWarnings(cramerV(tbl))
    }
  }
  print(cramerV_matrix)
}

# Check correlation of variables that to have similar distribution
cramerv(df, c("prop", "nitr", "pro", "diuretic", "exang"))
```

Cramer’s V association test

```
##          prop   nitr   pro diuretic  exang
```

```
## prop      1.0000 0.7140 0.7072    0.6344 0.6277
## nitr      0.7140 1.0000 0.7479    0.6319 0.6450
## pro       0.7072 0.7479 1.0000    0.6259 0.6482
## diuretic  0.6344 0.6319 0.6259    1.0000 0.5616
## exang     0.6277 0.6450 0.6482    0.5616 1.0000

cramerv(df, c("ladprox", "laddist", "cxmain", "om1", "rcaprox", "rcadist"))

##          ladprox laddist cxmain    om1 rcaprox rcadist
## ladprox  1.0000  0.5490 0.5943 0.6370  0.6245  0.6277
## laddist  0.5490  1.0000 0.5914 0.6084  0.5994  0.6215
## cxmain   0.5943  0.5914 1.0000 0.6150  0.6324  0.6126
## om1      0.6370  0.6084 0.6150 1.0000  0.6385  0.6851
## rcaprox  0.6245  0.5994 0.6324 0.6385  1.0000  0.6128
## rcadist  0.6277  0.6215 0.6126 0.6851  0.6128  1.0000

drop_colinearity_cols <- c("prop", "nitr", "pro", "diuretic", "laddist",
                           "cxmain", "om1", "rcaprox", "rcadist")
```

Based on the results, each set of variables are moderate to highly correlated. To reduce multicollinearity, the following variables will be kept due to clinical significance: * **exang** - exercise-induced angina * **ladprox** - proximal LAD artery

Summary for categorical variables

```
# recommendation code generated with the help of ChatGPT.
cat_summary <- data.frame(
  Variable = character(),
  p_value = numeric(),
  Recommendation = character(),
  stringsAsFactors = FALSE
)

for (var in cat_vars) {
  tbl <- table(df[[var]], df$has_disease)
  p_val <- tryCatch(chisq.test(tbl)$p.value, error = function(e) NA)

  rec <- if (!is.na(p_val) && p_val < 0.05) {
    "Include (predictive)"
  } else {
    "Optional (evaluate with model)"
  }

  cat_summary <- rbind(
    cat_summary,
    data.frame(Variable = var, p_value = round(p_val, 5), Recommendation = rec)
  )
}

## Warning in chisq.test(tbl): Chi-squared approximation may be incorrect
## Warning in chisq.test(tbl): Chi-squared approximation may be incorrect

cat_summary
```

```
##          Variable p_value          Recommendation
## 1          sex 0.00000      Include (predictive)
```

```
## 2      painloc 0.00001      Include (predictive)
## 3      painexer 0.00000      Include (predictive)
## 4      relrest 0.00000      Include (predictive)
## 5      cp 0.00000      Include (predictive)
## 6      htn 0.00051      Include (predictive)
## 7      fbs 0.00000      Include (predictive)
## 8      famhist 0.61273 Optional (evaluate with model)
## 9      restecg 0.00567      Include (predictive)
## 10     prop 0.00000      Include (predictive)
## 11     nitr 0.00000      Include (predictive)
## 12     pro 0.00000      Include (predictive)
## 13     diuretic 0.00082      Include (predictive)
## 14     exang 0.00000      Include (predictive)
## 15     slope 0.00000      Include (predictive)
## 16     ladprox 0.00000      Include (predictive)
## 17     laddist 0.00000      Include (predictive)
## 18     cxmain 0.00000      Include (predictive)
## 19     om1 0.00000      Include (predictive)
## 20     rcaprox 0.00000      Include (predictive)
## 21     rcadist 0.00000      Include (predictive)
## 22     location 0.00000      Include (predictive)
## 23     ca_missing 0.00002      Include (predictive)
## 24     thal_missing 0.49284 Optional (evaluate with model)
## 25     thaltime_missing 0.00000      Include (predictive)
## 26     smoke_profile 0.22461 Optional (evaluate with model)
## 27     rldv5_missing 0.12282 Optional (evaluate with model)
```

```
# create list of all variables recommended as Include
```

```
selected_cat_vars <- cat_summary %>%
  filter(str_detect(Recommendation, "Include")) %>%
  pull(Variable)
```

```
selected_cat_vars
```

```
## [1] "sex"      "painloc"  "painexer" "relrest"
## [5] "cp"       "htn"      "fbs"       "restecg"
## [9] "prop"     "nitr"     "pro"       "diuretic"
## [13] "exang"    "slope"    "ladprox"   "laddist"
## [17] "cxmain"   "om1"      "rcaprox"   "rcadist"
## [21] "location" "ca_missing" "thaltime_missing"
```

Final Variable Sets

```
# combine selected variables
```

```
included_vars <- c(selected_num_vars, selected_cat_vars)
```

```
model_df <- df %>%
  select(all_of(included_vars), has_disease) %>%
  select(-all_of(drop_colinearity_cols))

str(model_df)
```

```
## tibble [899 x 39] (S3: tbl_df/tbl/data.frame)
## $ age      : num [1:899] 63 67 67 37 41 56 62 57 63 53 ...
## $ ca_imputed : num [1:899] 0 3 2 0 0 0 2 0 1 0 ...
```

```
## $ thal_imputed : num [1:899] 6 3 7 3 3 3 3 7 7 ...
## $ trestbps_missing: num [1:899] 0 0 0 0 0 0 0 0 0 ...
## $ trestbps_imputed: num [1:899] 145 160 120 130 130 120 140 120 130 140 ...
## $ chol_missing : num [1:899] 0 0 0 0 0 0 0 0 0 ...
## $ chol_imputed : num [1:899] 233 286 229 250 204 236 268 354 254 203 ...
## $ thaldur_missing : num [1:899] 0 0 0 0 0 0 0 0 0 ...
## $ thaldur_imputed : num [1:899] 10.5 9.5 8.5 13 7 11.3 6 9 8 5.5 ...
## $ met_missing : num [1:899] 0 0 0 0 0 0 0 0 0 ...
## $ met_imputed : num [1:899] 13 13 10 17 9 16 7 10 9 7 ...
## $ thalach_missing : num [1:899] 0 0 0 0 0 0 0 0 0 ...
## $ thalach_imputed : num [1:899] 150 108 129 187 172 178 160 163 147 155 ...
## $ thalrest_missing: num [1:899] 0 0 0 0 0 0 0 0 0 ...
## $ thalrest_imputed: num [1:899] 60 64 78 84 71 73 83 84 75 86 ...
## $ tpeakbps_missing: num [1:899] 0 0 0 0 0 0 0 0 0 ...
## $ tpeakbps_imputed: num [1:899] 190 160 140 195 160 165 180 165 120 185 ...
## $ tpeakbpd_missing: num [1:899] 0 0 0 0 0 0 0 0 0 ...
## $ trestbpd_missing: num [1:899] 0 0 0 0 0 0 0 0 0 ...
## $ oldpeak_missing : num [1:899] 0 0 0 0 0 0 0 0 0 ...
## $ oldpeak_imputed : num [1:899] 2.3 1.5 2.6 3.5 1.4 0.8 3.6 0.6 1.4 3.1 ...
## $ rldv5e_missing : num [1:899] 0 0 0 0 0 0 0 0 0 ...
## $ rldv5e_imputed : num [1:899] 172 185 150 167 40 127 122 122 75 68 ...
## $ lmt_missing : num [1:899] 0 0 0 0 0 0 0 0 0 ...
## $ sex : Factor w/ 2 levels "Female","Male": 2 2 2 2 1 2 1 1 2 2 ...
## $ painloc : Factor w/ 3 levels "Missing","Otherwise",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ painexer : Factor w/ 3 levels "Missing","Otherwise",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ relrest : Factor w/ 3 levels "Missing","Otherwise",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ cp : Ord.factor w/ 4 levels "Typical Angina"<...: 1 4 4 3 2 2 4 4 4 4 ...
## $ htn : Factor w/ 3 levels "0","1","Missing": 2 2 2 1 2 2 1 2 2 1 ...
## $ fbs : Factor w/ 3 levels "Missing","No",...: 3 2 2 2 2 2 2 2 2 3 ...
## $ restecg : Factor w/ 4 levels "Abnormal","Hypertrophy",...: 2 2 2 4 2 4 2 4 2 2 ...
## $ exang : Factor w/ 3 levels "Missing","No",...: 2 3 3 2 2 2 2 3 2 3 ...
## $ slope : Factor w/ 4 levels "Downsloping",...: 1 2 2 1 4 4 1 4 2 1 ...
## $ ladprox : Factor w/ 3 levels "1","2","Missing": 1 2 1 1 1 1 2 1 2 1 ...
## $ location : chr [1:899] "Cleveland" "Cleveland" "Cleveland" "Cleveland" ...
## $ ca_missing : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
## $ thaltime_missing: Factor w/ 2 levels "0","1": 1 1 1 1 2 2 1 1 1 1 ...
## $ has_disease : Factor w/ 2 levels "No","Yes": 1 2 2 1 1 1 2 1 2 2 ...
```

Export dataset for modeling

```
# export modeling dataset to RDS file
saveRDS(model_df, here("Data/Clean", "heart_disease_modeling.rds"))

model_df %>% dim()
```

```
## [1] 899 39
```

The final modeling set has 899 observations and 39 features including the target variable.